Bioinformatics

KB8029

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Highlights:

Databases, records and evidence

Choosing databases and connecting DNA, RNA and protein records

Using identifiers and cross-references and accessing evidence and versions

Database access and retrieval

Command-line/programmatic access and database web pages

Genome browsers: DNA, RNA and protein in one interface

Reading literature online

Database searches and AI-assisted discovery

Building queries and checking original sources

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Databases

A database is an organised collection of data that supports searching and retrieval.

Database management software stores, updates and retrieves the data.

Hardware provides storage and computing power.

A database website provides an interface for searching and viewing records.

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Growing data volumes

Decimal units and dated examples

Unit	Bytes per unit	Symbol	Example	Size
Byte	10^0	B	One ASCII character	1 B
Kilobyte	10^3	kB	1,000 ASCII characters	1 kB
Megabyte	10^6	MB	Swiss-Prot, compressed text (2026_03)	~700 MB
Gigabyte	10^9	GB	TrEMBL, compressed text (2026_03)	~121 GB
Terabyte	10^{12}	TB	PDB Core snapshot (1 Jan 2026)	~1.5 TB
Petabyte	10^{15}	PB	SRA unique data (April 2026)	>50 PB

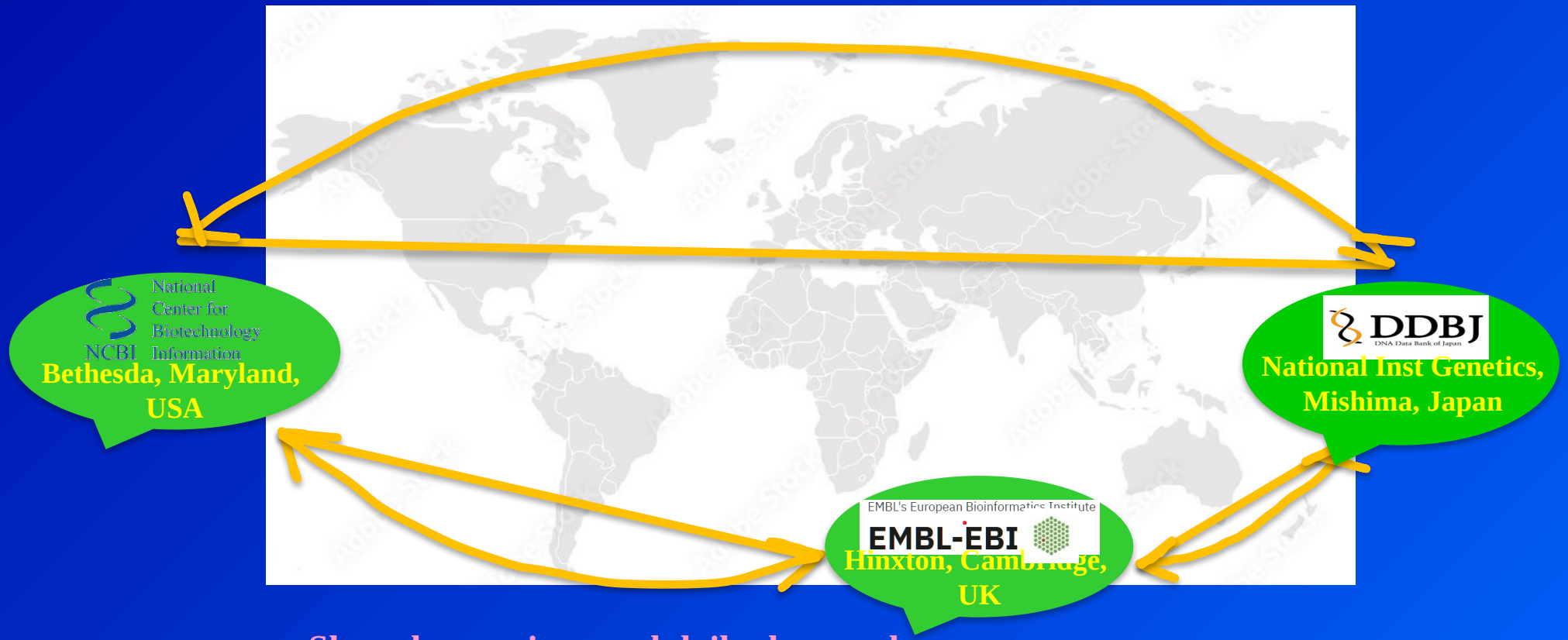
Full copies of the largest archives are impractical.

Use shared databases efficiently.

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Shared sequence archives

International Nucleotide Sequence Database Collaboration (INSDC)



Shared accessions and daily data exchange

Common standards, different search interfaces

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Shared sequence archives

Access data through any partner.

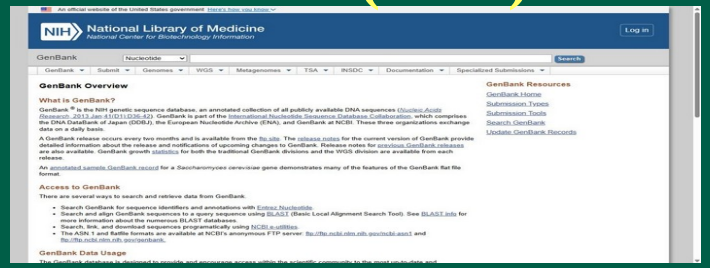
Submit data to only one.





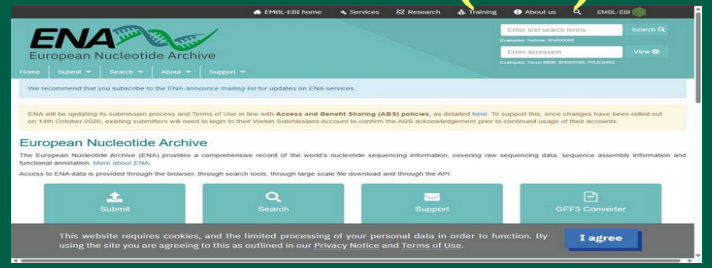


NCBI (USA)



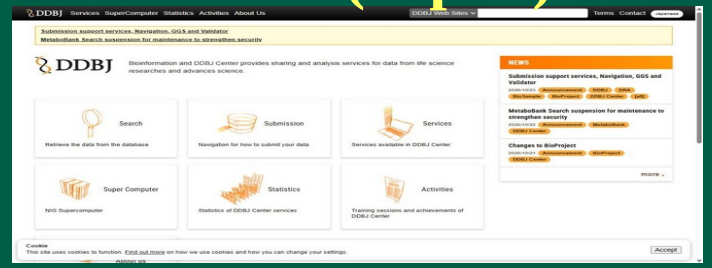
ncbi.nlm.nih.gov/genbank

EMBL-EBI (UK)



ebi.ac.uk/ena/browser

DDBJ (Japan)



ddbj.nig.ac.jp

Daily exchange

Daily exchange

Connected records

Read archives
SRA / ENA reads / DRA
Reads and quality data

Assembled sequences and annotations
GenBank / ENA / DDBJ

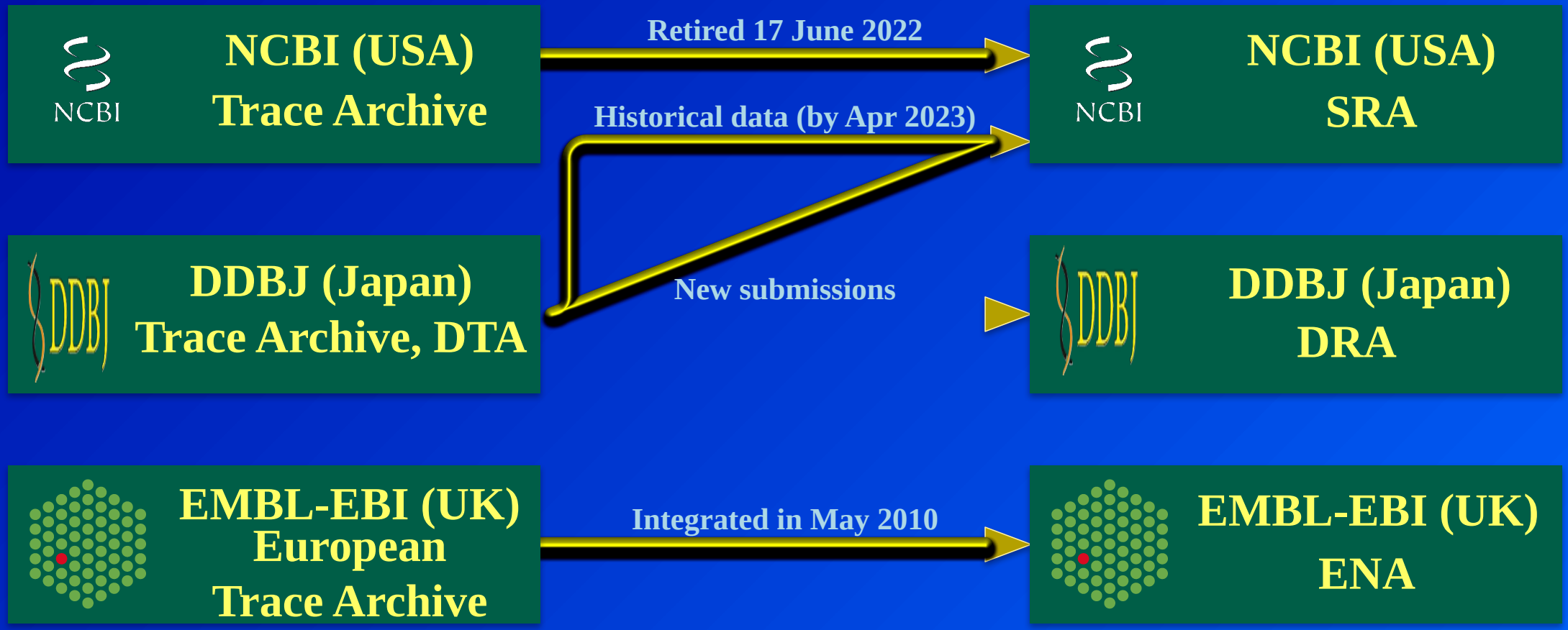
BioProject / Study
Project description

BioSample / Sample
Biological material

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Shared sequence archives

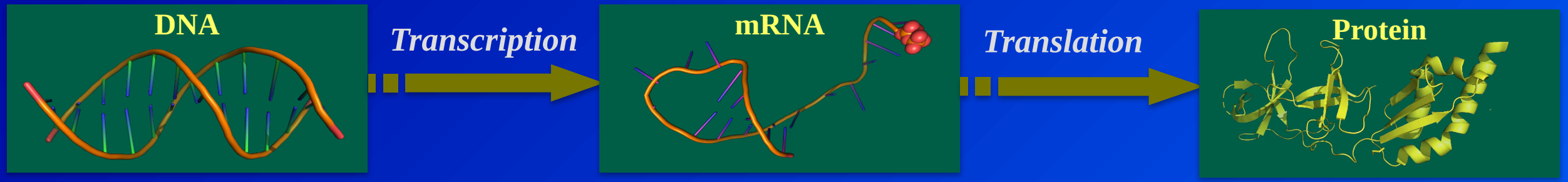
Archives merge and access routes change.



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DNA, RNA and protein data

Central dogma of biology



Linked database records

DNA-level data

Genomic sequences and gene context

NIH GenBank

EMBL's European Bioinformatics Institute EMBL-EBI

DDBJ

Cross-references

RNA-level data

RefSeq

mRNAsequences

RNAcentral

Noncoding RNA sequences

Rfam

RNA families

Cross-references

Protein-level data

Sequence and structure

Function and interactions

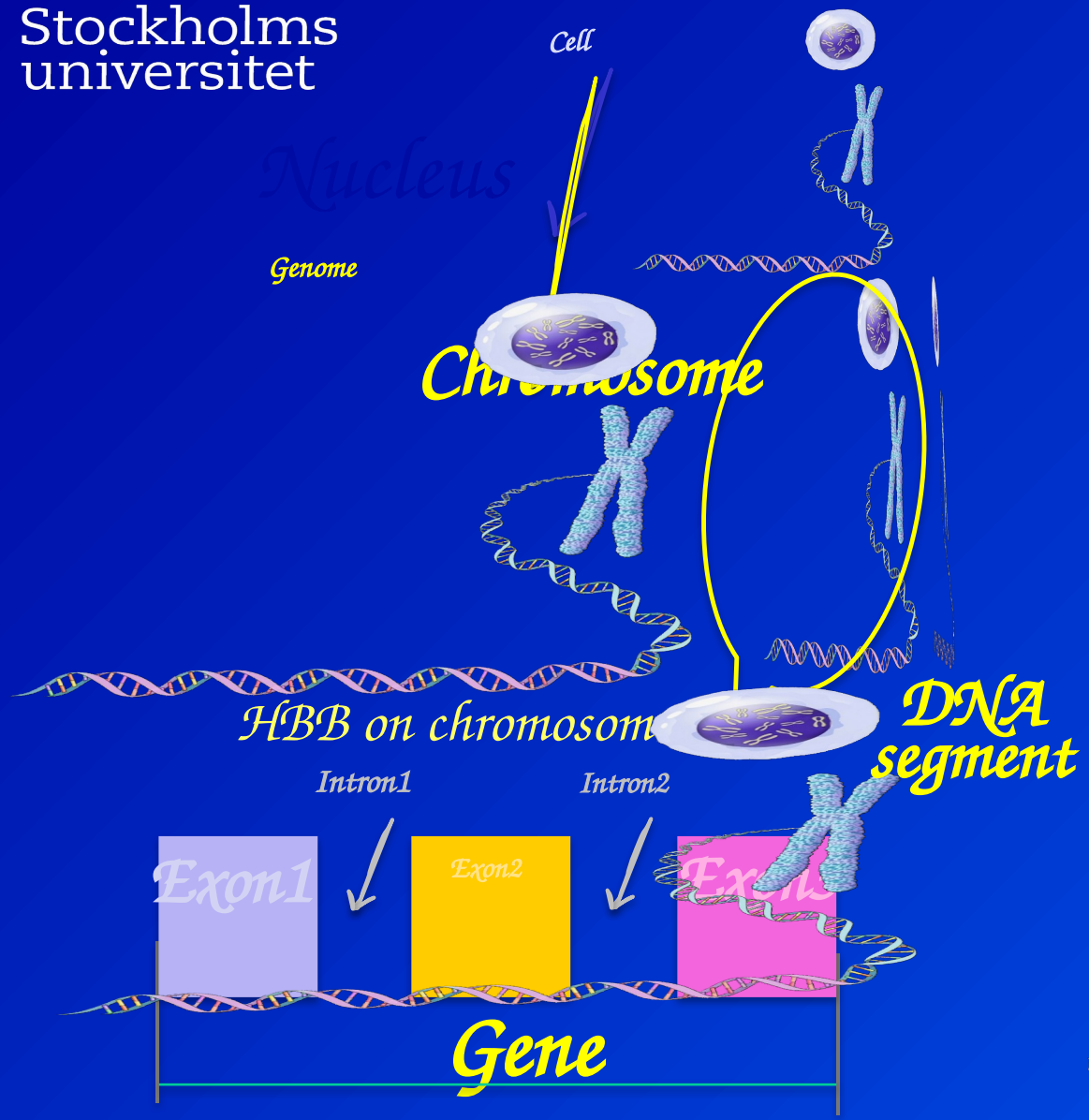
UniProt

RCSB PDB PROTEIN DATA BANK

IntAct

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DNA-level data



Genome

Genome assemblies and their annotations

Chromosome overview

Chromosome sequences and mapped information

Selected DNA region

Genes and non-coding DNA within a region

Gene

NCBI Gene

Gene structure and links to RNA and proteins

NCBI GenBank

ENA ENA

DDBJ DDBJ Annotated/ Assembled sequences

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RNA-level data



RNA transcripts

Sequences, exon composition and transcript variants

RefSeq

Ensembl

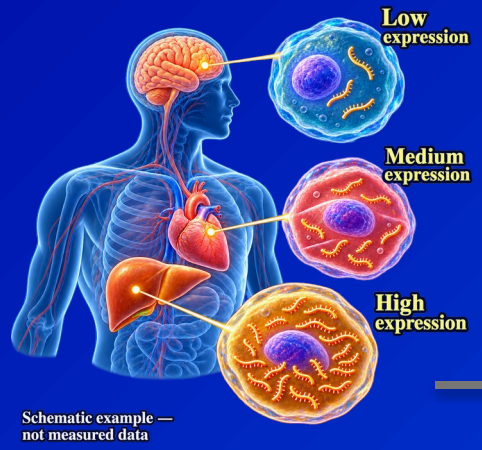


Noncoding RNAs

RNA sequences, families and structure

RNAcentral

Rfam



RNA expression

Where RNA is detected and how abundant it is

Expression Atlas

NCBI GEO DDBJ GEA

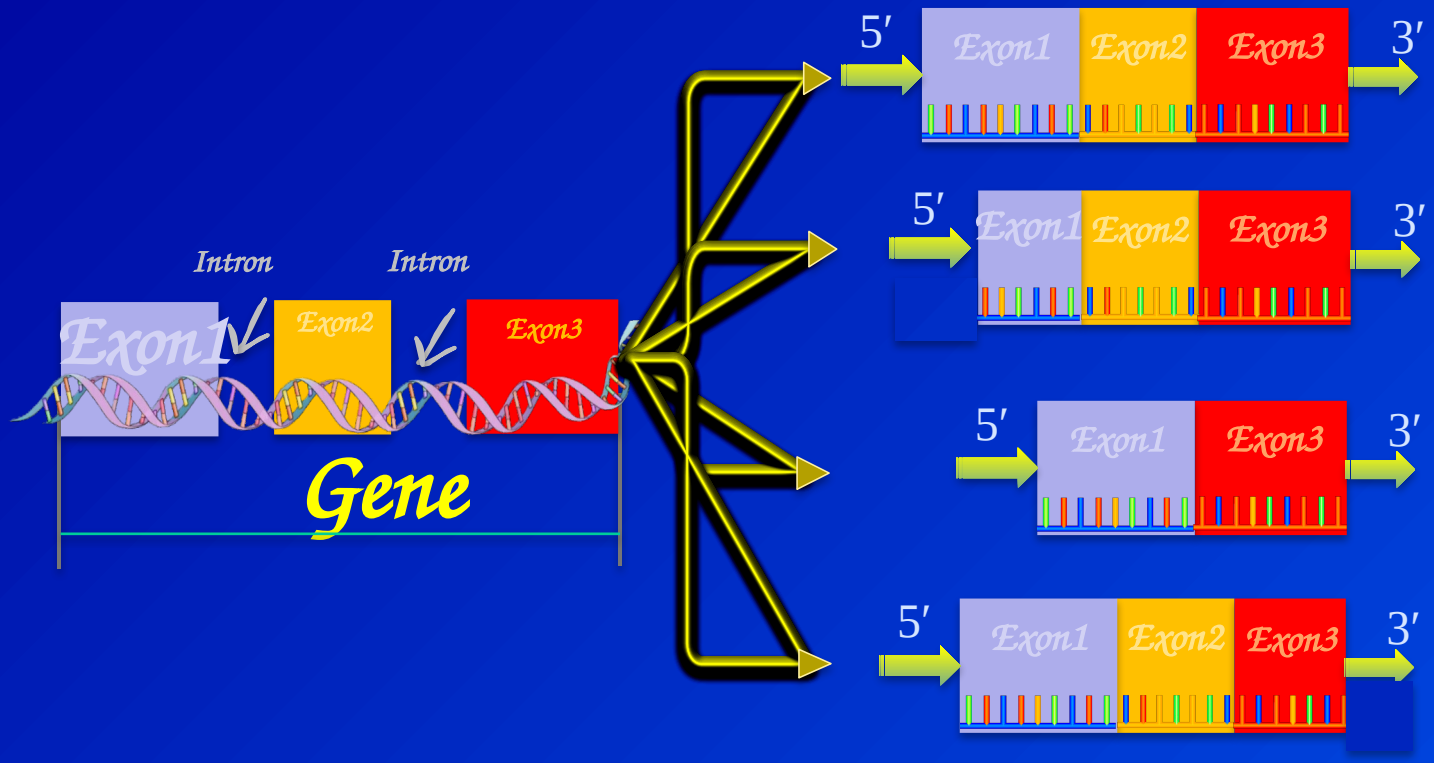
Same RNA sequence, two writing conventions

RNA notation	5′	A	G	C	U	U	C	G	A	3′
DNA notation	5′	A	G	C	T	T	C	G	A	3′

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RNA-level data

One gene, several RNA transcripts



Baseline transcript

Transcription start at different sites

Alternative splicing

Alternative cleavage/polyadenylation sites

Two transcripts, same gene?

Transcripts originating from the same gene share a linked gene identifier.

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Protein-level data

Amino-acid sequence

Sequences and annotated variants

MVHLTPEEKS
AVTALWGKVN

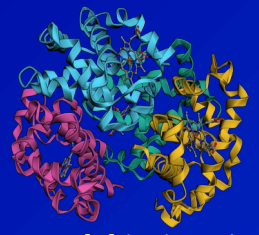
β-globin sequence

UniProtKB Protein Knowledgebase

NCBI PROTEIN

Three-dimensional structure

Experimental (PDB) or predicted (AlphaFold)



Haemoglobin (PDB)

RCSB PDB PROTEIN DATA BANK

PDBj Protein Data Bank Japan

PDBe Protein Data Bank in Europe

AlphaFold

Domains and families

Conserved regions and functional sites



Domain map: schematic

Pfam

InterPro

SCOP

Interactions and associations

Physical / genetic interactions and functional links



HBB network (STRING)

BioGRID

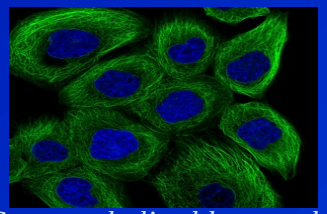
BioGRID

STRING

IntAct

Abundance and location

Where a protein occurs and how much is detected



Green: tubulin; blue: nuclei

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Behind database records

Experiments generate raw data

- Sequencing signals
- Instrument readings
- Experimental images
- Phenotypic observations



Analysis produces results and predictions

- Assembled sequences
- Domain assignments
- Translated protein sequences
- Three-dimensional structures



Curation evaluates and connects the information

- Assess supporting evidence
- Compare findings across studies
- Create and update annotations
- Link related records

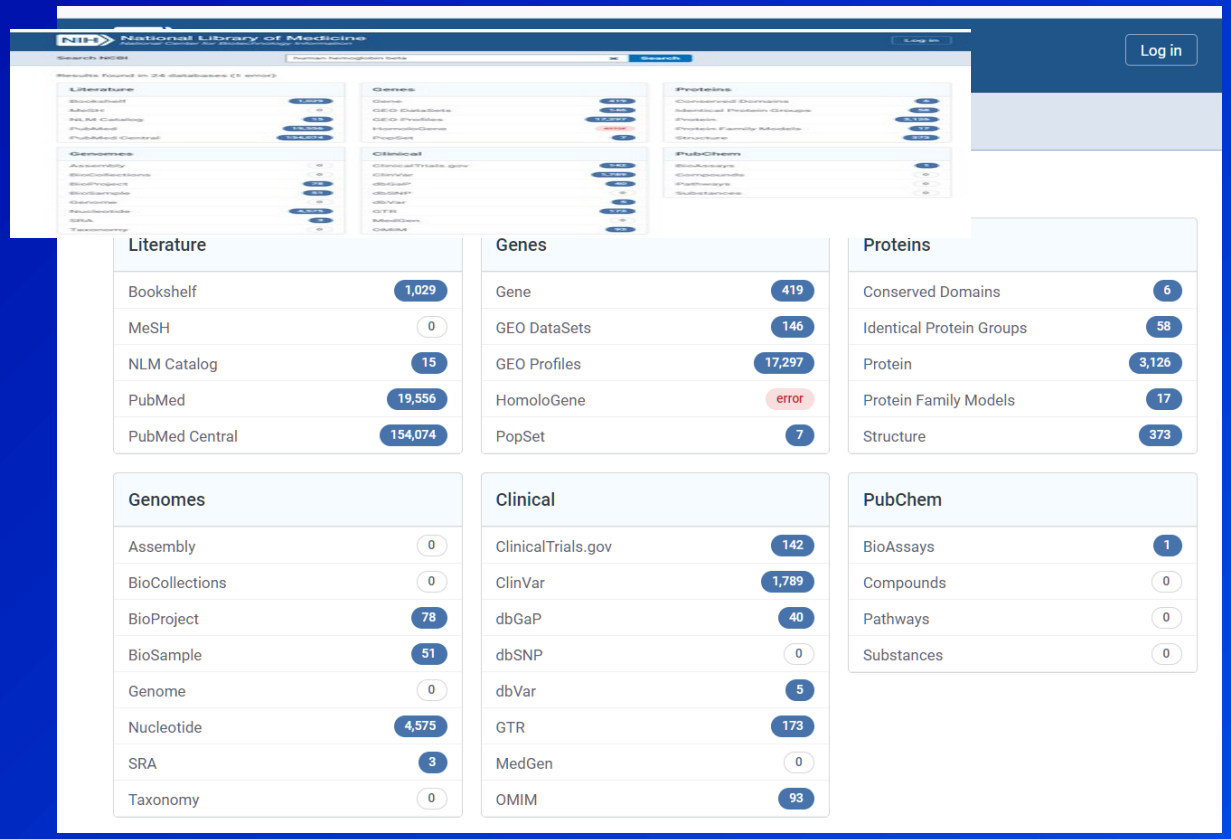


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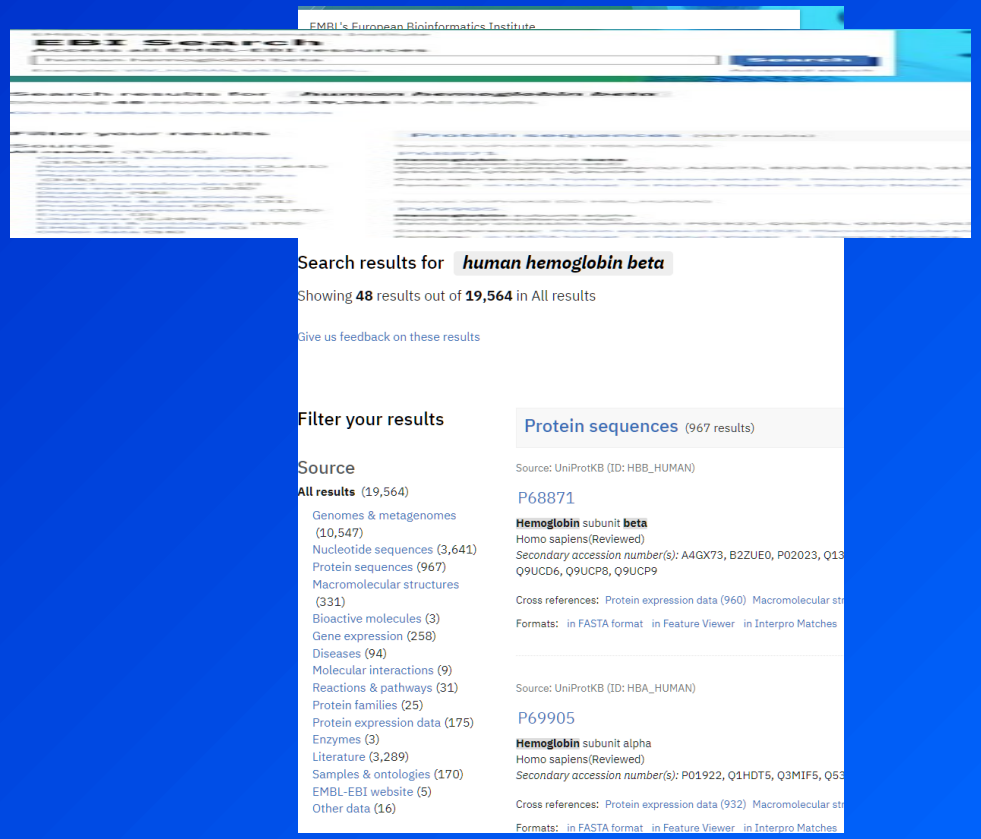
Search by name

One biological name, different databases

NCBI



EMBL-EBI

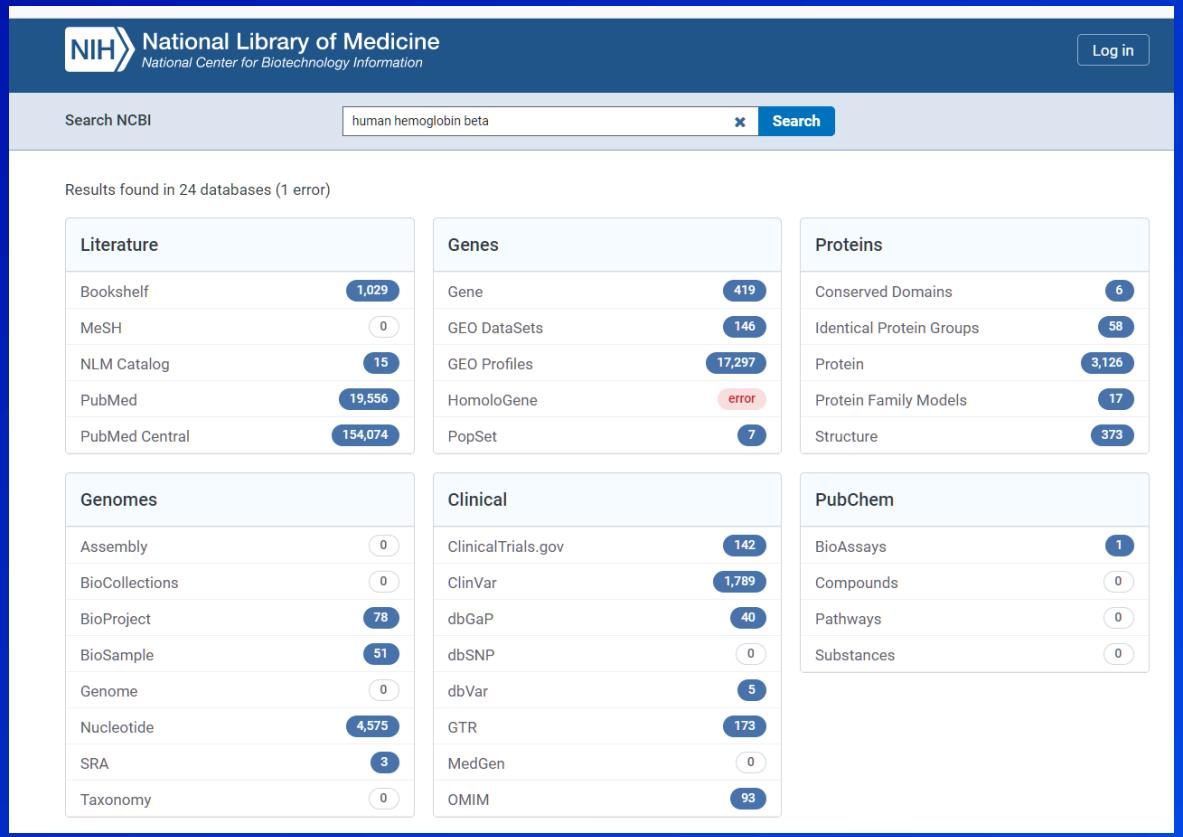


Stockholms universitet

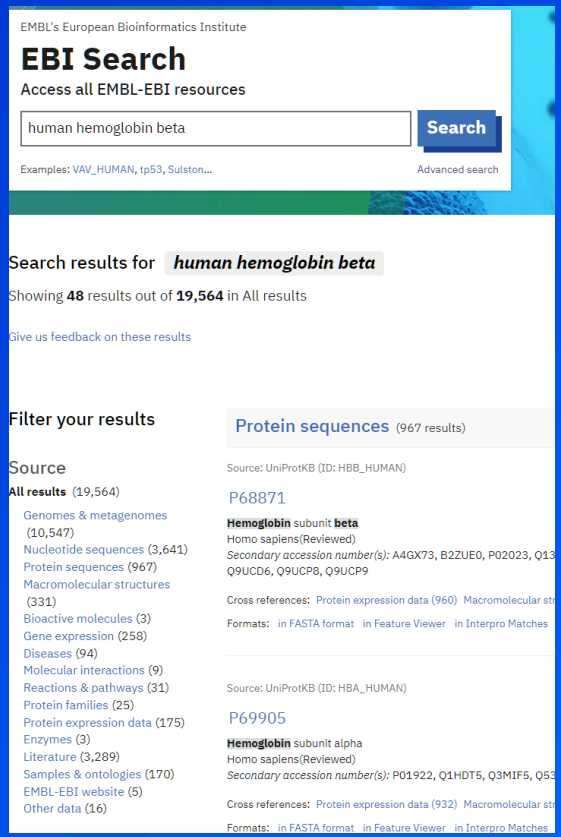
Search by name

One search, several kinds of records

NCBI



EMBL-EBI

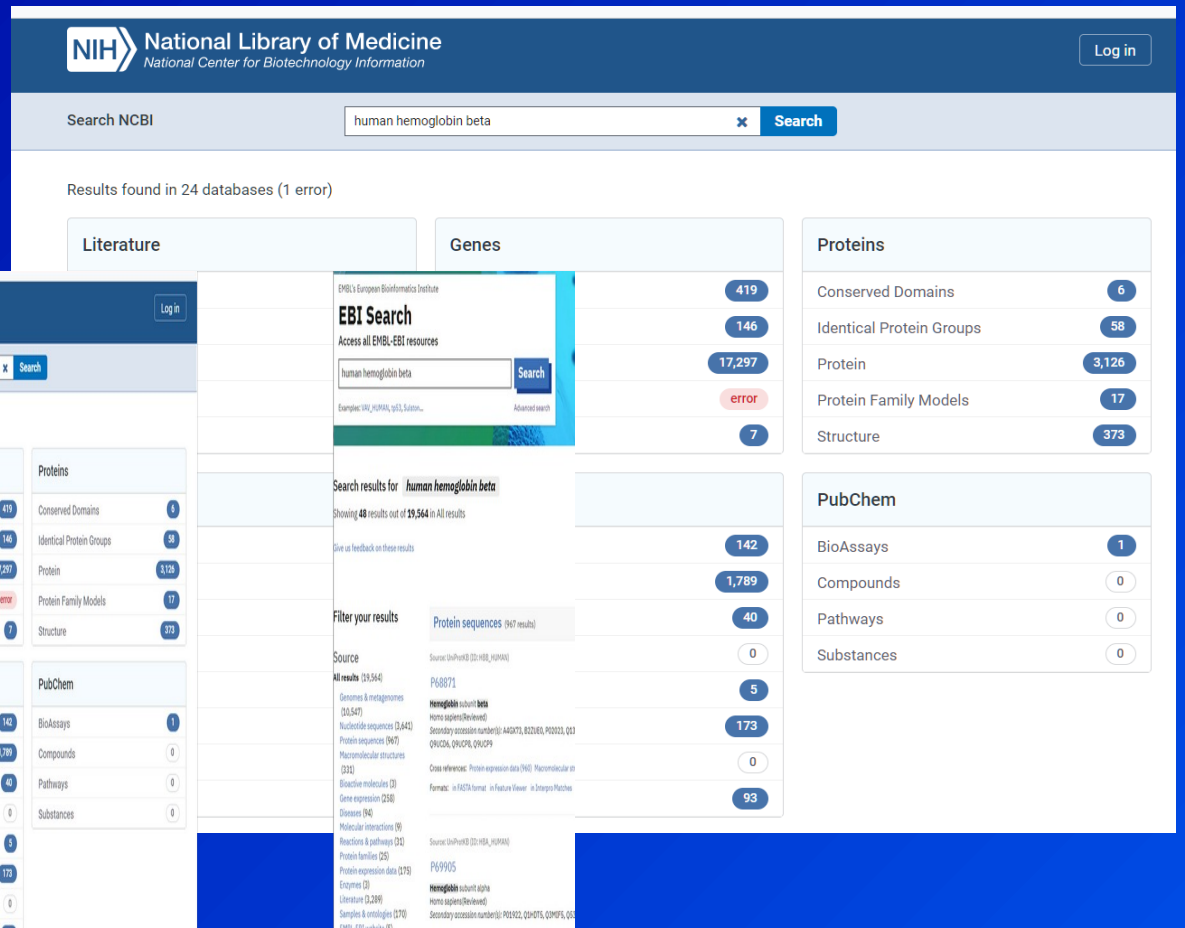


Stockholms universitet

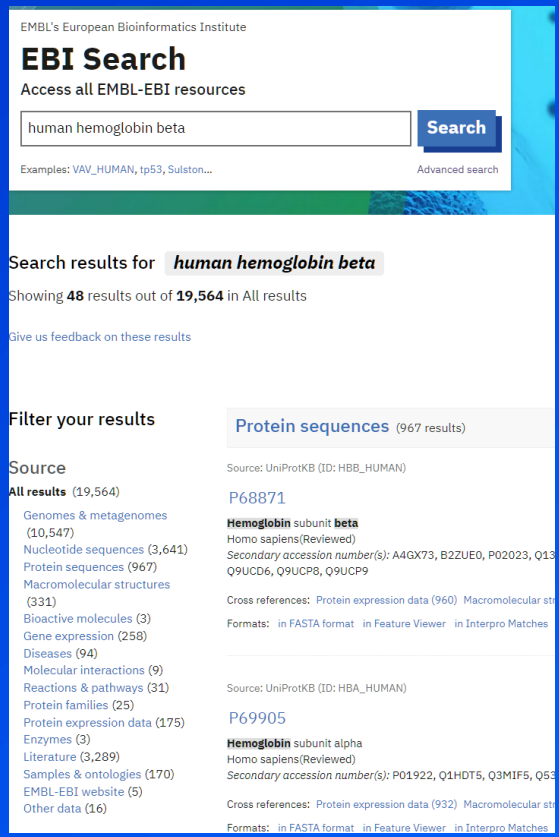
Search by name

Nucleotide sequences and their sources

NCBI



EMBL-EBI

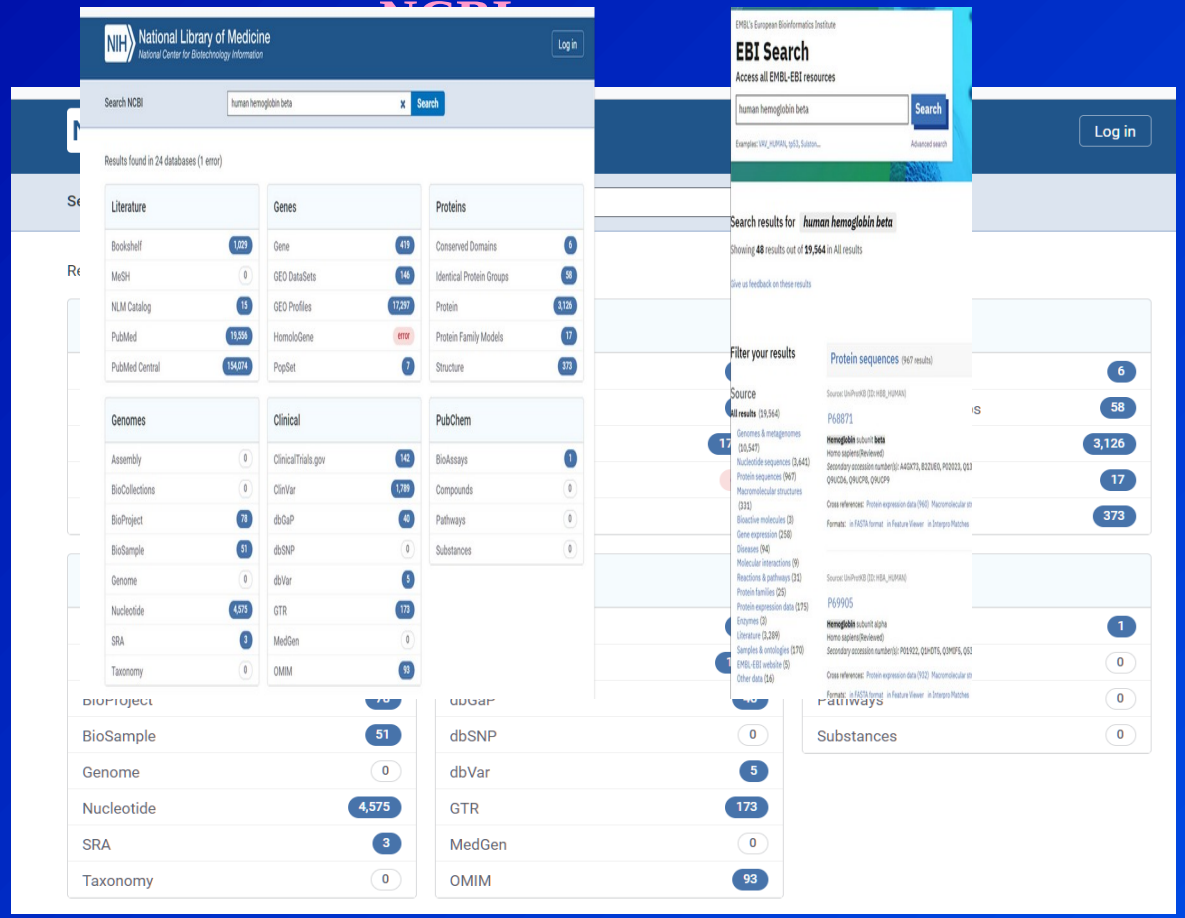


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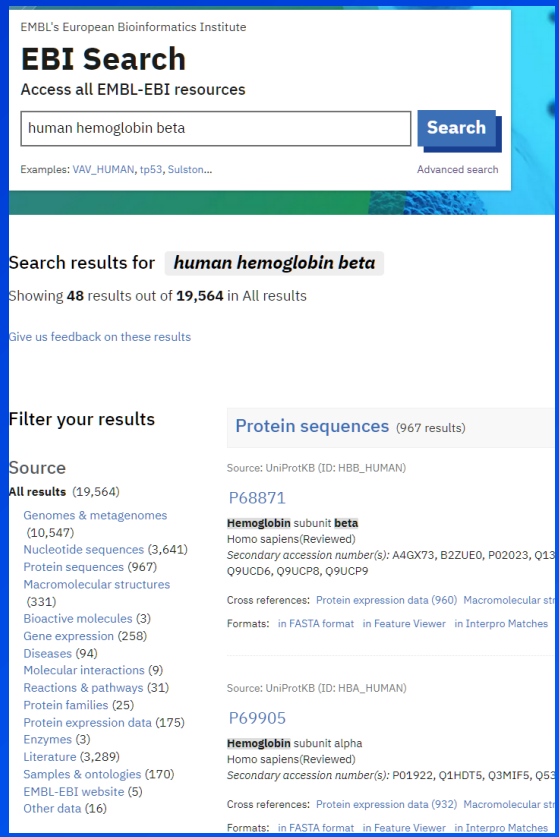
Search by name

Gene records and expression studies

NCBI



EMBL-EBI

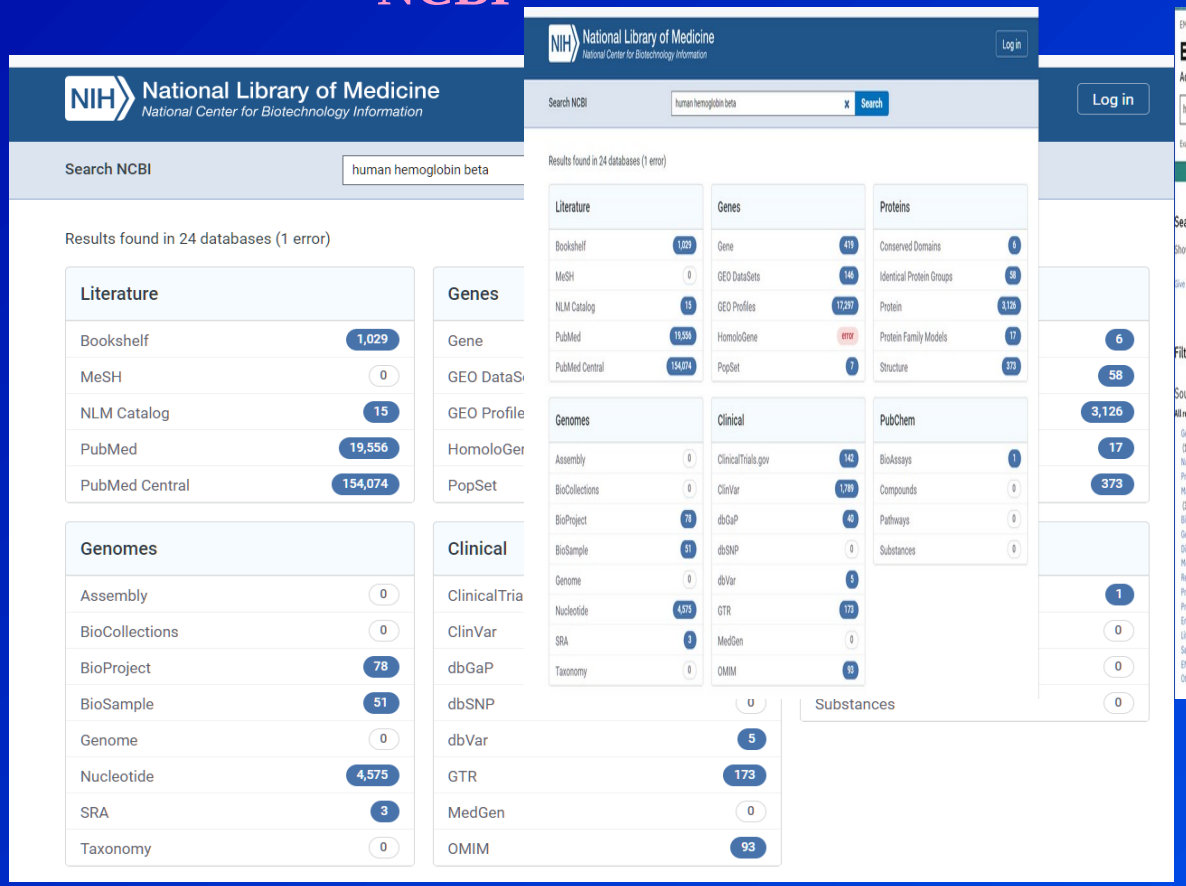


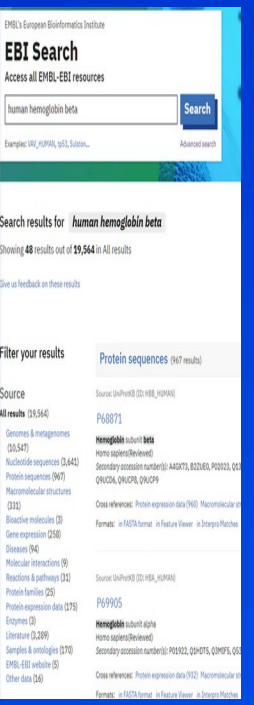
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Search by name

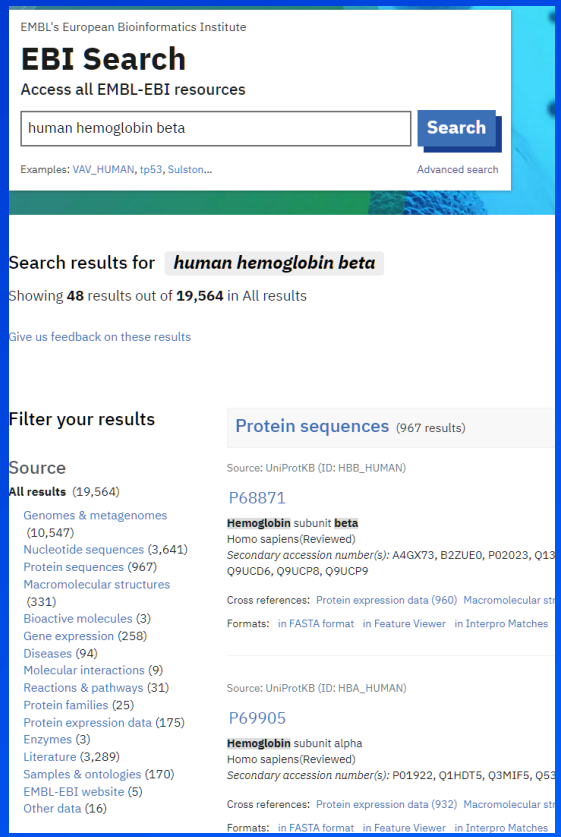
Different questions about a protein

NCBI





EMBL-EBI



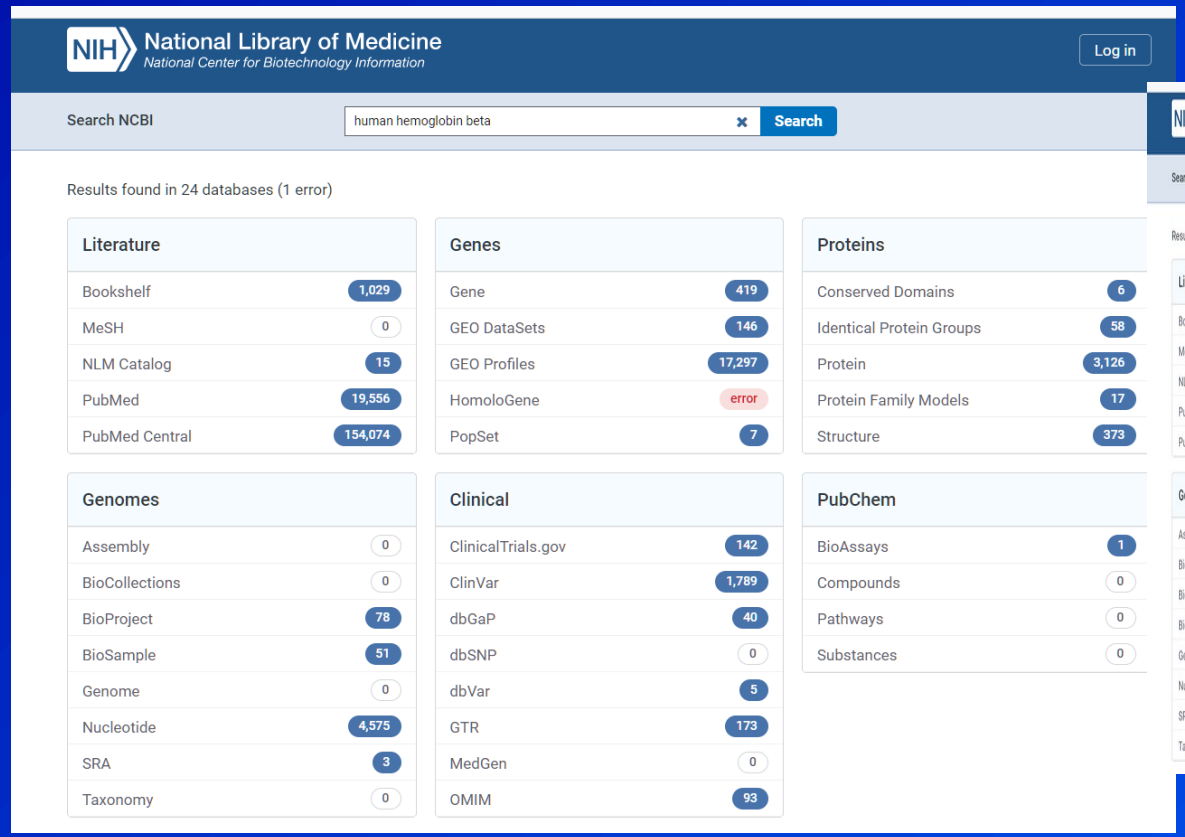
Stockholms universitet

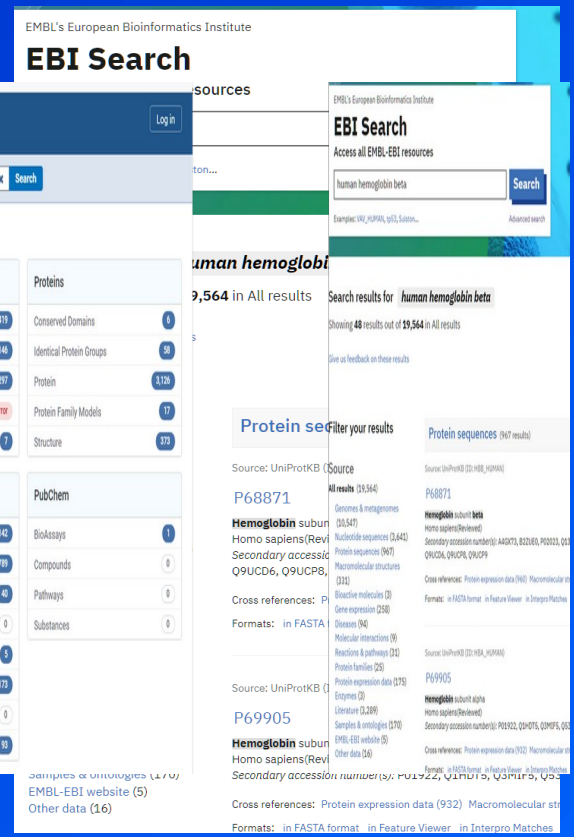
Search by name

Choosing and checking a result

NCBI

EMBL-EBI



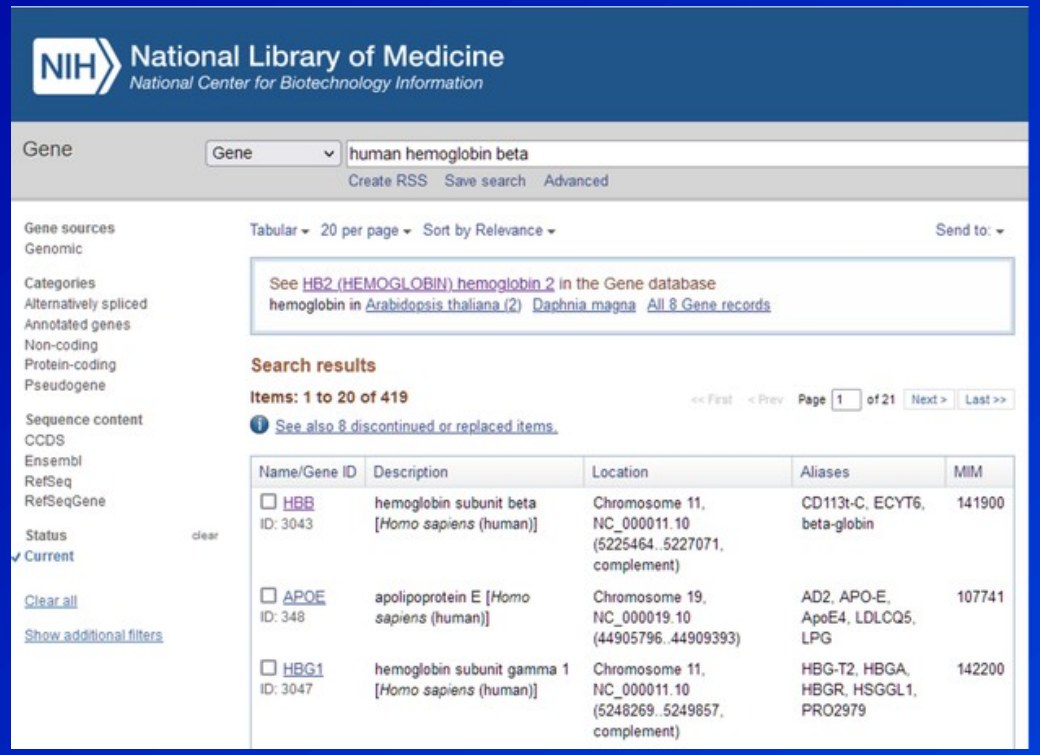


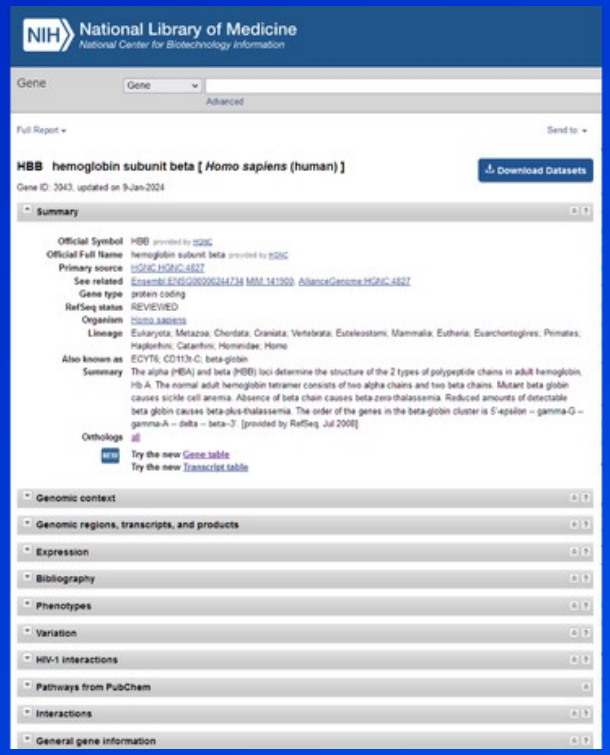
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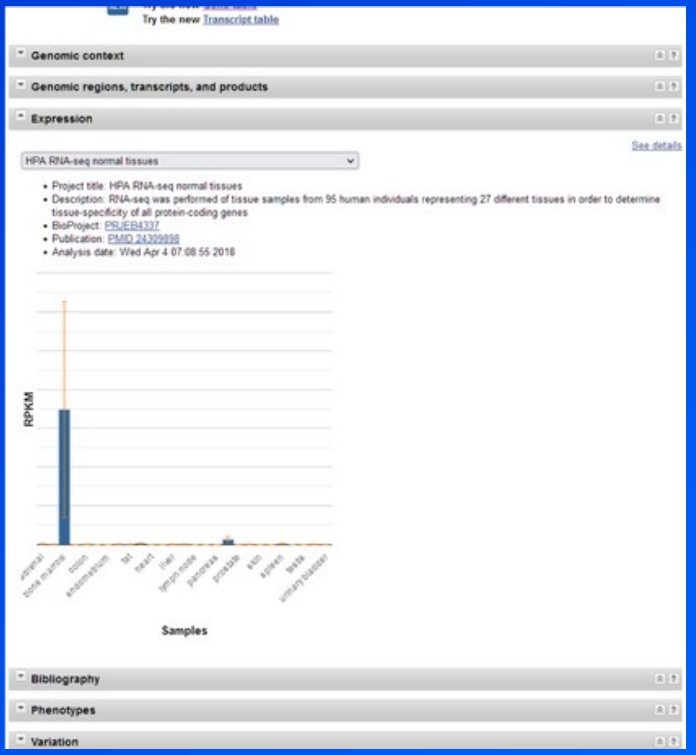
Search by name

Search within selected resource

NCBI Gene





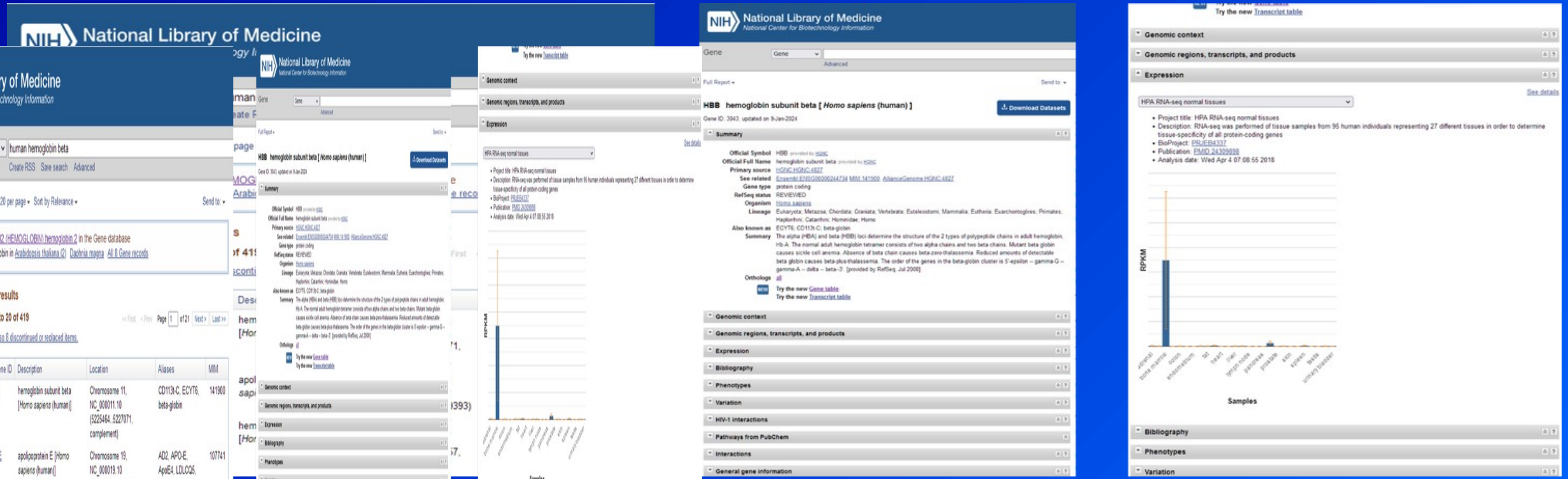


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Search by name

Check the name and organism

NCBI Gene

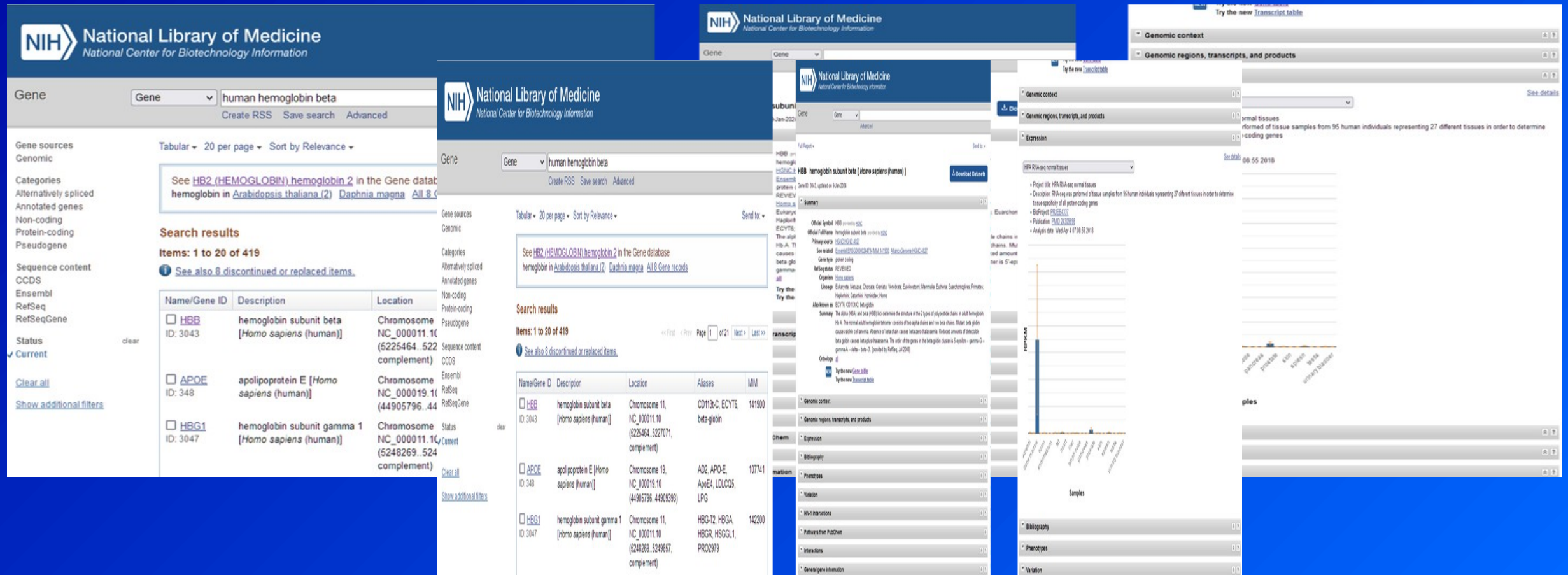


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Search by name

Explore the gene record

NCBI Gene

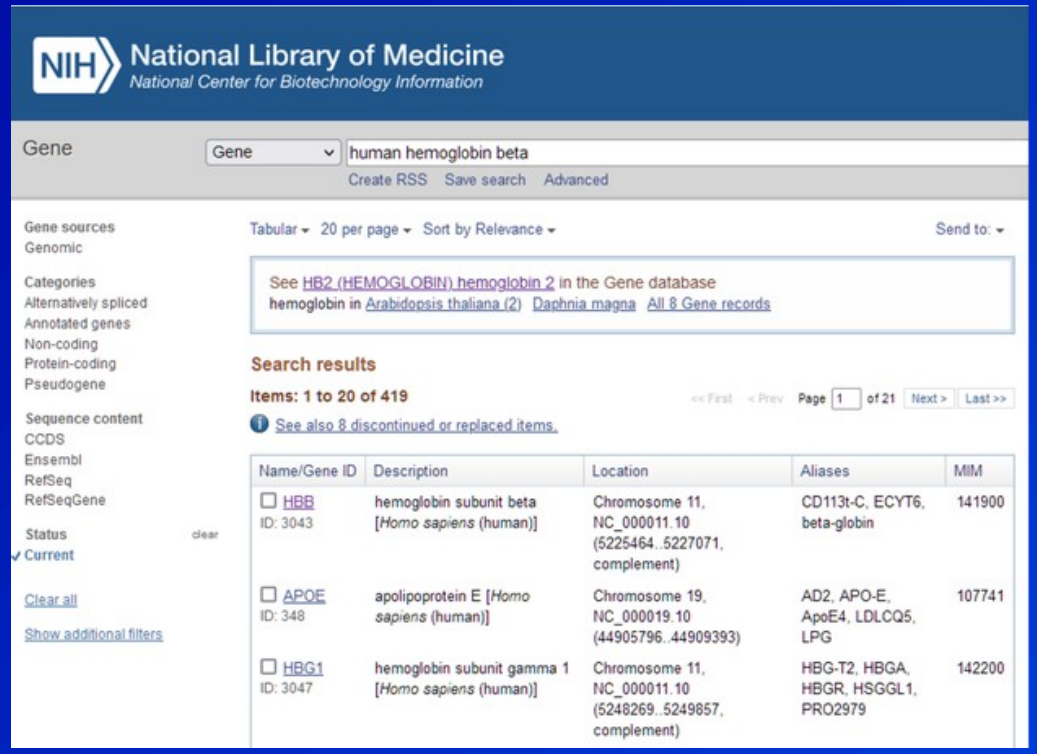


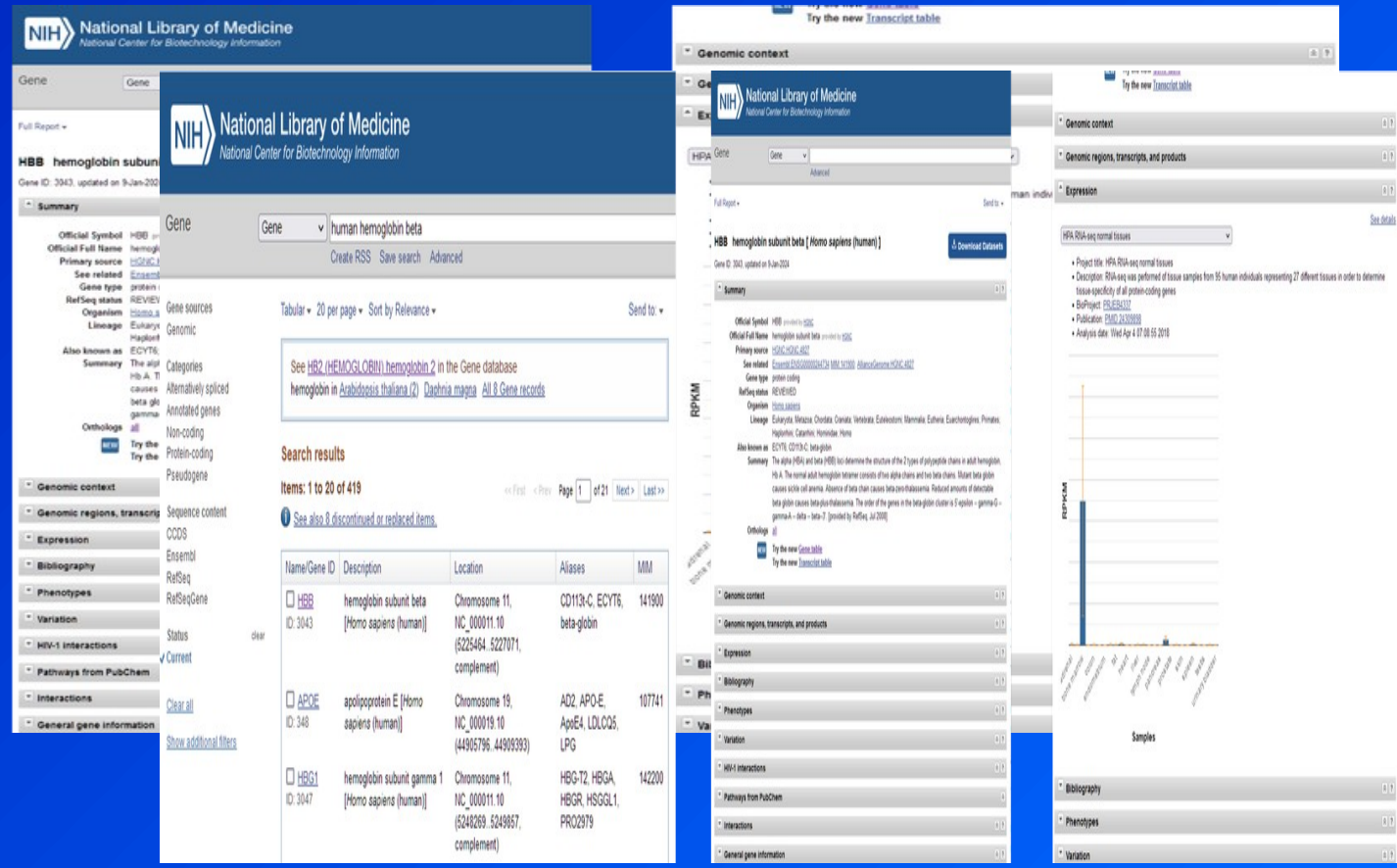
Stockholms universitet

Search by name

Trace expression data to the study

NCBI Gene

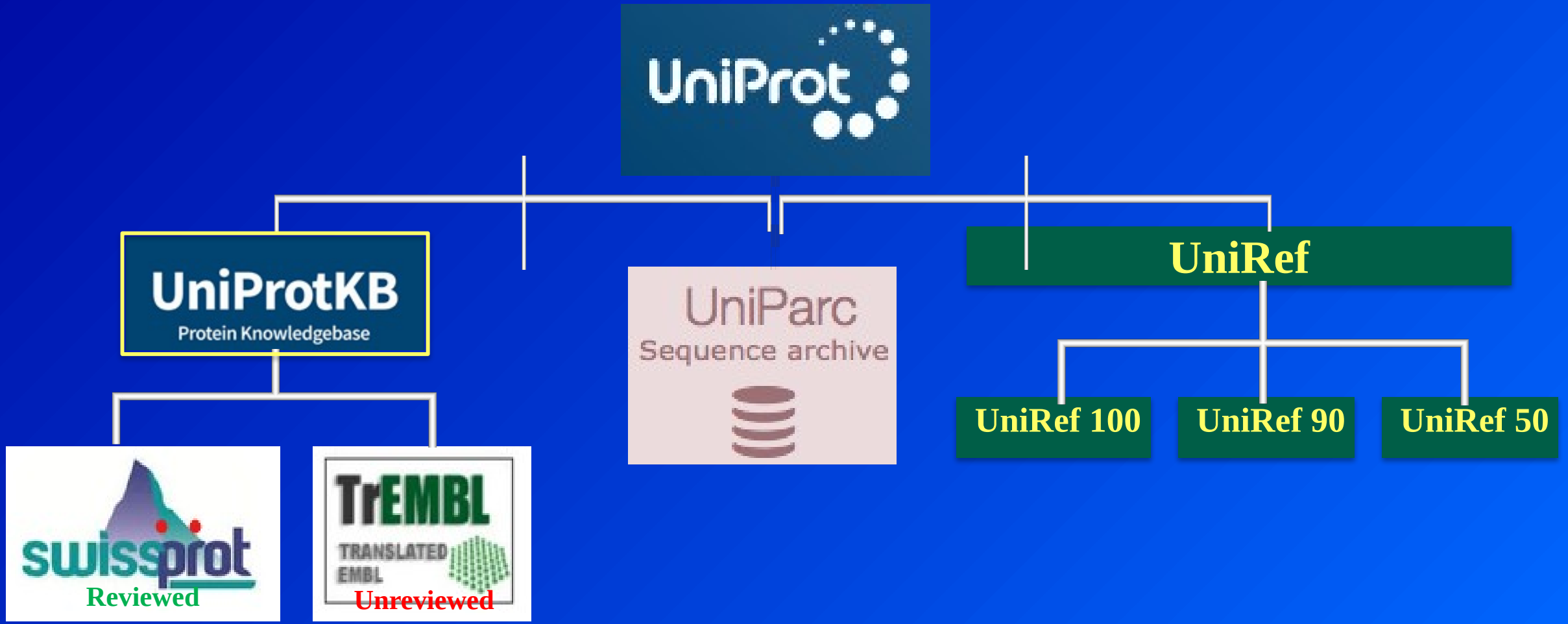




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Universal Protein Resource

Resource content

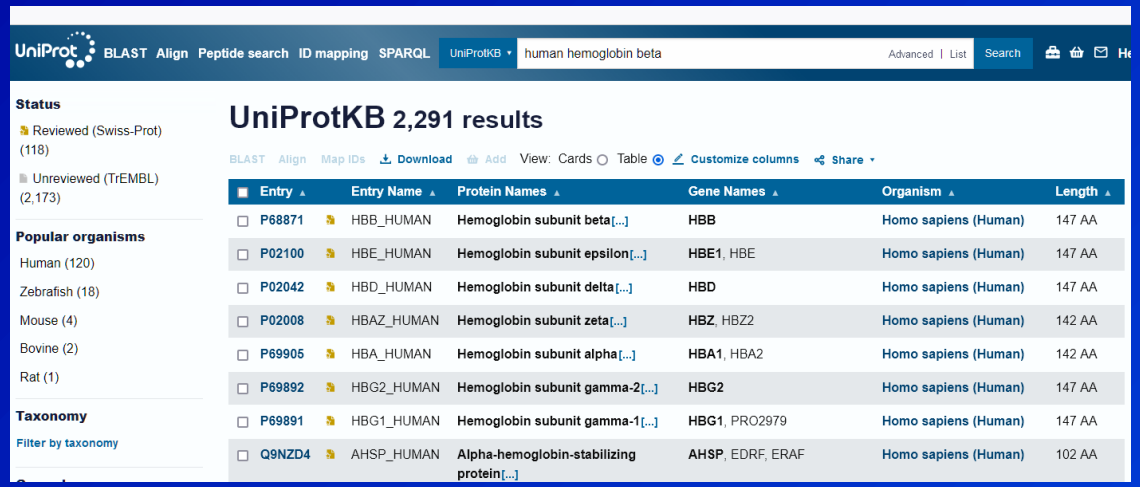


Stockholms universitet

Search by name

Search within UniProtKB

UniProtKB

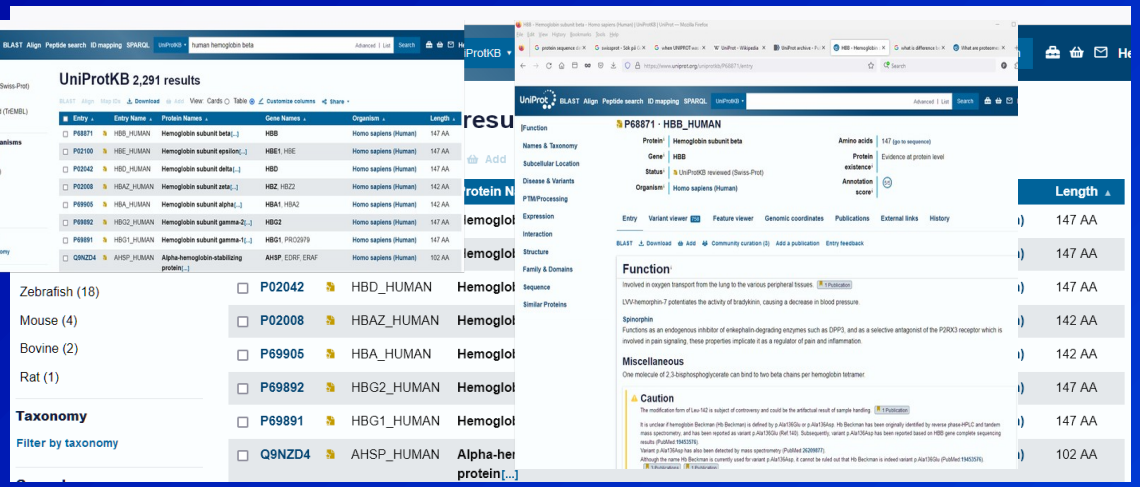


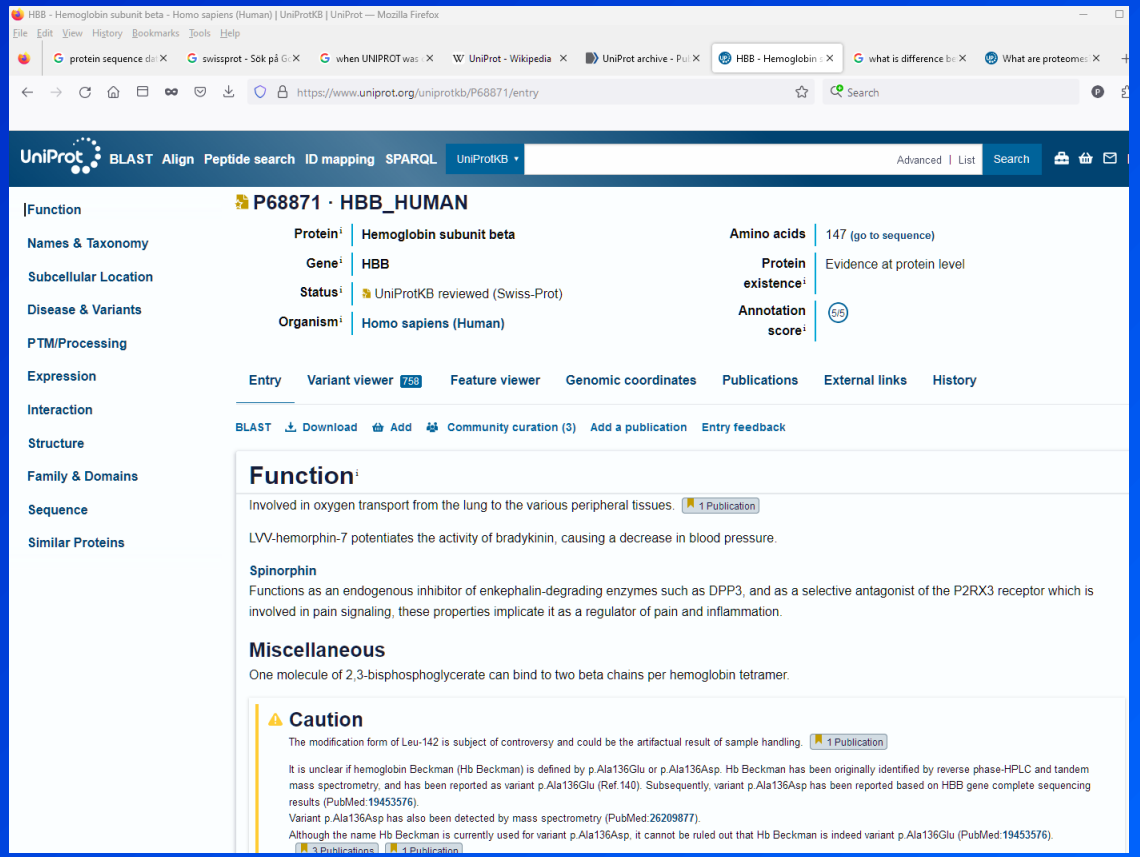
Stockholms universitet

Search by name

Filter by organism and review status

UniProtKB



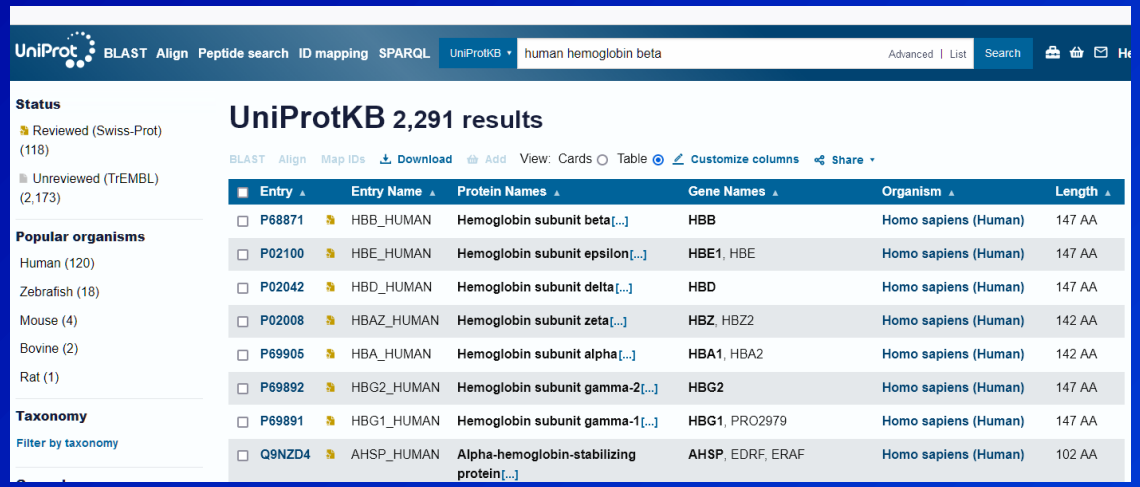


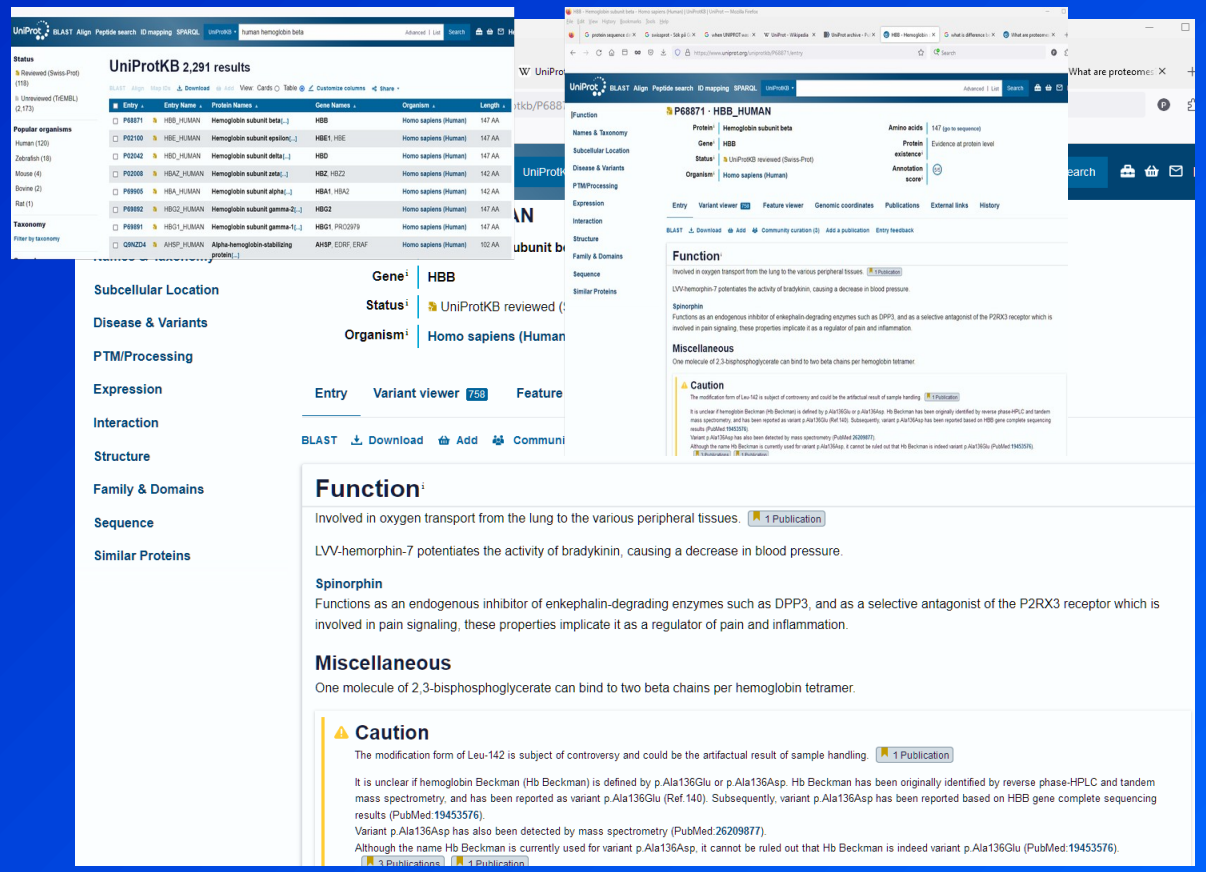
Stockholms universitet

Search by name

Check the protein name and organism

UniProtKB



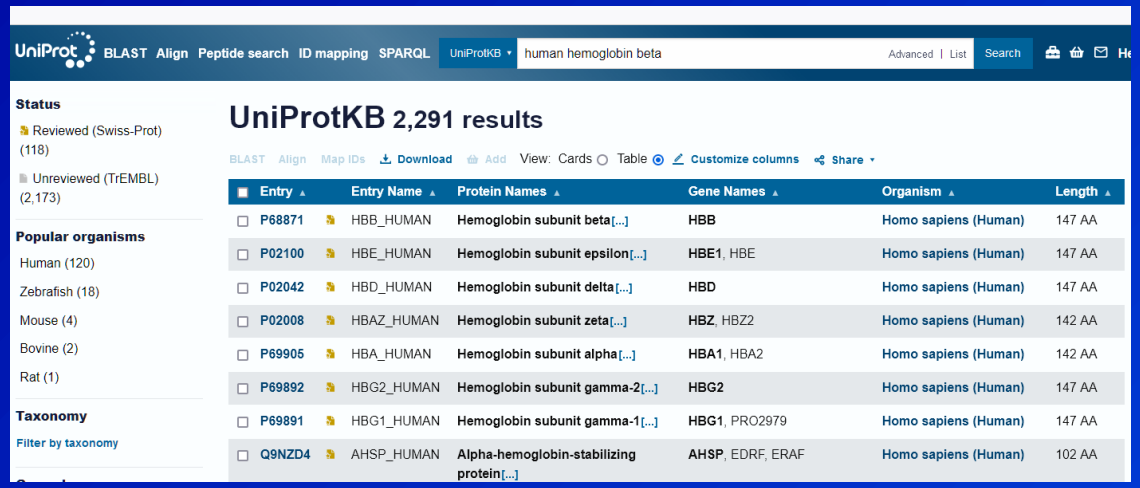


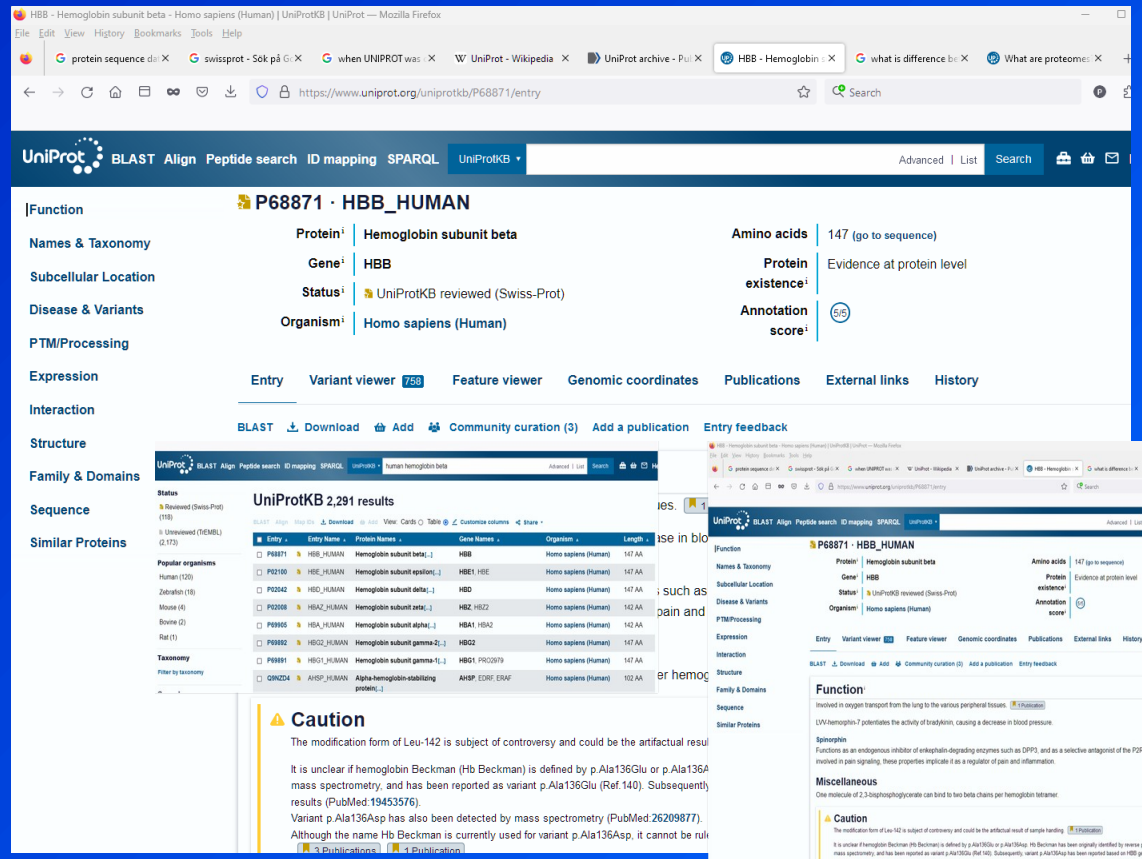
Stockholms universitet

Search by name

Check the function and its evidence

UniProtKB



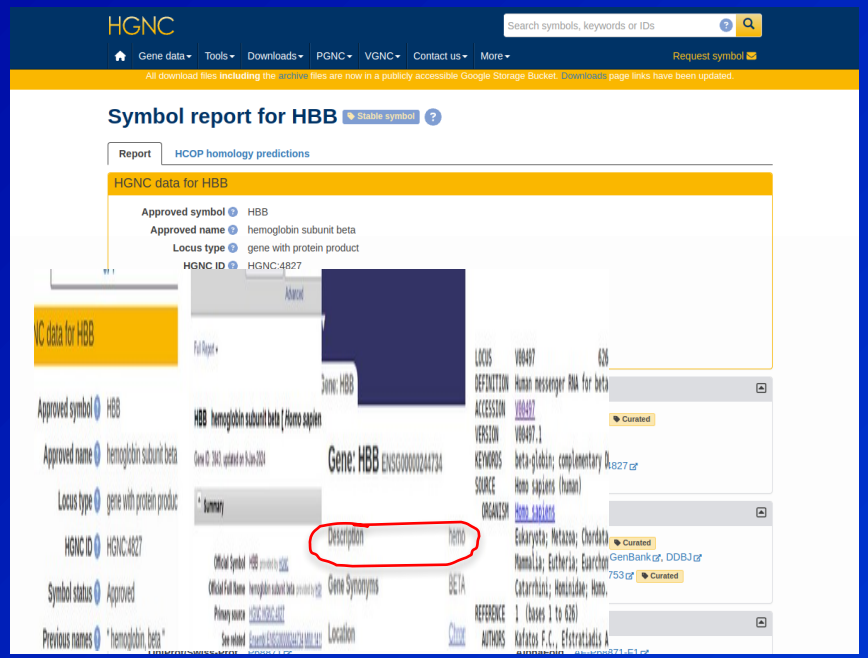


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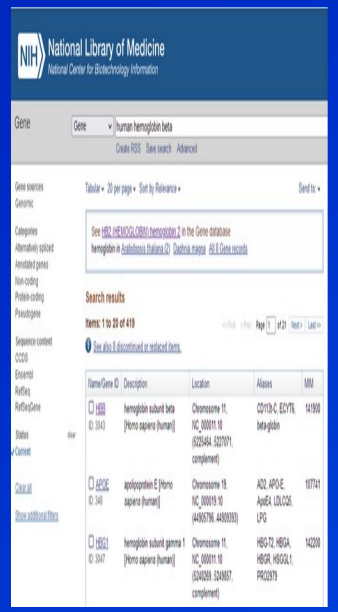
Search by official symbol

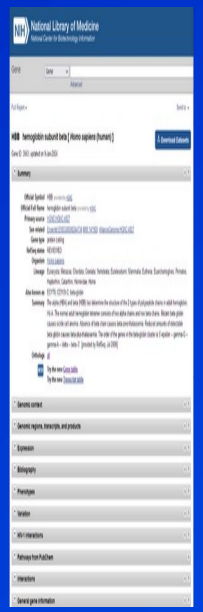
One symbol, different record IDs

Official gene symbol (HBB)



NCBI







EMBL-EBI



DDBJ



Stockholms universitet

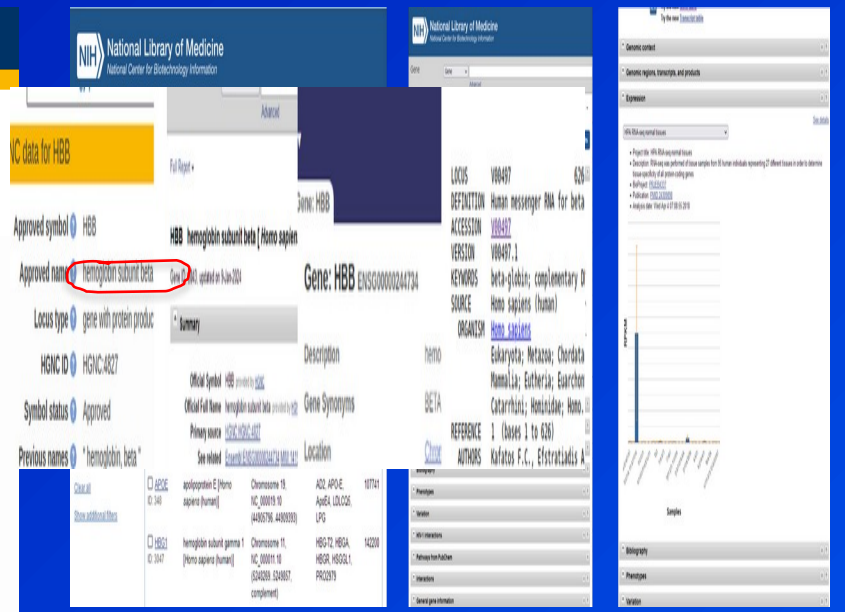
Search by official symbol

One symbol, different record IDs

Official gene symbol (HBB)



NCBI



EMBL-EBI



DDBJ

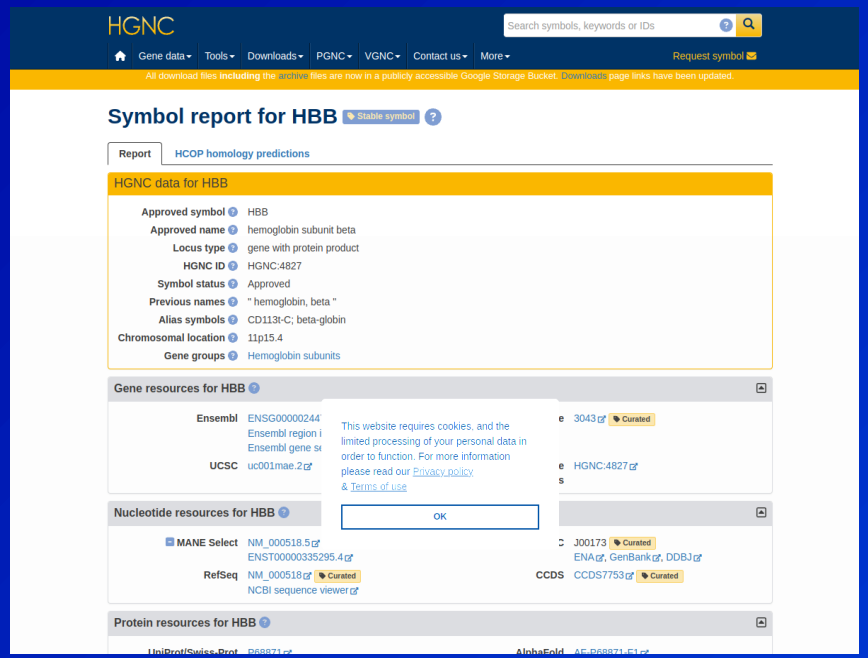


Stockholms universitet

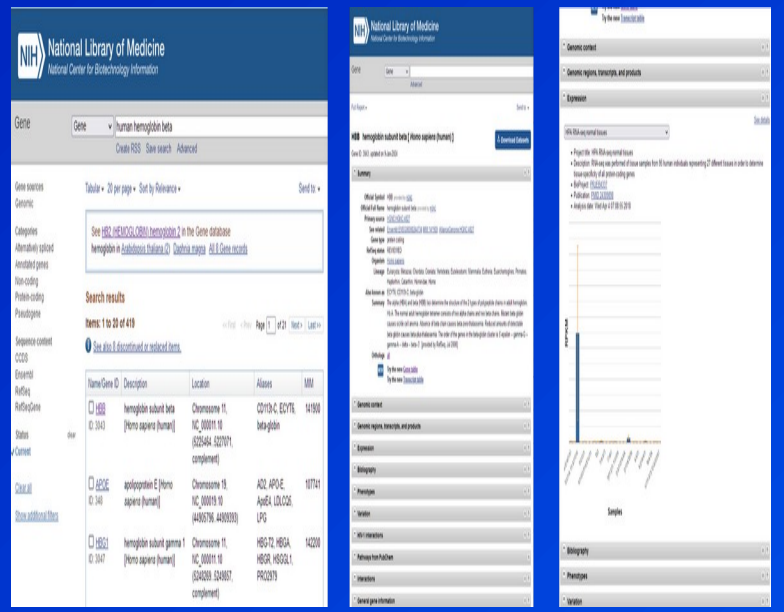
Search by official symbol

One symbol, different record IDs

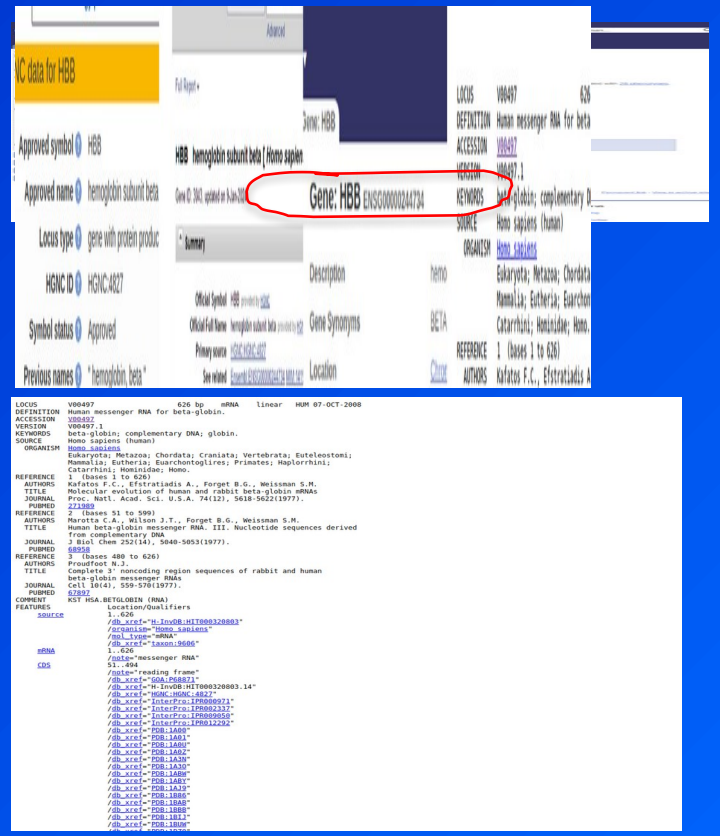
Official gene symbol (HBB)



NCBI



EMBL-EBI

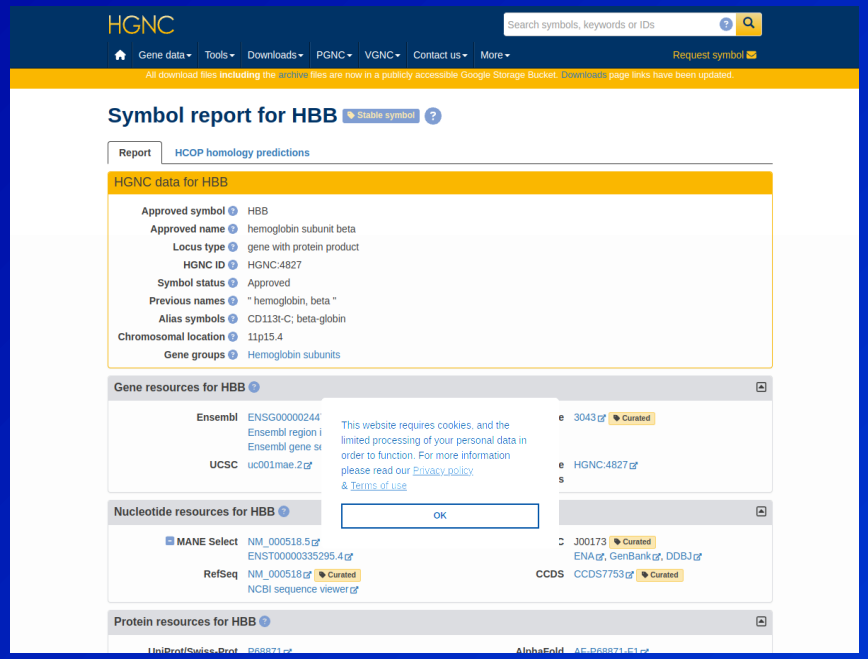


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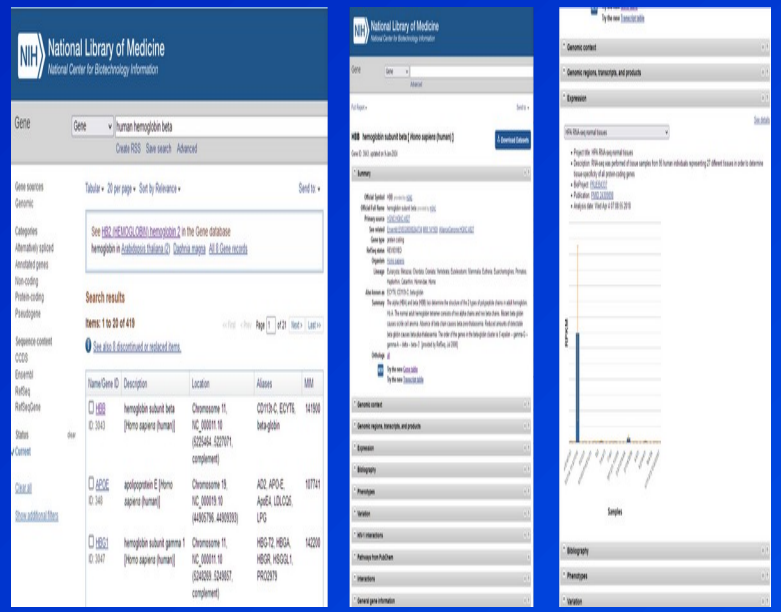
Search by official symbol

One symbol, different record IDs

Official gene symbol (HBB)



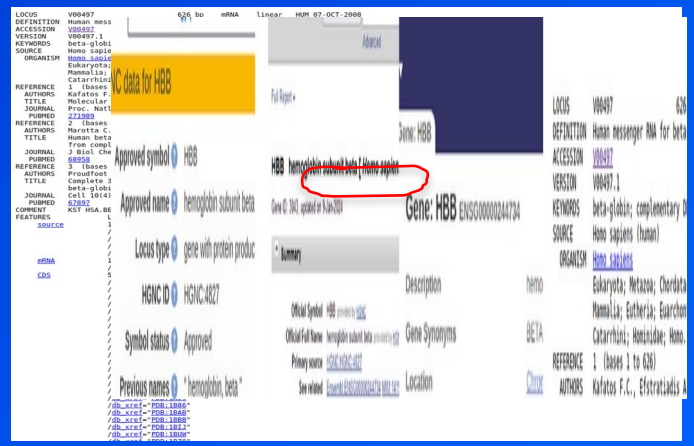
NCBI



EMBL-EBI



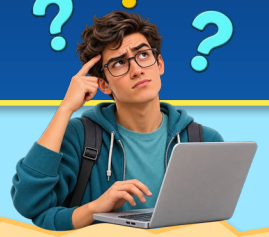
DDBJ



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From a symbol to a record

Approved human gene symbol
HBB



Nomenclature
HGNC:4827

Gene record
3043

Gene record
ENSG00000244734

Genomic sequence
NG_059281.1

ClinVar variant
VCV000015495.3

Reference mRNA
NM_000518.5

Submitted mRNA
V00497.1

Ensembl transcript
ENST00000335295.4

Ensembl protein
ENSP00000333994.3

RefSeq protein
NP_000509.1

Protein function
P68871

Haemoglobin complex
2HHB

Globin family
IPR000971

Globin domain
PF00042

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From a symbol to a record

What kind of sequence?

Which version did we use?

NCBI

RefSeq mRNA

NM_000518.5

NCBI PROTEIN

RefSeq protein

NP_000509.1

Save the full accession and version to retrieve the same sequence.

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Clues in the code

Prefixes and suffixes carry clues

Resource	Pattern or example	What the pattern tells us
RefSeq	**NM_000518.5** **NP_000509.1**	NM_ mRNA, NP_ protein .5 sequence version 5
Ensembl (human)	**ENSG… / ENST… / ENSP…** **ENST00000335295.4**	Gene / transcript / protein .4 version of this transcript
INSDC read runs	**SRR… / ERR… / DRR…** **Example SRR390728**	Sequencing run Prefix shows the originating archive
GEO	**GSE… / GSM…** **GSE10072 / GSM254625**	GSE study series GSM sample
UniProtKB isoforms	**P04150-2**	-2 identifies a protein isoform
UniProtKB features	**VAR_002863** **PRO_0000052976**	VAR_ natural sequence variant PRO_ processed chain or peptide
Rfam / Pfam	**RF00005 / PF00042**	RF RNA family PF protein family or domain



Check theresource's rules

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Links across resources

Linked identifiers are cross-references

They lead to relevant information in another resource.

UniProtKB
Protein Knowledgebase

UniProtKB
P68871

Human beta-globin

Cross-references

NCBI Gene 3043

RefSeq NP_000509.1

PDB 2HHB

NCBI

Human HBB gene
3043

NCBI PROTEIN

Reference protein
NP_000509.1

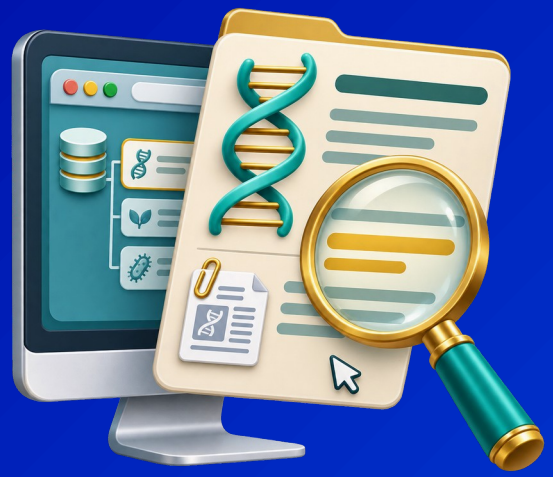
RCSB PDB PROTEIN DATA BANK

Haemoglobin complex
2HHB

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Routes to database info

Databaserecord pages



Inspect records andcheck their evidence.

Programmaticaccess



Automate queries usingcode or command-line tools.

Genomebrowsers

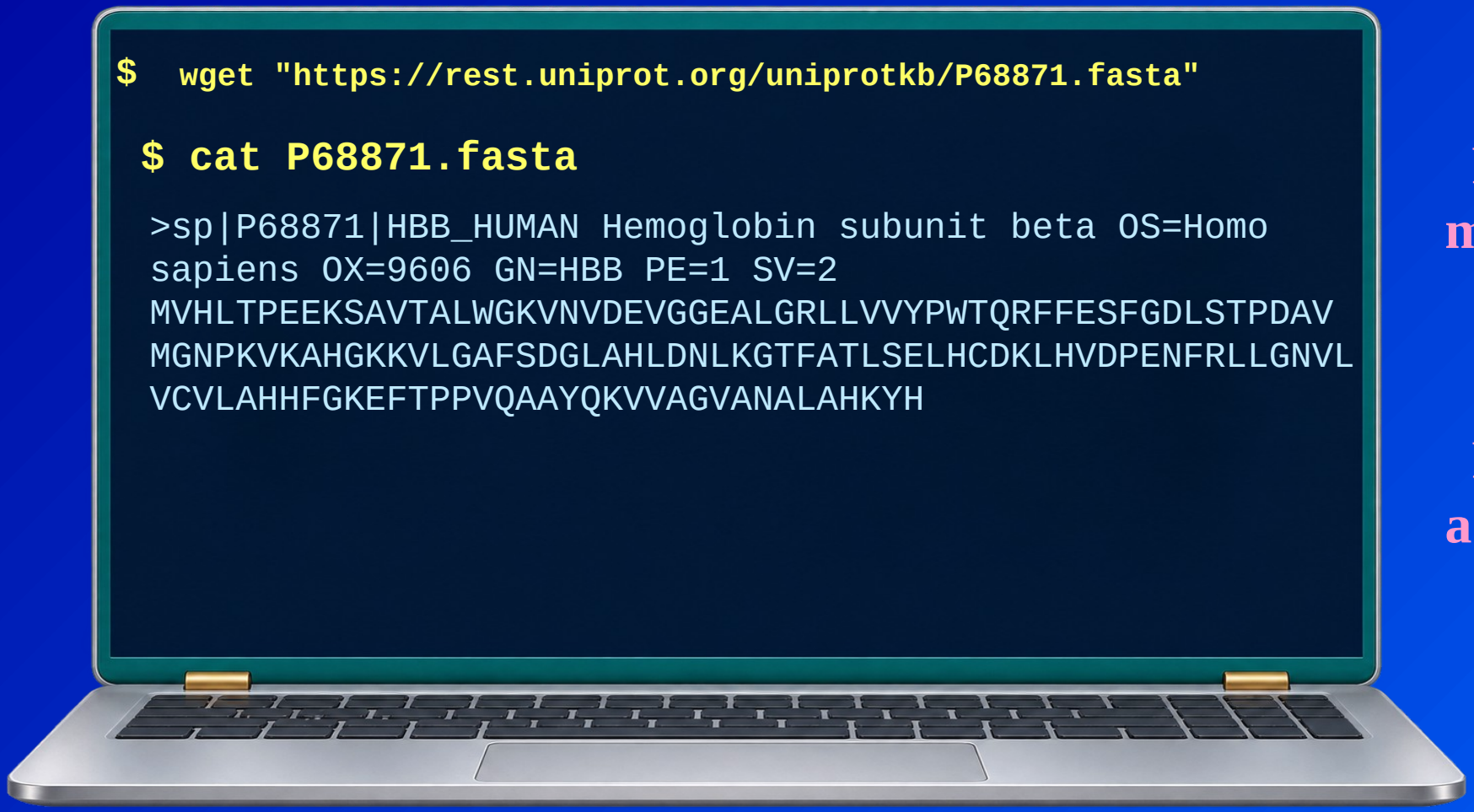


Explore DNA, transcripts and proteins through one web interface

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Programmatic access

One command, one file



File may contain many records

File processing needs additional tools or code

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Programmatic access

Download first, process next

Python script

```
```



Output

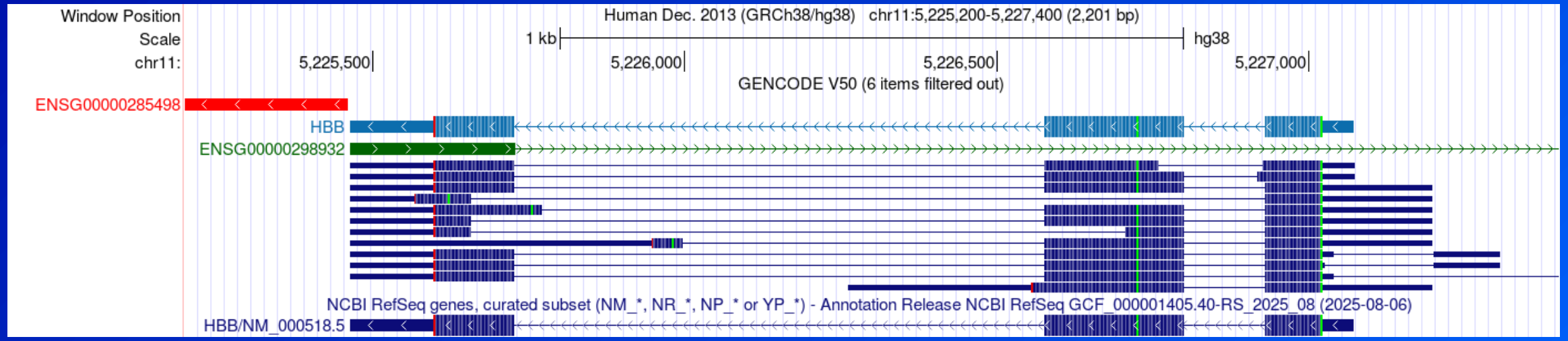
```
```

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Genome browsers

Everything in one interface

HBB in UCSC Genome Browser: https://genome.ucsc.edu



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Genome browsers

Reference assembly and displayed region

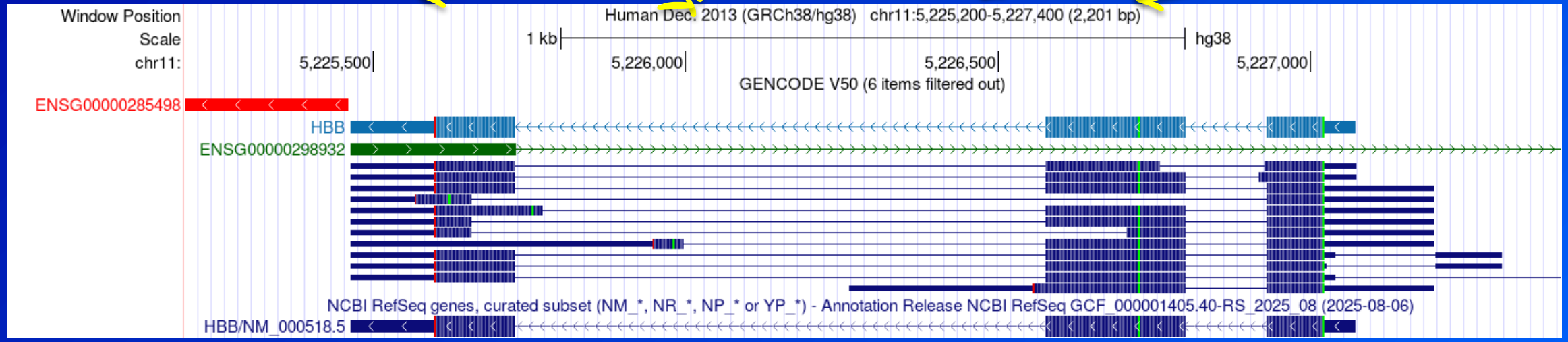
Assembly organism

Assembly release date

Assembly name

Chromosome interval

Window length



HBB in UCSC Genome Browser: https://genome.ucsc.edu

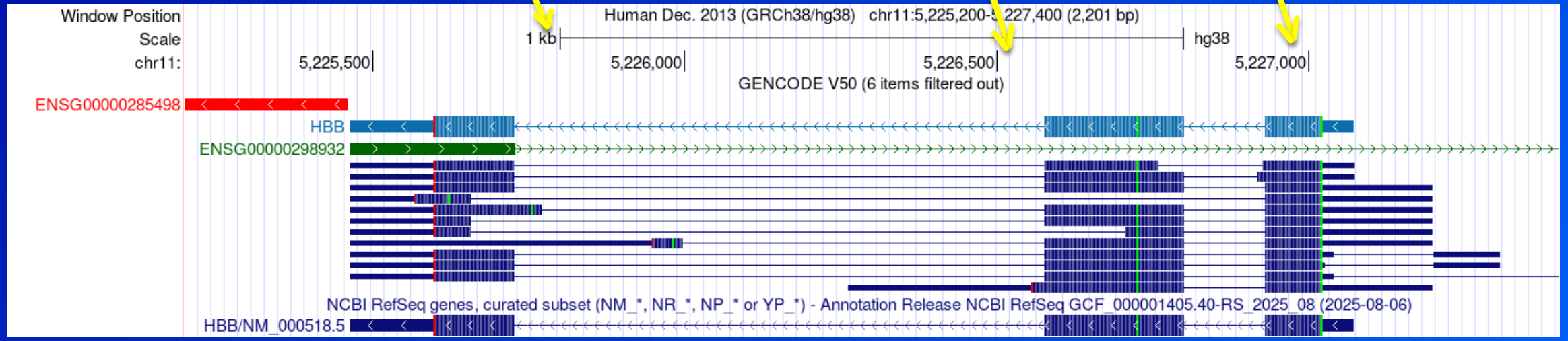
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Genome browsers

Distances along chromosome 11

Scale bar (1,000 bases)

Positions along the chromosome 11



HBB in UCSC Genome Browser: https://genome.ucsc.edu

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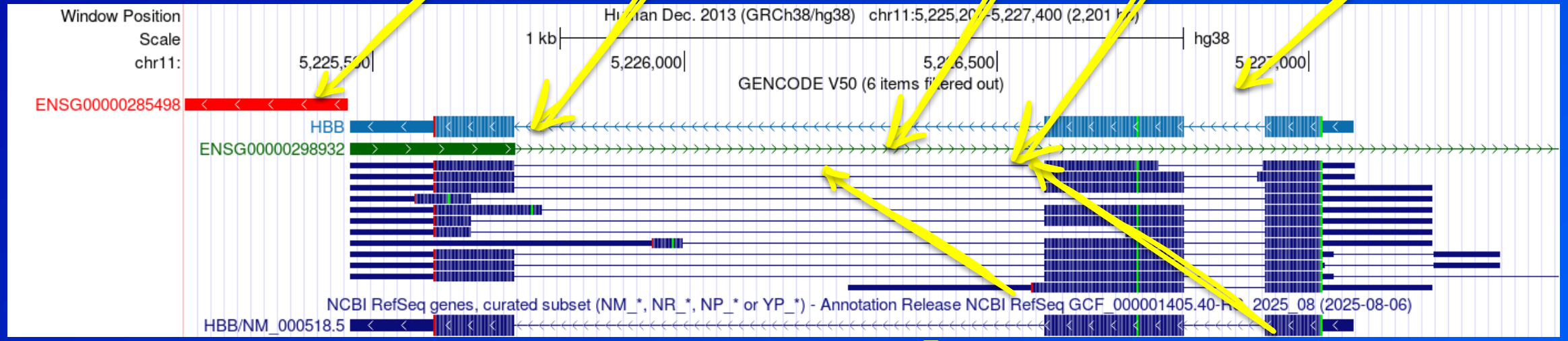
Genome browsers

Exons and Introns

Non-coding exon part (UTR)

Coding exon parts

UTR



Introns

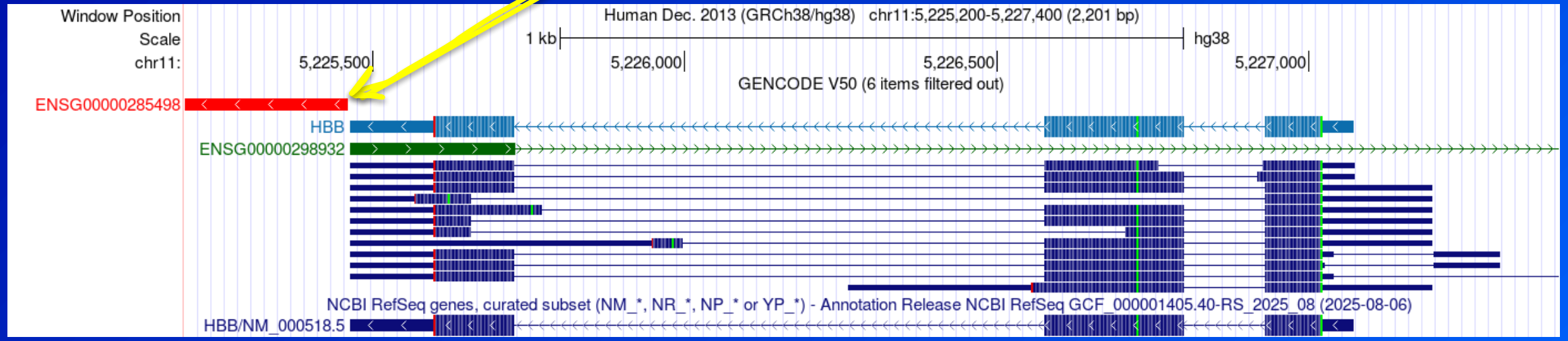
HBB in UCSC Genome Browser: https://genome.ucsc.edu

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Genome browsers

Some annotations still need confirmation

Possible transcript



HBB in UCSC Genome Browser: https://genome.ucsc.edu

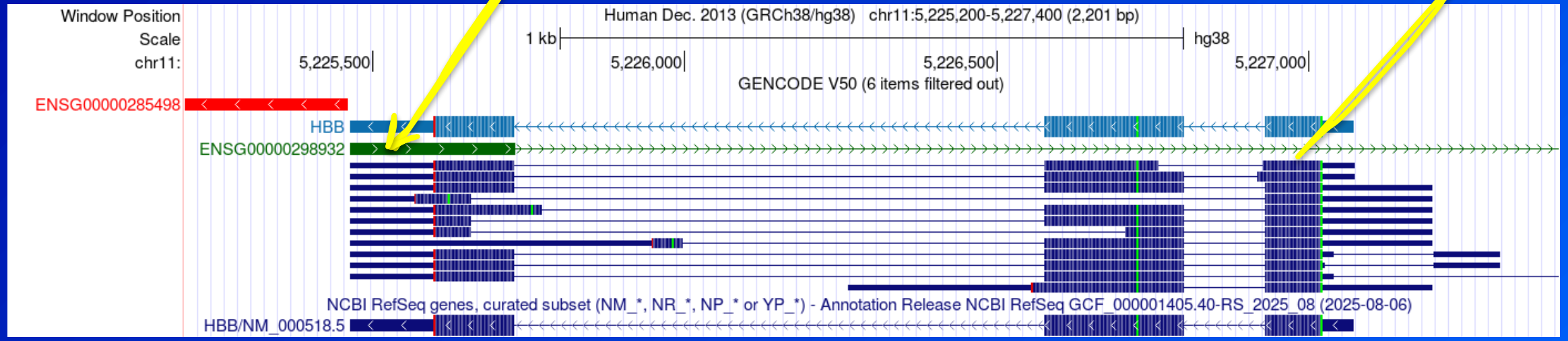
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Genome browsers

A neighbouring long non-coding RNA

Next exon outside this window

Non-coding RNA exon on opposite DNA strand



HBB in UCSC Genome Browser: https://genome.ucsc.edu

Stockholms universitet

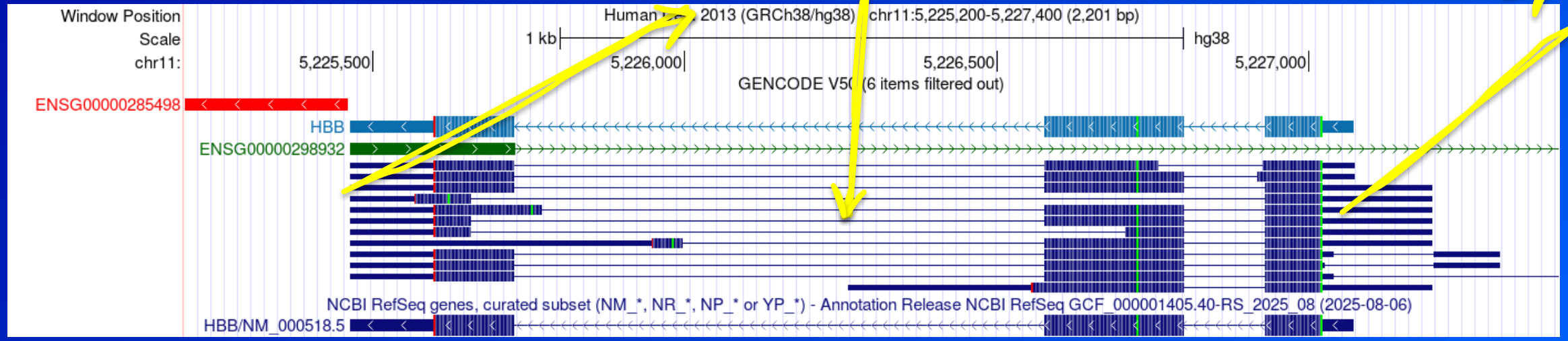
Genome browsers

One gene, several transcripts

Different exon boundaries

Alternative transcripts

Different transcript ends



HBB in UCSC Genome Browser: https://genome.ucsc.edu

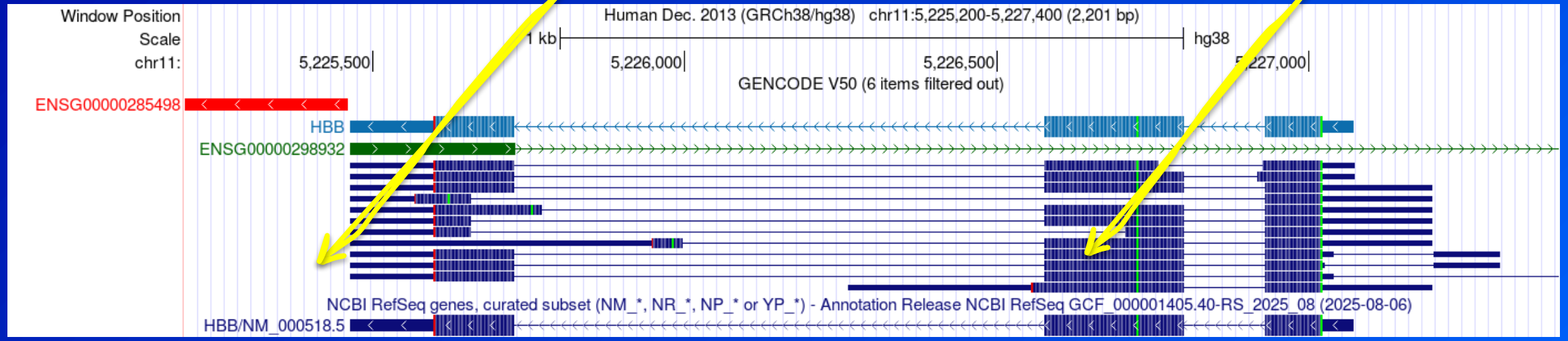
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Genome browsers

One curated HBB reference transcript

Curated RefSeq annotation

Annotation release identification



HBB in UCSC Genome Browser: https://genome.ucsc.edu

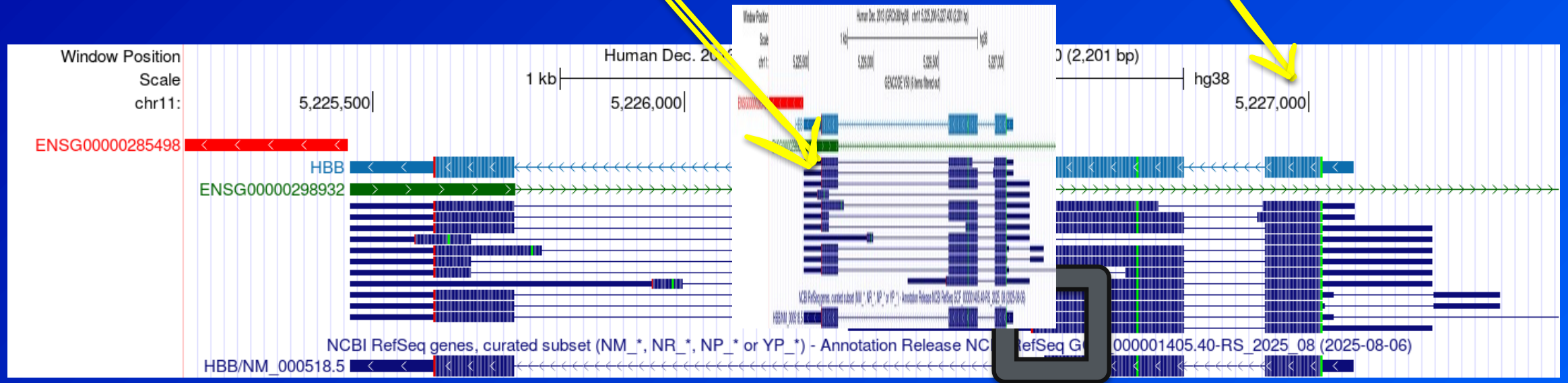
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Genome browsers

Transcription direction

HBB transcription runs to the left

Coordinates increase to the right



HBB in UCSC Genome Browser: https://genome.ucsc.edu

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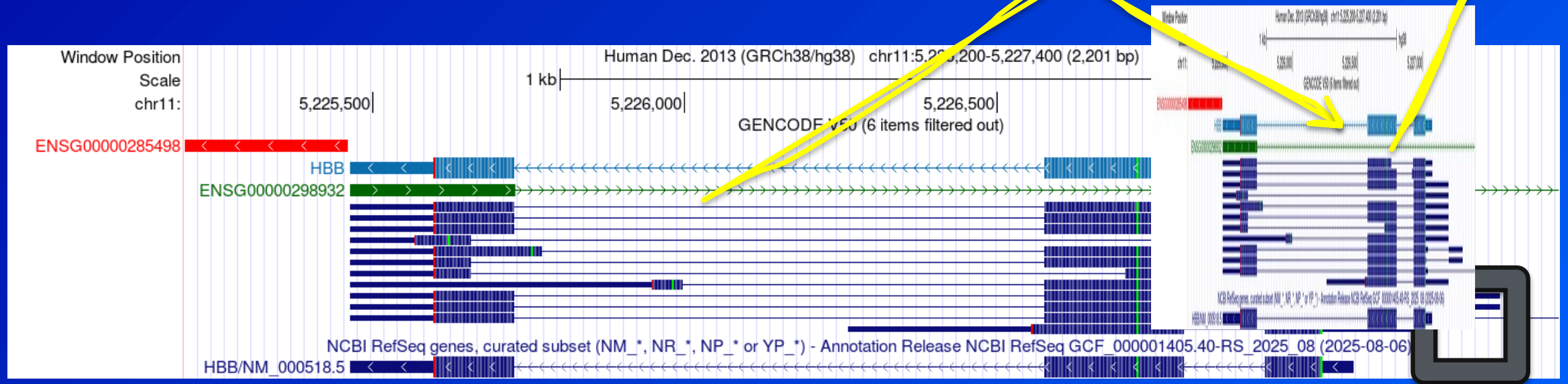
Genome browsers

Exon structure and codon colours

Blue shades: successive codons

Green: start codon (methionine)

Thin part: untranslated region



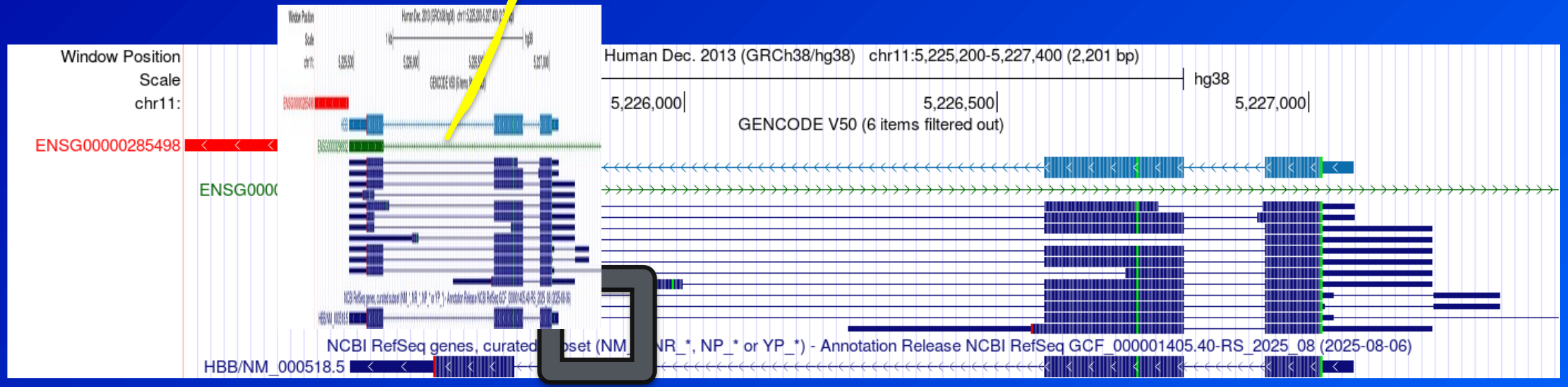
HBB in UCSC Genome Browser: https://genome.ucsc.edu

Stockholms universitet

Genome browsers

Exon structure and codon colours

Red: stop codon



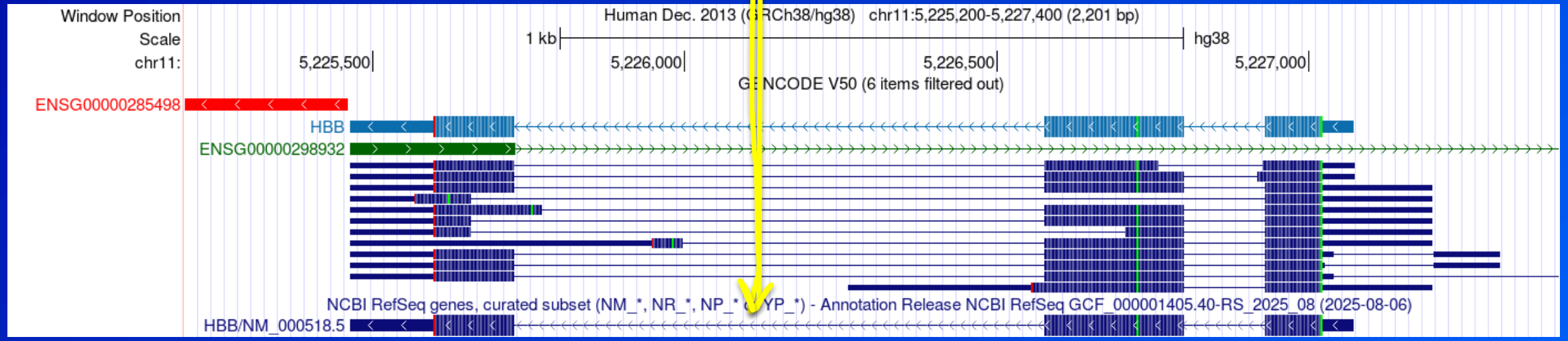
HBB in UCSC Genome Browser: https://genome.ucsc.edu

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Genome browsers

One transcript, different sequences

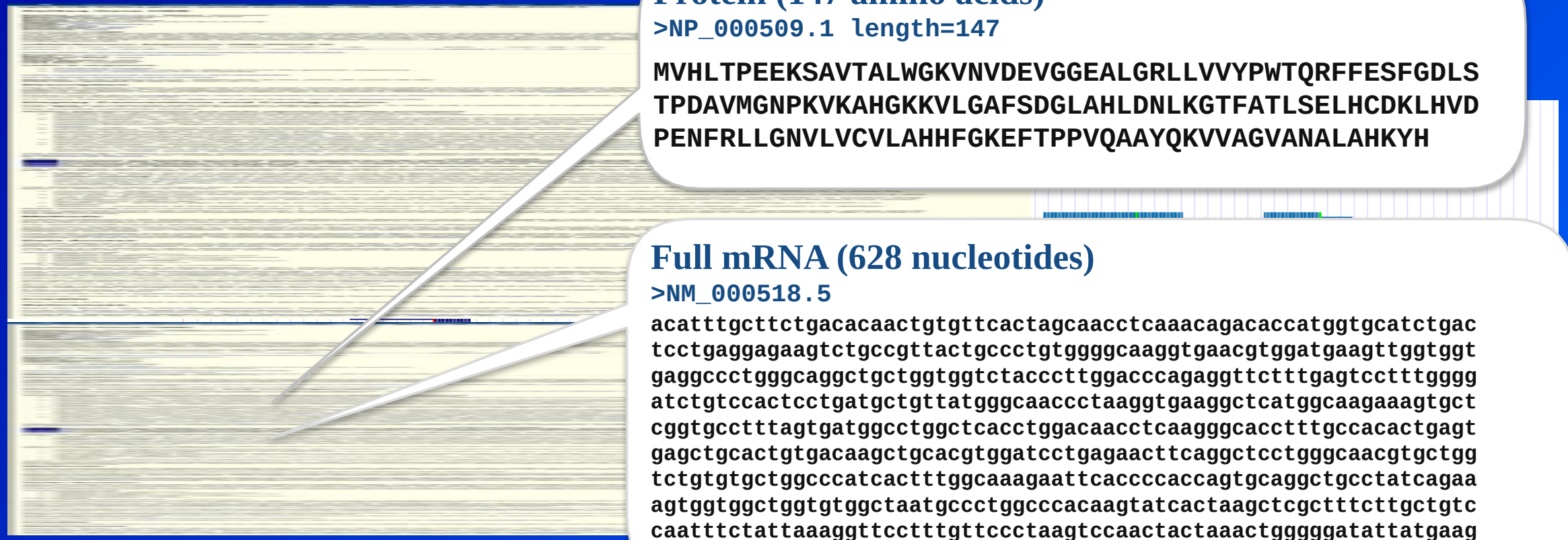
Click this transcript



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Genome browsers

One transcript, different sequences



Protein (147 amino acids)

>NP_000509.1 length=147

MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLS
TPDAVMGNPKVKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVD
PENFRLLGNVLVCVLAHHFGKEFTPPVQAAYQKVVAGVANALAHKYH

Full mRNA (628 nucleotides)

>NM_000518.5

acatttgcttctgacacaactgtgttcactagcaacctcaaacagacaccatggtgcatctgac
tcctgaggagaagtctgccgttactgccctgtggggcaaggtgaacgtggatgaagttggtggt
gaggccctgggcaggctgctggtggtctacccttggacccagaggttctttgagtcctttgggg
atctgtccactcctgatgctgttatgggcaaccctaaggtgaaggctcatggcaagaaagtgct
cggtgcctttagtgatggcctggctcacctggacaacctcaagggcacctttgccacactgagt
gagctgcactgtgacaagctgcacgtggatcctgagaacttcaggctcctgggcaacgtgctgg
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agtggtggctggtgtggctaatgccctggcccacaagtatcactaagctcgctttcttgctgtc
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ggccttgagcatctggattctgcctaataaaaaacatttattttcattgcaa

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Genome browsers

One transcript, different sequences



Genomes | Genome Browser | Tools | Mirrors | Downloads | My Data | Projects | Help | About Us | Login

Get DNA in Window (hg38/Human)

Get DNA for

Position chr11:5,225,464-5,227,071

Note: if you would prefer to get DNA for more than one feature of this track at a time, try the Table Browser using the output format sequence.

Sequence Retrieval Region Options:

- ☐ Promoter/Upstream by 1000 bases
- ☑ 5' UTR Exons
- ☑ CDS Exons
- ☑ 3' UTR Exons
- ☑ Introns
- ☐ Downstream by 1000 bases
- ◉ One FASTA record per gene.
- ○ One FASTA record per region (exon, intron, etc.) with 0 extra bases upstream (5') and 0 extra downstream (3')
- ☐ Split UTR and CDS parts of an exon into separate FASTA records

Note: if a feature is close to the beginning or end of a chromosome and upstream/downstream bases are added, they may be truncated in order to avoid extending past the edge of the chromosome.

Sequence Formatting Options:

- ◉ Exons in upper case, everything else in lower case.
- ○ CDS in upper case, UTR in lower case.
- ○ All upper case.
- ○ All lower case.
- ☐ Mask repeats: ◉ to lower case ○ to N

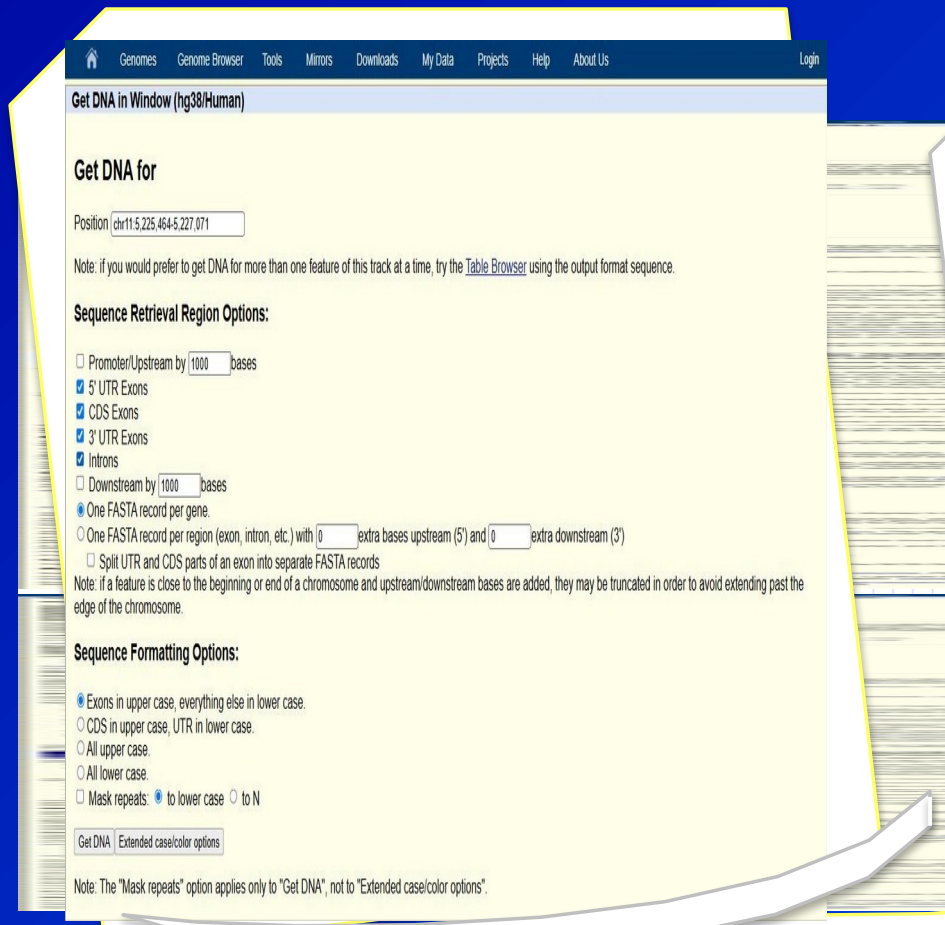
Get DNA | Extended case/color options

Note: The "Mask repeats" option applies only to "Get DNA", not to "Extended case/color options".

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Genome browsers

One transcript, different sequences



Genomic DNA: 1,608 nt

```
>hg38_ncbiRefSeqCurated_NM_000518.5
range=chr11:5225464-5227071 5'pad=0 3'pad=0 strand=- repeatMasking=none

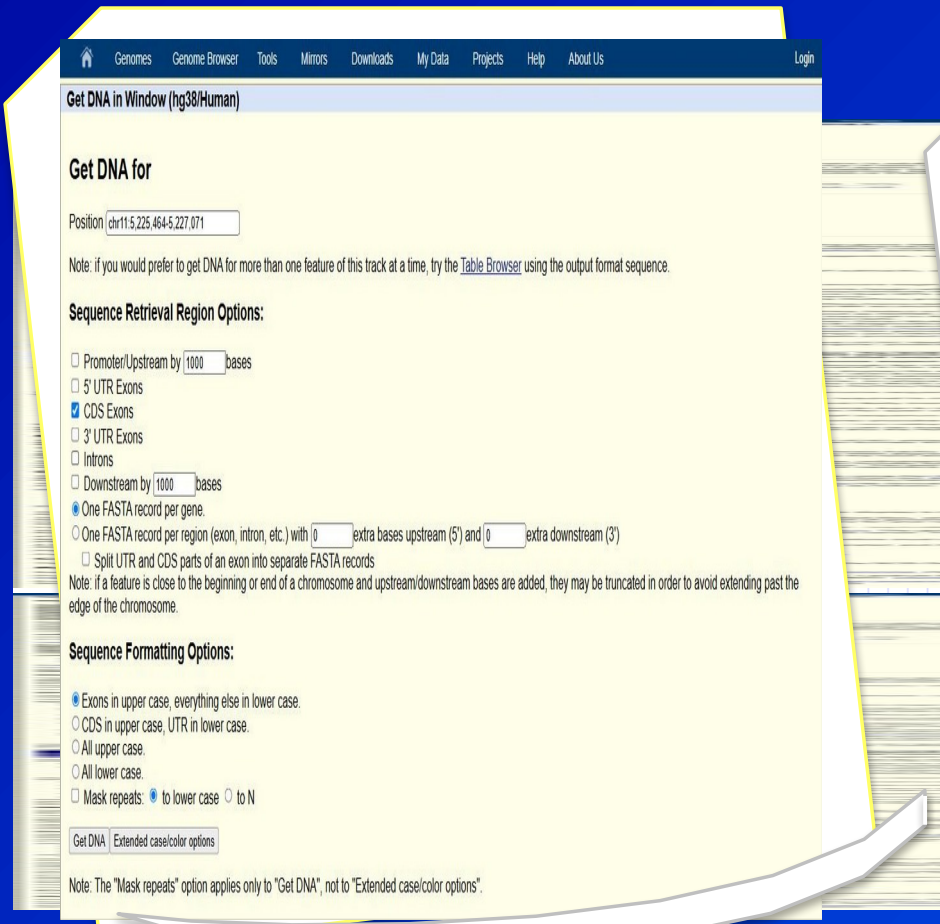
ACATTTGCTTCTGACACAACTGTGTTCACTAGCAACCTCAAACAGACACCATGGTGCATCTGACTCCTGAGGAGAAGTCTGCCG
TTACTGCCCTGTGGGGCAAGGTGAACGTGGATGAAGTTGGTGGTGAGGCCCTGGGCAGgttggtatcaaggttacaagacaggt
ttaaggagaccaatagaaactgggcatgtggagacagagaagactcttgggtttctgataggcactgactctctctgcctattg
gtctattttcccaccctagGCTGCTGGTGGTCTACCCTTGGACCCAGAGGTTCTTTGAGTCCTTTGGGGATCTGTCCACTCCT
GATGCTGTTATGGGCAACCCTAAGGTGAAGGCTCATGGCAAGAAAGTGCTCGGTGCCTTTAGTGATGGCCTGGCTCACCTGGAC
AACCTCAAGGGCACCTTTGCCACACTGAGTGAGCTGCACTGTGACAAGCTGCACGTGGATCCTGAGAACTTCAGGgtgagtcta
tgggacgcttgatgttttctttcccttctttttctatggttaagttcatgtcataggaaggggataagtaacagggtacagttt
agaatgggaaacagacgaatgattgcatcagtgtggaagtctcaggatcgttttagtttcttttatttgctgttcataacaatt
gttttcttttgtttaattcttgctttctttttttttcttctccgcaatttttactattatacttaatgccttaacattgtgtat
aacaaaaggaaatatctctgagatacattaagtaacttaaaaaaaaactttacacagtctgcctagtacattactatttggaat
atatgtgtgcttatttgcatattcataatctccctactttatttcttttatttttaattgatacataatcattatacatattt
atgggttaaagtgtaatgttttaatatgtgtacacatattgaccaaatcagggtaattttgcatttgtaatttttaaaaaatgct
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tatcatgcctctttgcaccattctaaagaataacagtgataatttctgggttaaggcaatagcaatatctctgcatataaaatat
ttctgcatataaattgtaactgatgtaagaggtttcatattgctaatagcagctacaatccagctaccattctgcttttatttt
atggttgggataaggctggattattctgagtccaagctaggccctttgctaatcatgttcatacctcttatcttcctcccaca
gCTCCTGGGCAACGTGCTGGTCTGTGTGCTGGCCCATCACTTTGGCAAAGAATTCACCCCACCAGTGCAGGCTGCCTATCAGAA
AGTGGTGGCTGGTGTGGCTAATGCCCTGGCCCACAAGTATCACTAAGCTCGCTTTCTTGCTGTCCAATTTCTATTAAAGGTTCC
TTTGTTCCCTAAGTCCAACTACTAAACTGGGGGATATTATGAAGGGCCTTGAGCATCTGGATTCTGCCTAATAAAAAACATTTA
TTTTCATTGCAA
```

Uppercase: exons Lowercase: introns

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Genome browsers

One transcript, different sequences



Coding sequence: 444 nt

```
>hg38_ncbiRefSeqCurated_NM_000518.5
range=chr11:5225598-5227021 5'pad=0 3'pad=0 strand=- repeatMasking=none

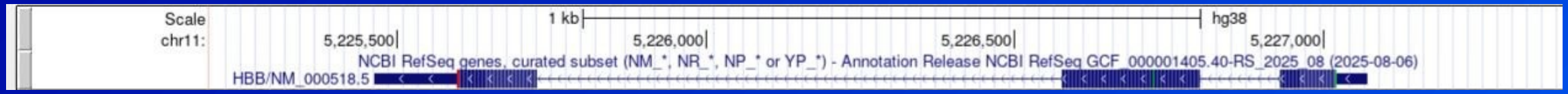
ATGGTGCATCTGACTCCTGAGGAGAAGTCTGCCGTTACTGCCCTGTGGGGCAAGGTGA
ACGTGGATGAAGTTGGTGGTGAGGCCCTGGGCAGGCTGCTGGTGGTCTACCCTTGGAC
CCAGAGGTTCTTTGAGTCCTTTGGGGATCTGTCCACTCCTGATGCTGTTATGGGCAAC
CCTAAGGTGAAGGCTCATGGCAAGAAAGTGCTCGGTGCCTTTAGTGATGGCCTGGCTC
ACCTGGACAACCTCAAGGGCACCTTTGCCACACTGAGTGAGCTGCACTGTGACAAGCT
GCACGTGGATCCTGAGAACTTCAGGCTCCTGGGCAACGTGCTGGTCTGTGTGCTGGCC
CATCACTTTGGCAAAGAATTCACCCCACCAGTGCAGGCTGCCTATCAGAAAGTGGTGG
CTGGTGTGGCTAATGCCCTGGCCCACAAGTATCACTAA
```

Joined CDS exons, including terminal stop codon

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Genome browsers

One transcript, different sequences



Genomic DNA, 1,608 nt

```
ACATTTGCTTCTGACACAACTGTGTTCACTAGCAACCTCAAACAGACACCATGGTGCATCTGACTCCTGAGGAGAAGTCTGCCGT
TACTGCCCTGTGGGGCAAGGTGAACGTGGATGAAGTTGGTGGTGAGGCCCTGGGCAGgttggtatcaaggttacaagacaggttt
aaggagaccaatagaaactgggcatgtggagacagagaagactcttgggtttctgataggcactgactctctctgcctattggtc
tattttcccaccccttagGCTGCTGGTGGTCTACCCTTGGACCCAGAGGTTCTTTGAGTCCTTTGGGGATCTGTCCACTCCTGATG
CTGTTATGGGCAACCCTAAGGTGAAGGCTCATGGCAAGAAAGTGCTCGGTGCCTTTAGTGATGGCCTGGCTCACCTGGACAACCT
CAAGGGCACCTTTGCCACACTGAGTGAGCTGCACTGTGACAAGCTGCACGTGGATCCTGAGAACTTCAGGgtgagtctatgggac
gcttgatgttttctttccccttctttttctatggttaagttcatgtcataggaaggggataagtaacagggtacagtttagaatgg
gaaacagacgaatgattgcatcagtgtggaagtctcaggatcgttttagtttcttttatttgctgttcataacaattgttttctt
ttgtttaattcttgctttctttttttttctttctccgcaatttttactattatacttaatgccttaacattgtgtataacaaaagg
aaatatctctgagatacattaagtaacttaaaaaaaaactttacacagtctgcctagtacattactatttggaatatatgtgtgc
ttatttgcatattcataatctccctactttatttttctttttatttttaattgatacataatcattatacatatttatgggttaaag
tgtaatgttttaatatgtgtacacatattgaccaaatcagggtaattttgcatttgtaatttttaaaaaatgctttcttcttttaa
tatacttttttgtttatctttatttctaatactttccctaatctctttctttcagggcaataatgatacaatgtatcatgcctctt
tgcaccattctaaagaataacagtgataatttctgggttaaggcaatagcaatatctctgcatataaaatatttctgcatataaat
tgtaactgatgtaagaggtttcatattgctaatagcagctacaatccagctaccattctgcttttatttatggttgggataagg
ctggattattctgagtccaagctaggcccttttgctaatcatgttcatacctcttatcttcctcccacagCTCCTGGGCAACGTG
CTGGTCTGTGTGCTGGCCCATCACTTTGGCAAAGAATTCACCCCACCAGTGCAGGCTGCCTATCAGAAAGTGGTGGCTGGTGTGG
CTAATGCCCTGGCCCACAAGTATCACTAAGCTCGCTTTCTTGCTGTCCAATTTCTATTAAAGGTTCCTTTGTTCCCTAAGTCCAA
CTACTAAACTGGGGGATATTATGAAGGGCCTTGAGCATCTGGATTCTGCCTAATAAAAAACATTTATTTTCATTGCAA
```

Join exons

Full transcript, 628 nt

```
acatttgcttctgacacaactgtgttcactagcaacctcaaacagacaccatggtgcatctgac
tcctgaggagaagtctgccgttactgccctgtggggcaaggtgaacgtggatgaagttggtggt
gaggccctgggcaggctgctggtggtctacccttggacccagaggttctttgagtcctttgggg
atctgtccactcctgatgctgttatgggcaaccctaaggtgaaggctcatggcaagaaagtgct
cggtgcctttagtgatggcctggctcacctggacaacctcaagggcacctttgccacactgagt
gagctgcactgtgacaagctgcacgtggatcctgagaacttcaggctcctgggcaacgtgctgg
tctgtgtgctggcccatcactttggcaaagaattcaccccaccagtgcaggctgcctatcagaa
agtggtggctggtgtggctaatgccctggcccacaagtatcactaagctcgctttcttgctgtc
caatttctattaaaggttcctttgttccctaagtccaactactaaactgggggatattatgaag
ggccttgagcatctggattctgcctaataaaaaacatttatttcattgcaa
```

Select CDS

Coding sequence, 444 nt

```
ATGGTGCATCTGACTCCTGAGGAGAAGTCTGCCGTTACTGCCCTGTGGGGCAAGGTGA
ACGTGGATGAAGTTGGTGGTGAGGCCCTGGGCAGGCTGCTGGTGGTCTACCCTTGGAC
CCAGAGGTTCTTTGAGTCCTTTGGGGATCTGTCCACTCCTGATGCTGTTATGGGCAAC
CCTAAGGTGAAGGCTCATGGCAAGAAAGTGCTCGGTGCCTTTAGTGATGGCCTGGCTC
ACCTGGACAACCTCAAGGGCACCTTTGCCACACTGAGTGAGCTGCACTGTGACAAGCT
GCACGTGGATCCTGAGAACTTCAGGCTCCTGGGCAACGTGCTGGTCTGTGTGCTGGCC
CATCACTTTGGCAAAGAATTCACCCCACCAGTGCAGGCTGCCTATCAGAAAGTGGTGG
CTGGTGTGGCTAATGCCCTGGCCCACAAGTATCACTAA
```

Translate

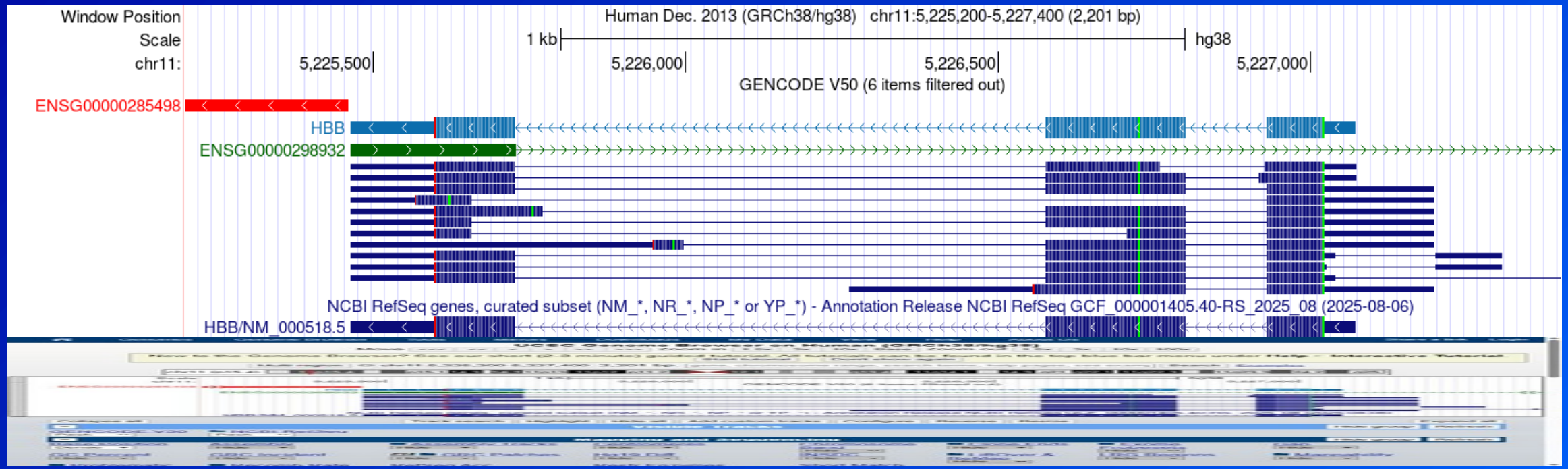
Protein, 147 aa

```
MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLS
TPDAVMGNPKVKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVD
PENFRLLGNVLVCVLAHHFGKEFTPPVQAAYQKVVAGVANALAHKYH
```

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Genome browsers

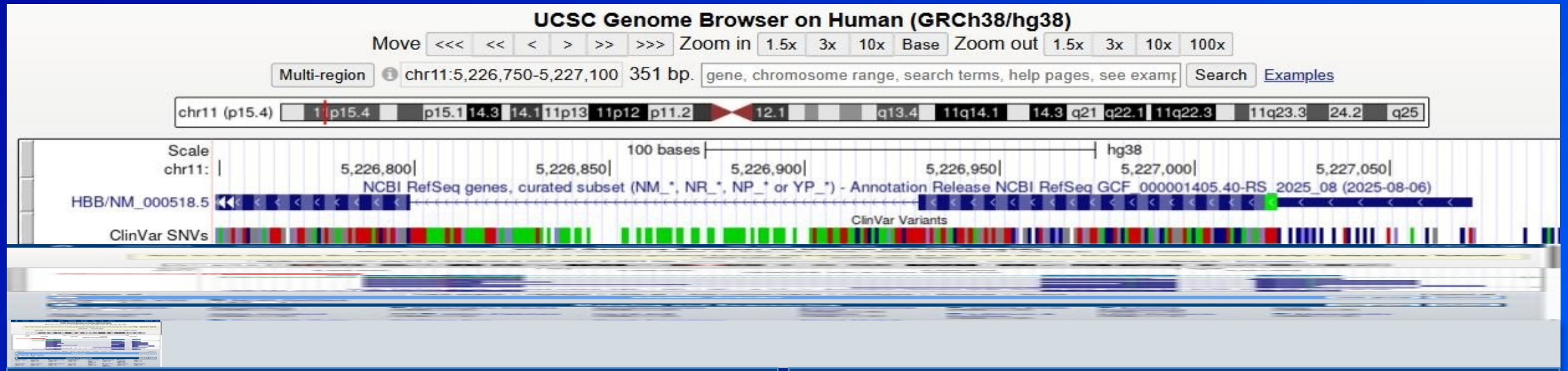
What is visible?



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Genome browsers

What is visible?

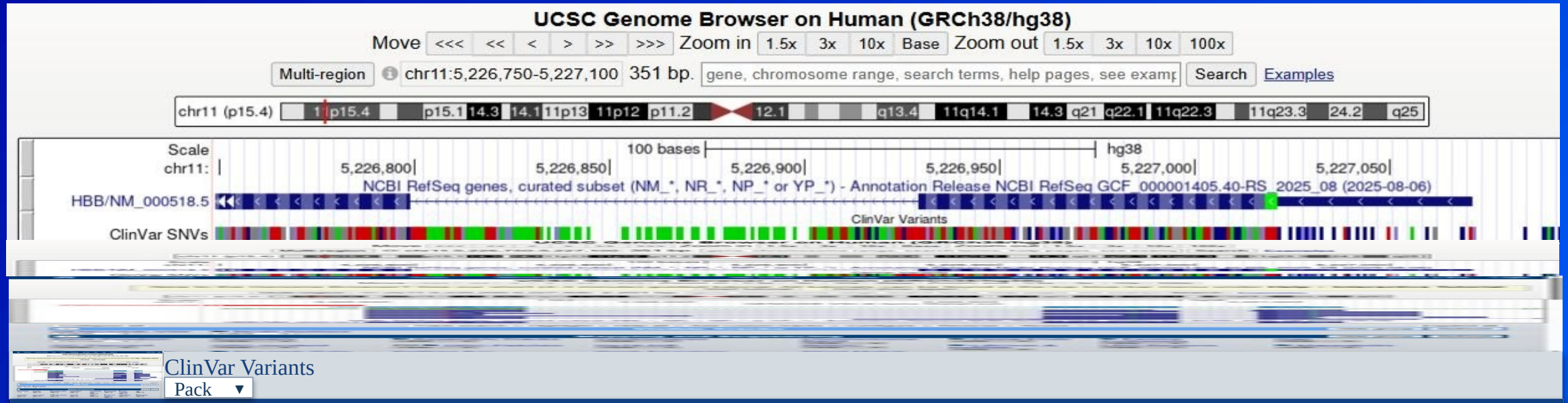


- + Mapping and Sequencing
- + Genes and Gene Predictions
- + Phenotypes, Variants, and Literature
- + Variation
- + Human Pangenome - HPRC
- + RNA and Transcriptome
- + Expression
- + Single Cell RNA-seq
- + Regulation
- + Comparative Genomics
- + Repeats

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Genome browsers

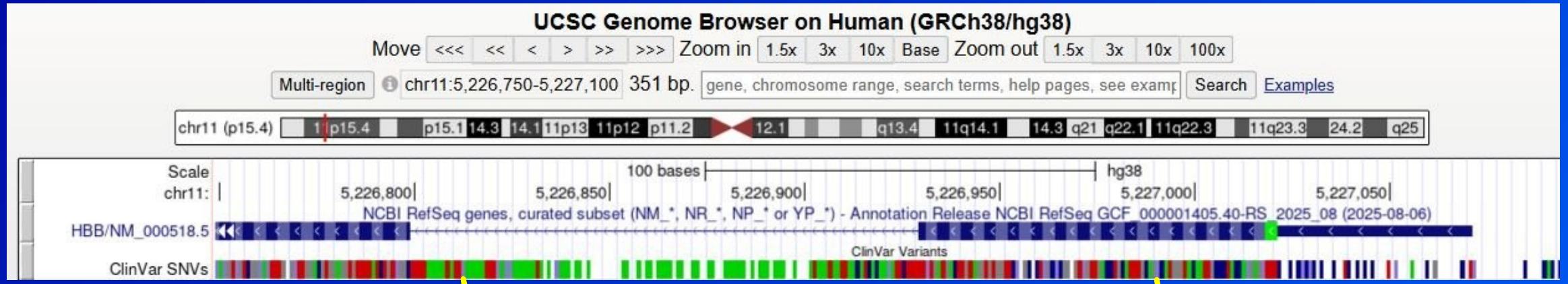
What is visible?



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Genome browsers

Where does the variant lie?



An intronic variant:

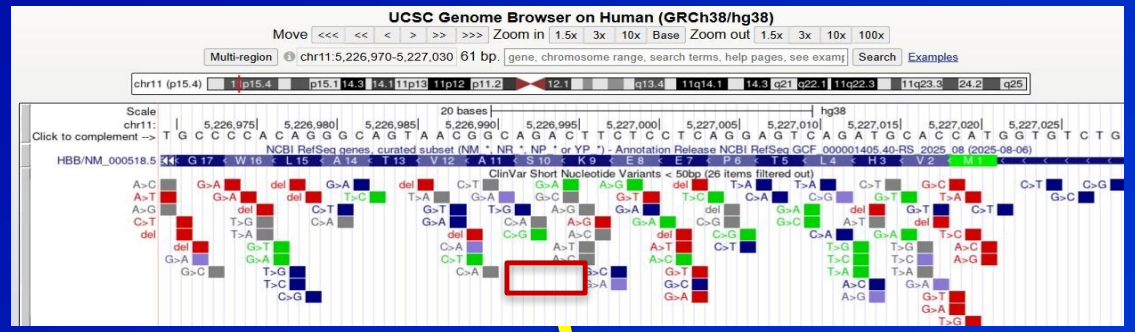
A coding variant:

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Genome browsers

What is consequence of a variant?

Coding variant

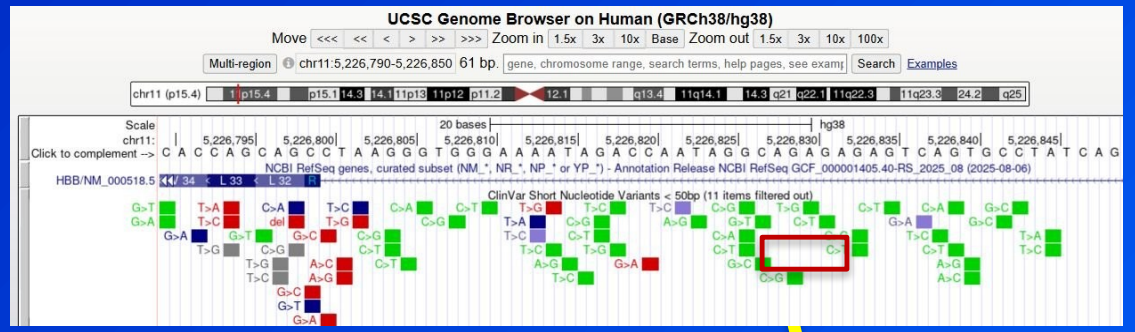


at DNA position 5,227,002 A changes to T

GAG becomes GTG

GLU7 in beta-globin becomes Val

Intronic variant



at DNA position 5,226,820 G changes to A

Leaves 19 intronic bases in mRNA

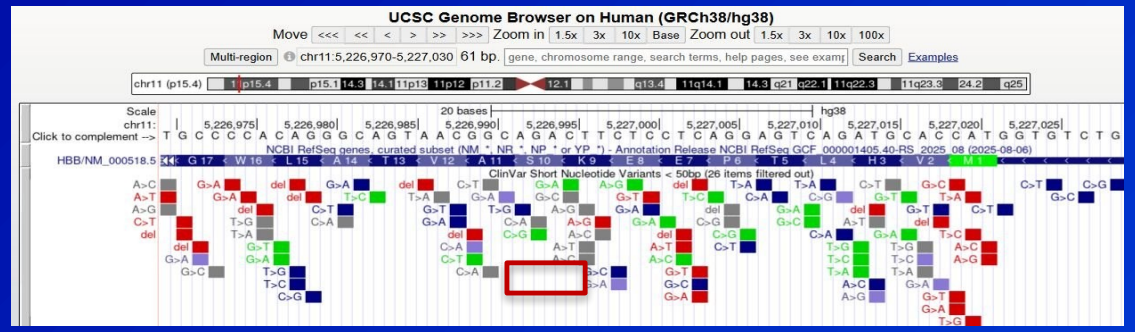
Premature stop codon

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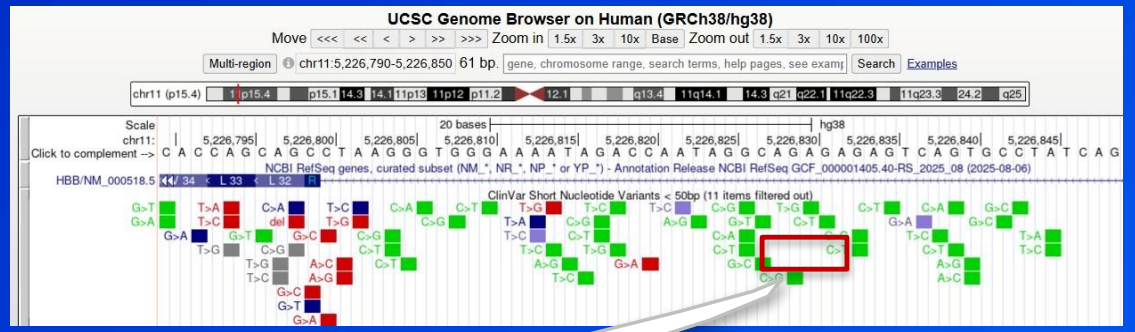
Genome browsers

Follow the evidence

Coding variant



Intronic variant



UCSC feature details

Link to ClinVar with Variant ID	VCV000015454
Clinical significance	Pathogenic
Review Status	★★★☆ based on: reviewed by expert panel

ClinVar record

Supporting submissions and cited studies

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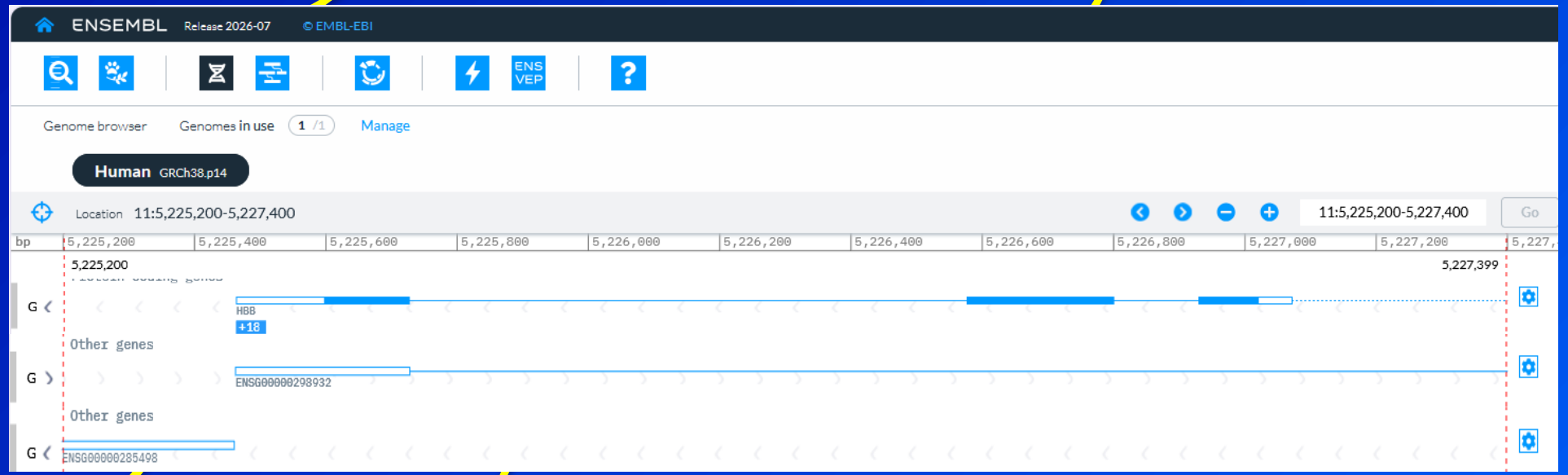
Genome browsers

What can we find in Ensembl?

Genome assembly

Exons separated by introns

Chromosome region



Possible transcript

Non-coding RNA exon on opposite DNA strand

https://www.ensembl.org/genome-browser/GCA_000001405.29?focus=location:11:5225200-5227400&location=11:5225200-5227400

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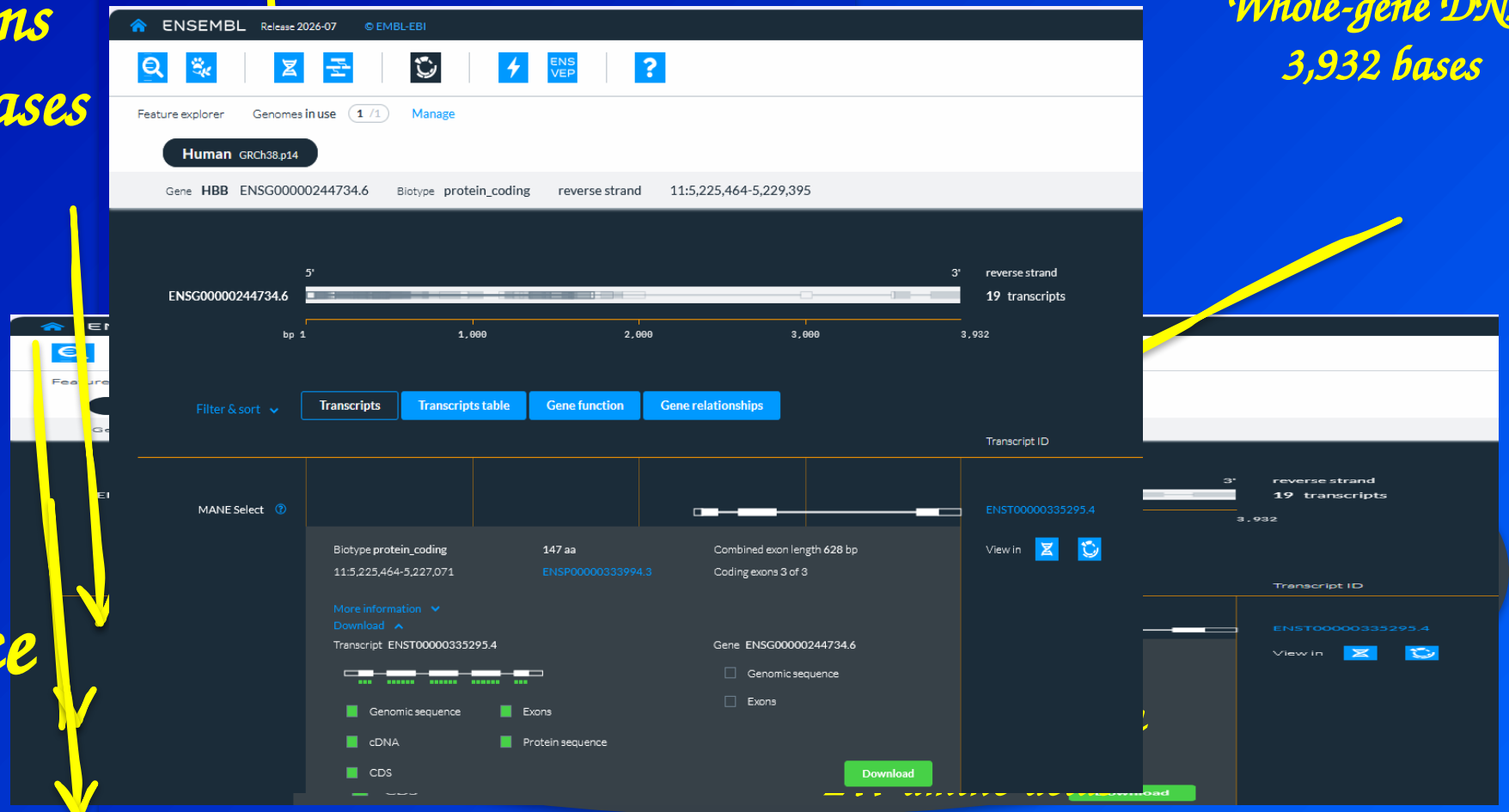
Genome browsers

One gene, many sequences

Three separate exons
142, 223 and 263 bases

Genomic DNA for that transcript
1,608 bases

Whole-gene DNA
3,932 bases



Spliced transcript
628 bases

Coding sequence
444 bases

https://www.ensembl.org/feature-explorer/GCA_000001405.29/gene:ENSG00000244734?view=transcripts

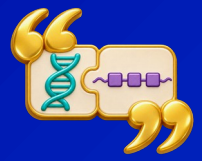
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Finding evidence online

We located an HBB splice variant. Which study measured its effect?



Literature databases

-  **Build efficient queries**
-  **Filter and open papers**
-  **Follow subject headings**



AI-based discovery

-  **Find candidate studies**
-  **Check claims in the paper**
-  **Spot AI search pitfalls**

Stockholms universitet

Building efficient queries

Logical operators

Operator	Schematic matches	X query	Y query	PubMed hits for		
AND		HBB	review	2,899	5,111,910	241
OR		HBB	"beta-globin"	2,899	8,884	11,067
NOT		(HBB AND thalassemia)	review	939	5,111,910	863

PubMed counts: 11 Nov 2026

Stockholms universitet

Building efficient queries

Restricting search space

Example	Unrestricted query	Hits	Restricted query	Hits
Gene	HBB	2,899	HBB[tiab]	2,701
Article type	review	5,111,910	review[pt]	4,100,845
AND	HBB AND review	241	HBB[tiab] AND review[pt]	148
OR	HBB OR "beta-globin"	11,067	HBB[tiab] OR "beta-globin"[tiab]	10,869
NOT	(HBB AND thalassemia) NOT review	863	(HBB[tiab] AND thalassemia[tiab]) NOT review[pt]	791

[tiab] Title, abstract and author keywords

[pt] Publication type

PubMed counts: 11 Nov 2026

Stockholms universitet

Building efficient queries

Quotation marks

Query

beta AND globin

beta AND globin

24,779 PubMed records

"beta globin"

8,885 PubMed records

Schematic text matches



Quotation marks keep the words together, in this order



PubMed, 11 Nov 2026. Counts change as the database updates.

Stockholms universitet

Building efficient queries

Use parentheses with care

Query	Schematic matches	Interpretation	PubMed hits
HBB AND thalassemia OR "beta globin"		“beta globin” alone is sufficient; HBB must be combined with thalassemia	9,357
HBB AND (thalassemia OR "beta globin")		requires HBB and at least one of the other two term	1,189
HBB OR "beta globin" AND thalassemia		Requires thalassemia, together with at least one of the two names.	3,636
HBB OR ("beta globin" AND thalassemia)		HBB alone is sufficient; “beta globin” must be combined with thalassemia	5,596

A: HBB B: "beta globin" C: thalassemia Pink: matching records

PubMed, 11 Nov 2026. Counts change as the database updates.

Stockholms universitet

Building efficient queries

Wildcards for word variants

An asterisk (*) can stand for zero or more characters

Search	Query	Matching word examples	PubMed hits
Exact word	HBB AND "thalassemia"	thalassemia	919
Wildcard	HBB AND thalass*	thalassemia, thalassaemia thalassemic, thalassaemic	941

PubMed, 11 Nov 2026. Counts change as the database updates.

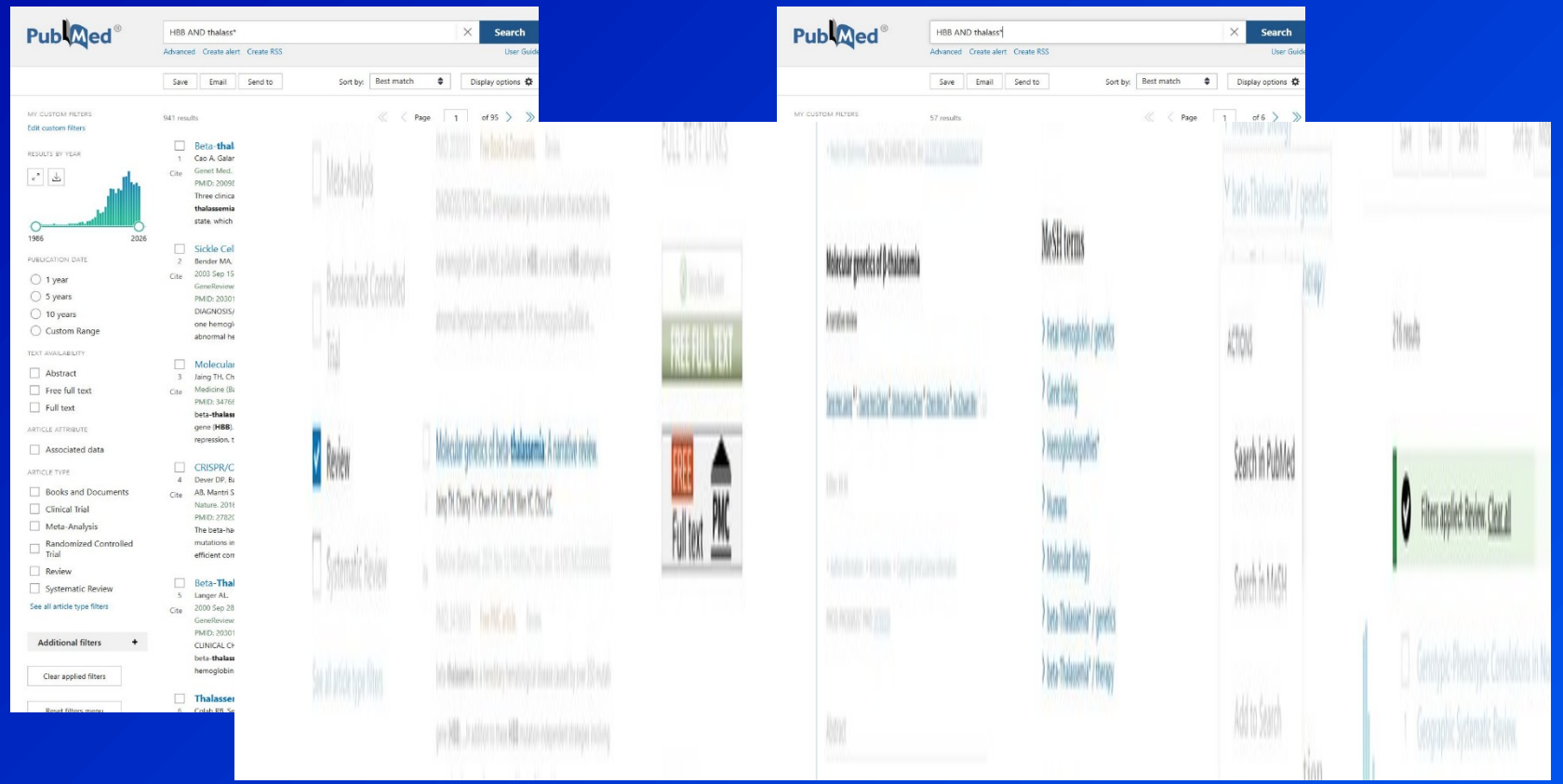
Stockholms universitet

Literature databases

Filter and open papers

Query

Filtered results

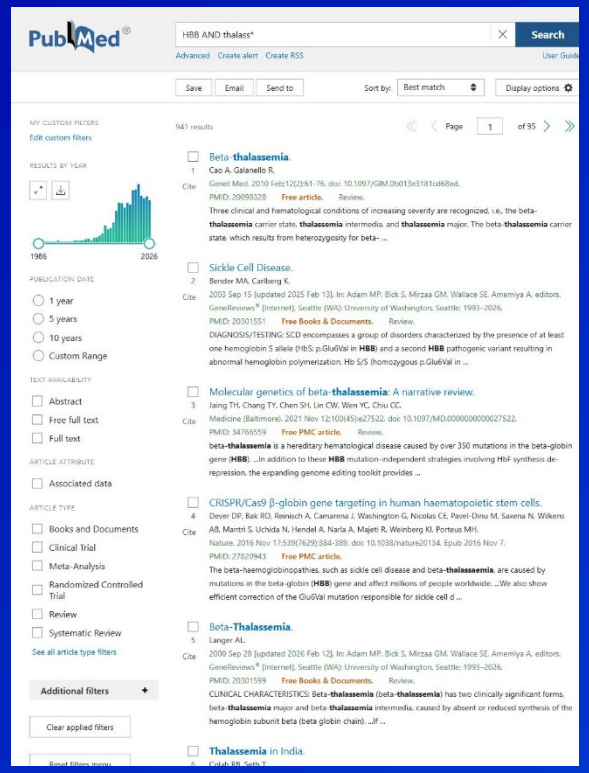


Stockholms universitet

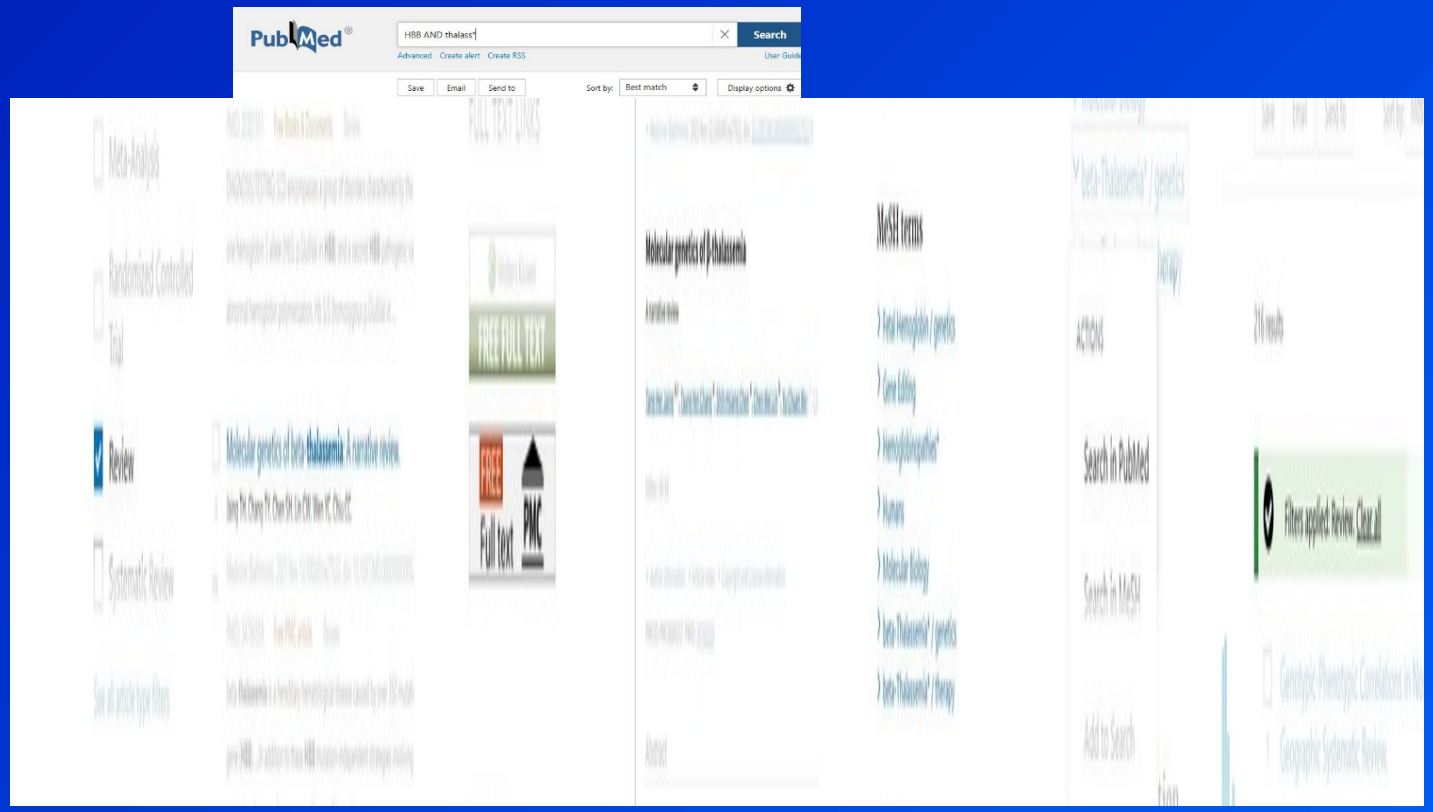
Literature databases

Filter and open papers

Query



Filtered results

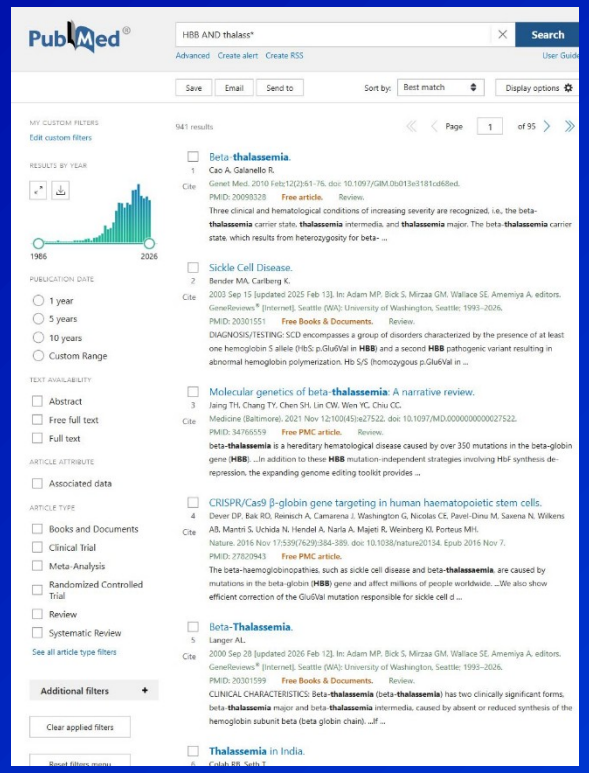


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Literature databases

Filter and open papers

Query



Filtered results



Article record

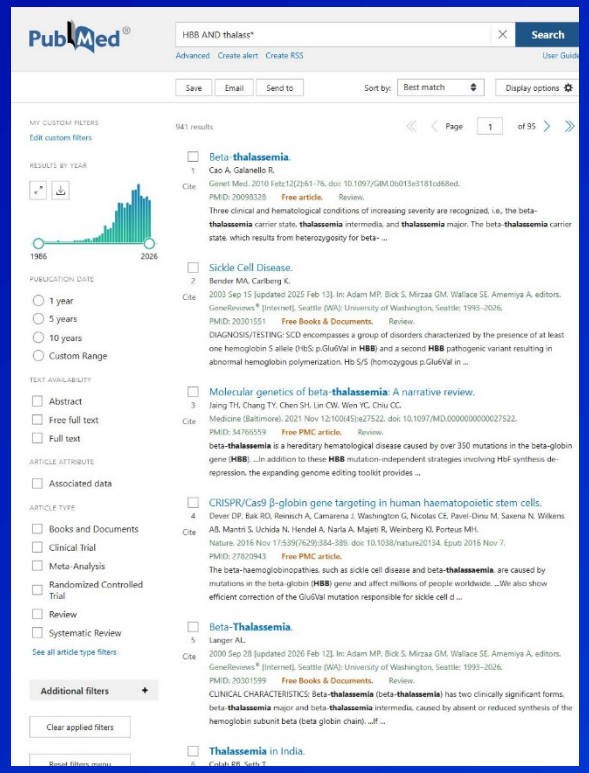


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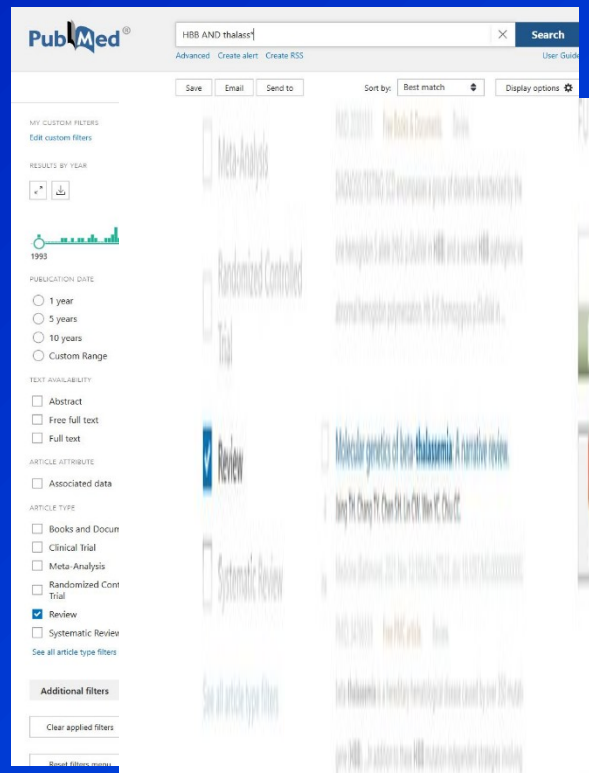
Literature databases

Filter and open papers

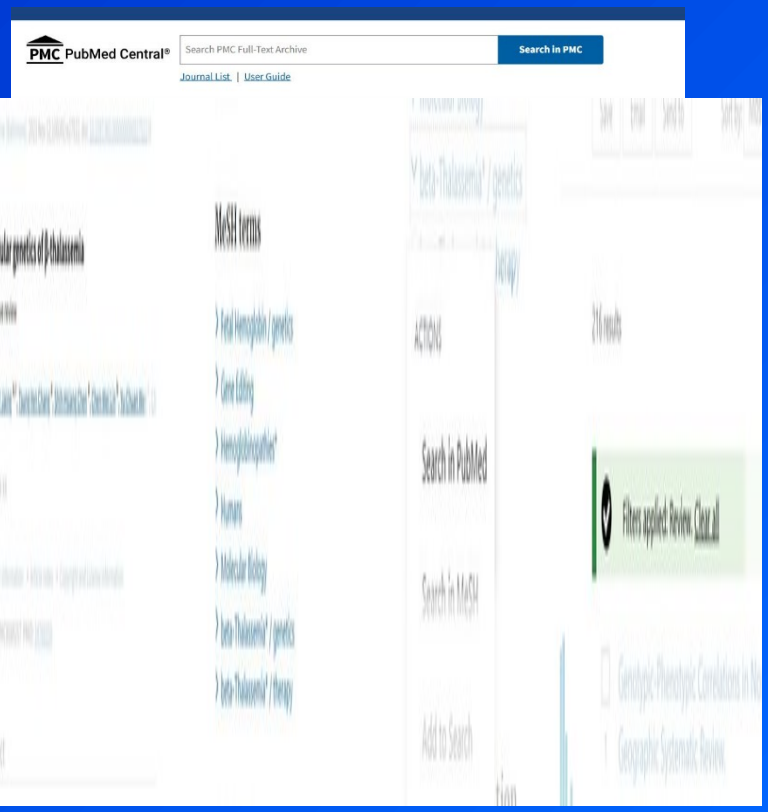
Query



Filtered results



Full articles



Stockholms universitet

Literature databases

Follow subject headings

Article record

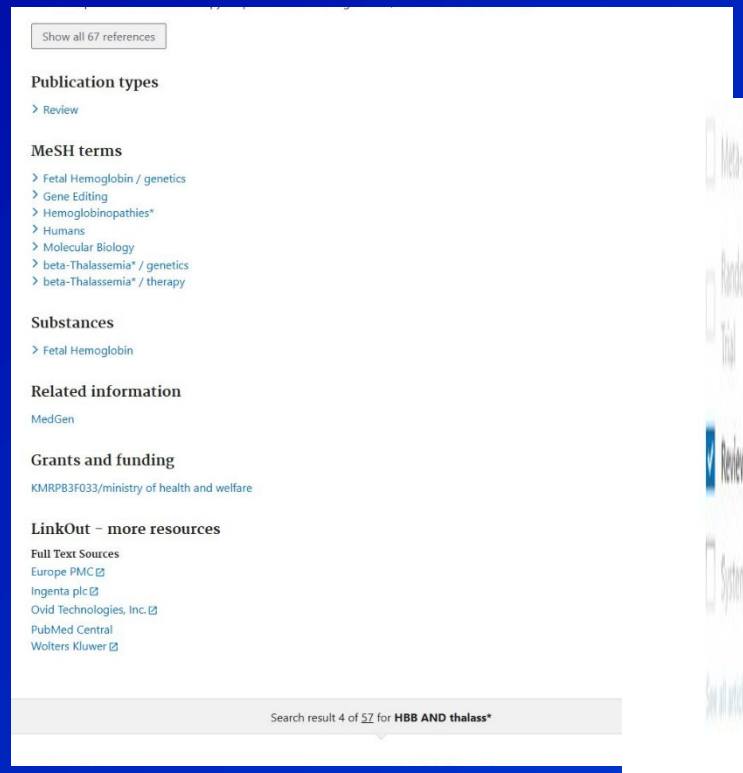


Stockholms universitet

Literature databases

Follow subject headings

Article record



Subject heading

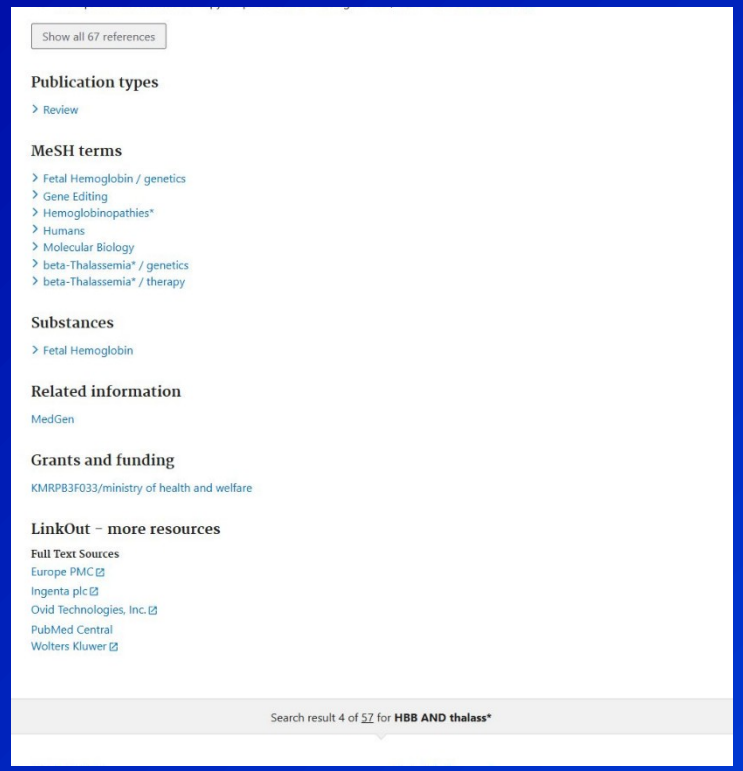


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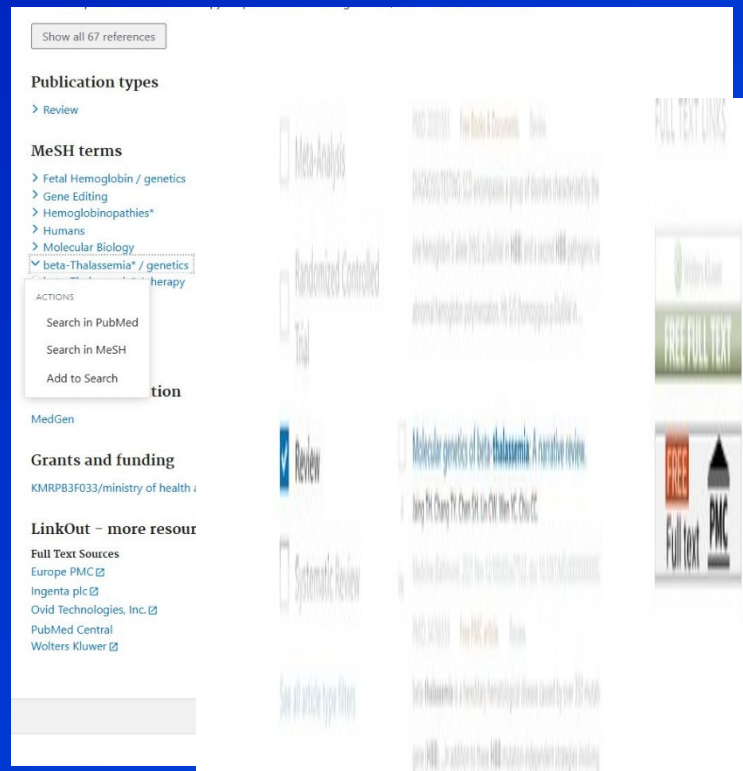
Literature databases

Follow subject headings

Article record



Subject heading



New results

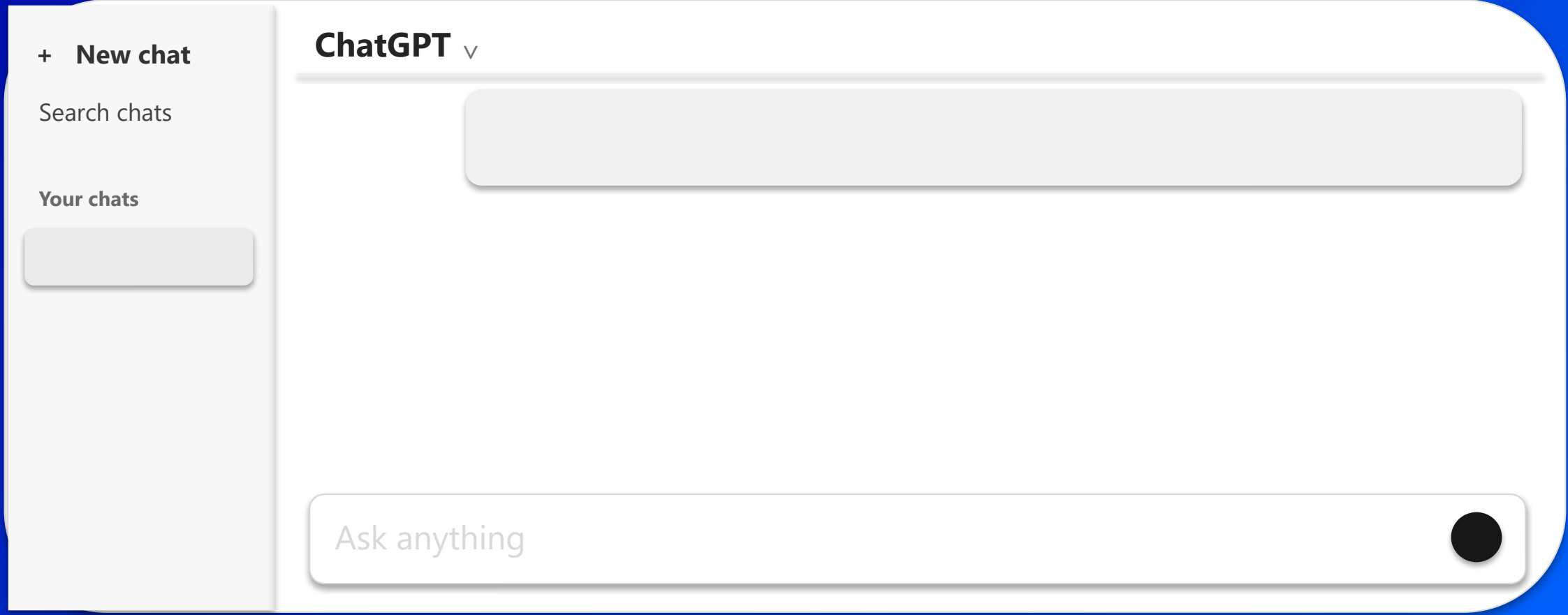


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AI-based discovery

Find candidate studies

Generic biological question



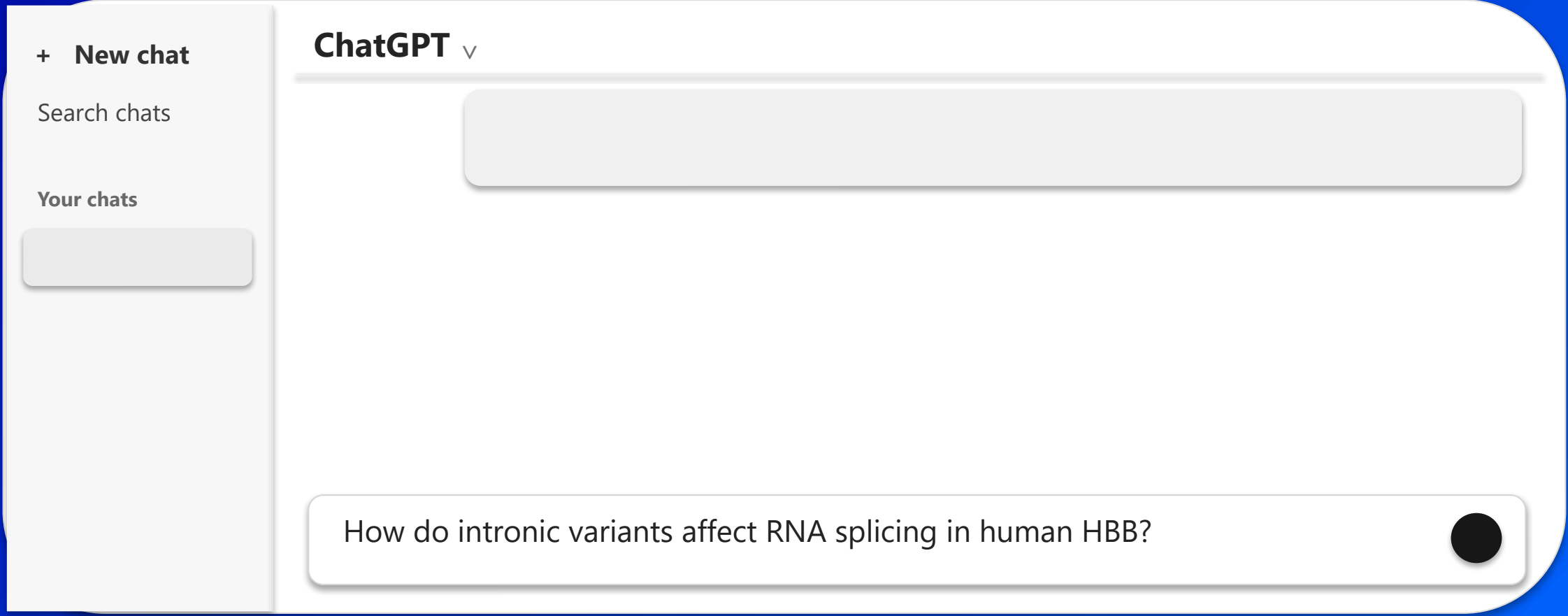
Abridged ChatGPT example

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AI-based discovery

Find candidate studies

Generic biological question



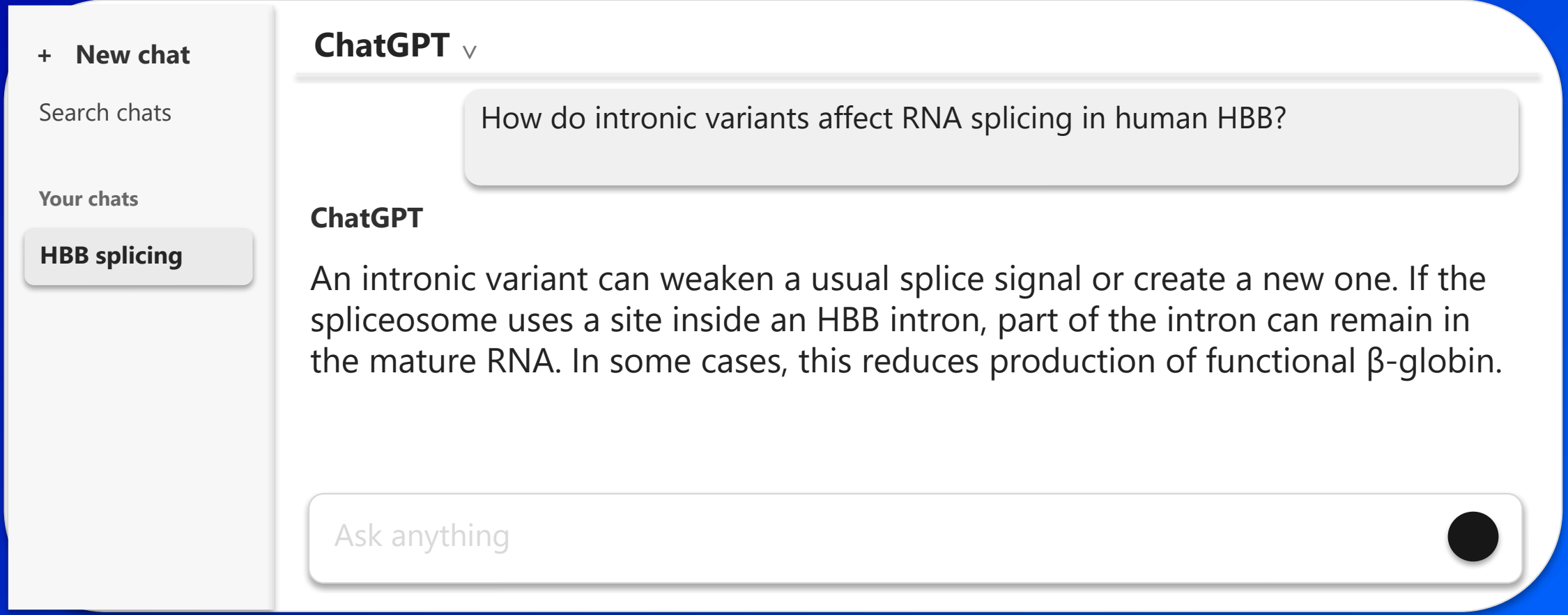
Abridged ChatGPT example

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AI-based discovery

Find candidate studies

Generic biological question



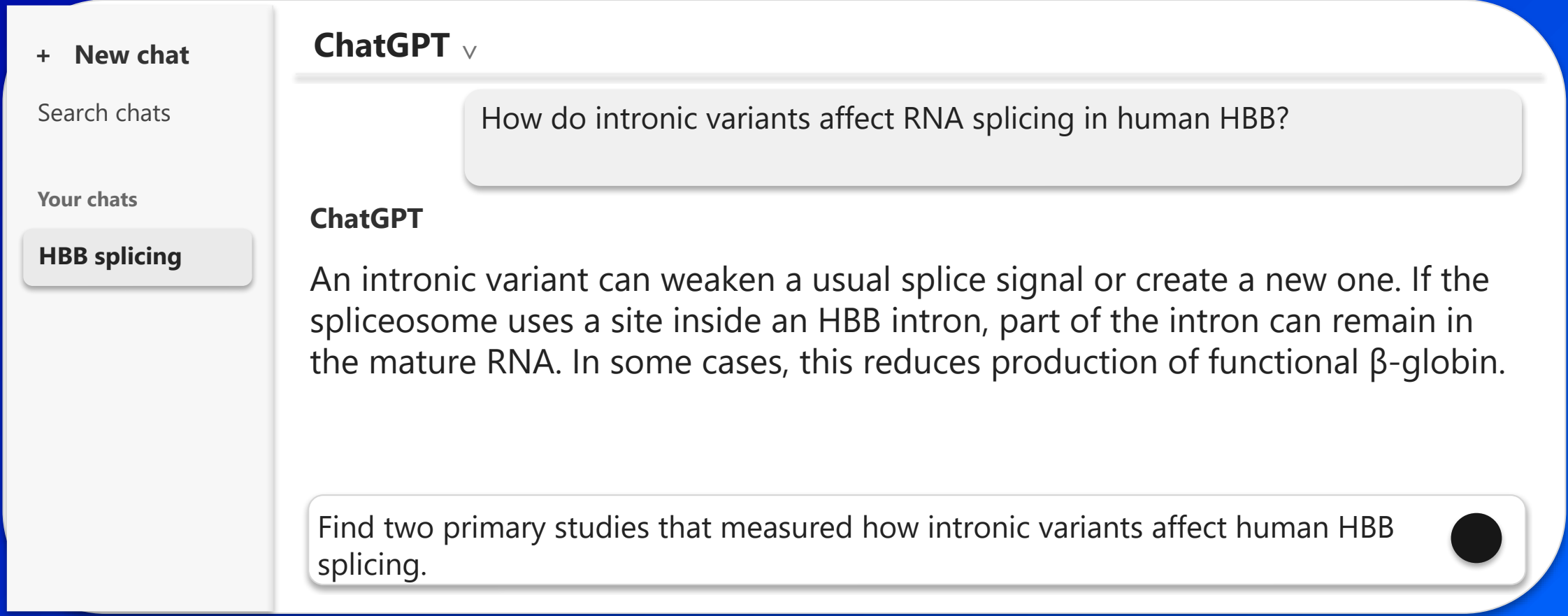
Abridged ChatGPT example

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AI-based discovery

Find candidate studies

Ask for primary studies



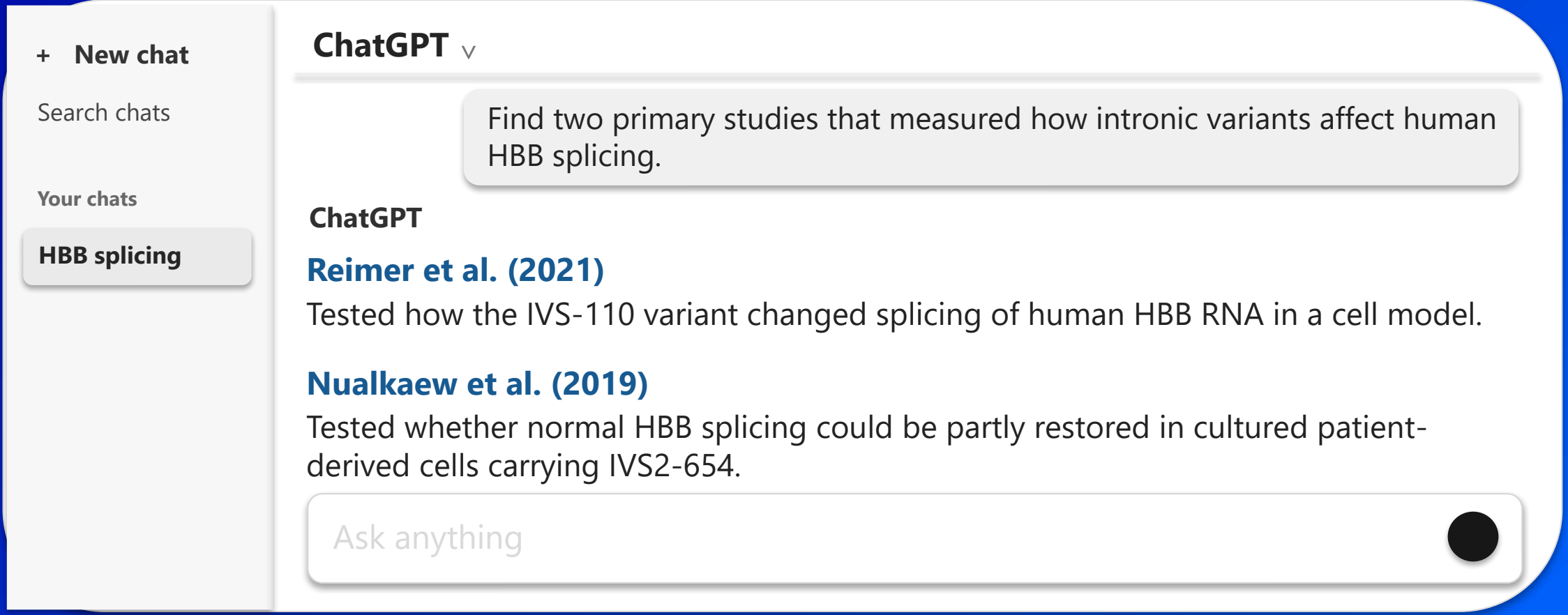
Abridged ChatGPT example

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AI-based discovery

Find candidate studies

Ask for primary studies



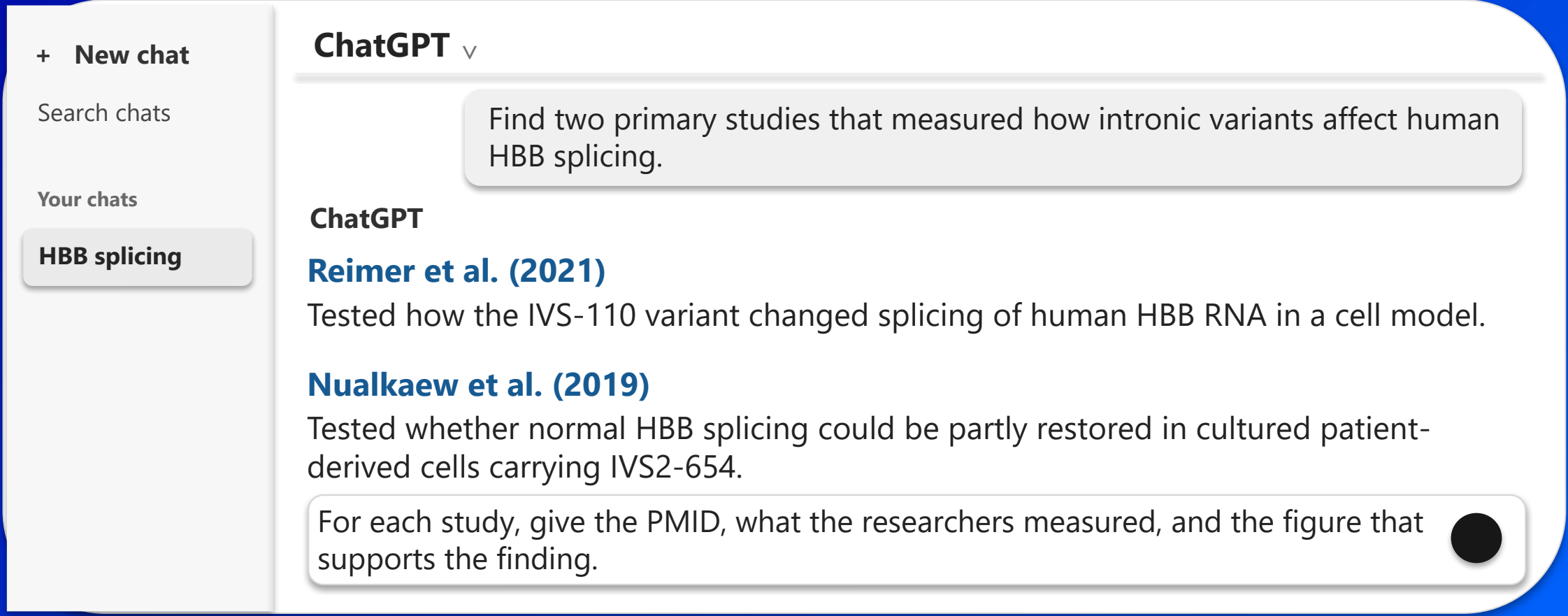
Abridged ChatGPT example

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AI-based discovery

Find candidate studies

Ask for supporting evidence



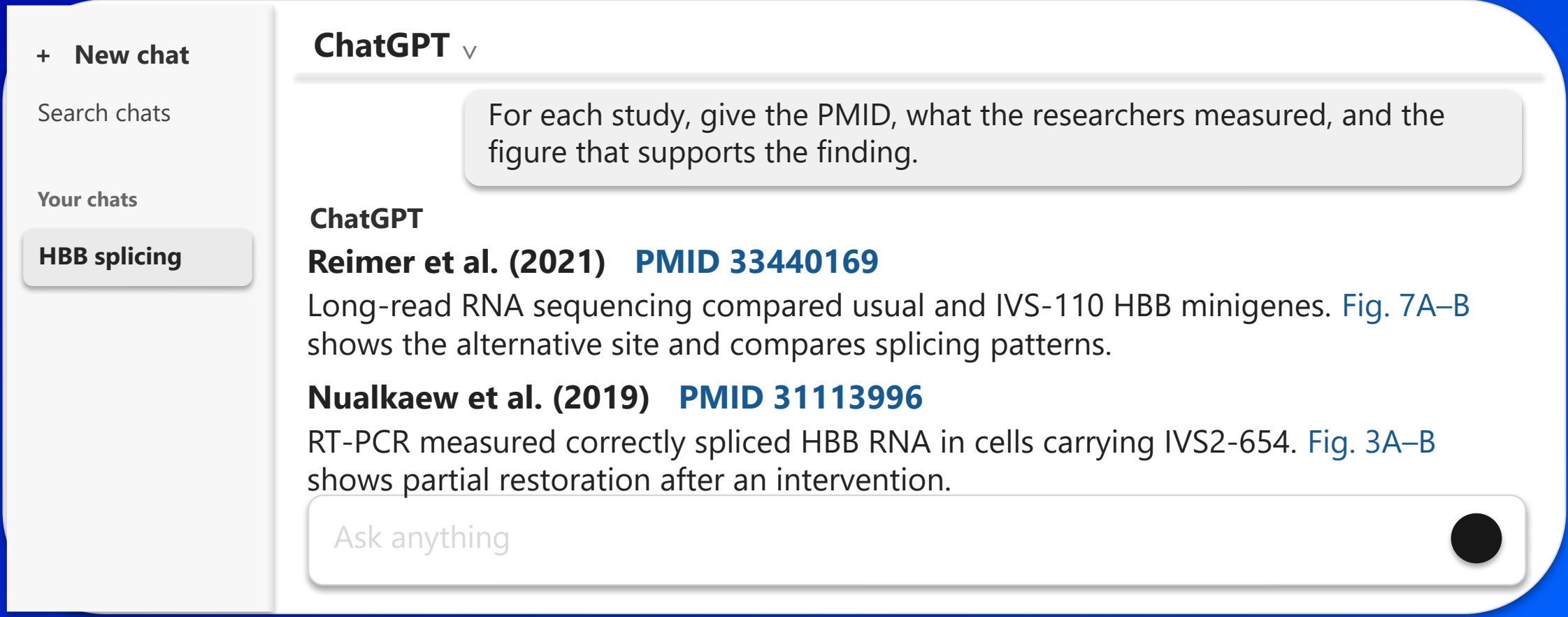
Abridged ChatGPT example

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AI-based discovery

Find candidate studies

Ask for supporting evidence



Abridged ChatGPT example

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AI-based discovery

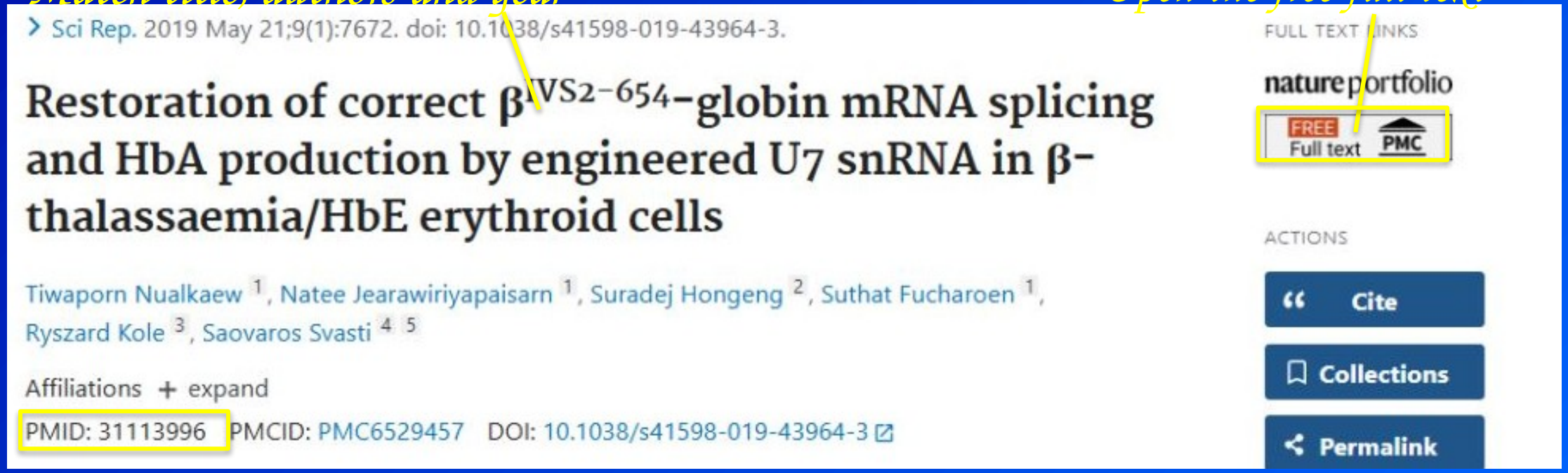
Check claims in the paper

Follow the citation **(Nualkaew et al. (2019), PMID 31113996)**

AI summary: partially restored HBB splicing in cultured cells

Match title, authors and year

Open the free full text

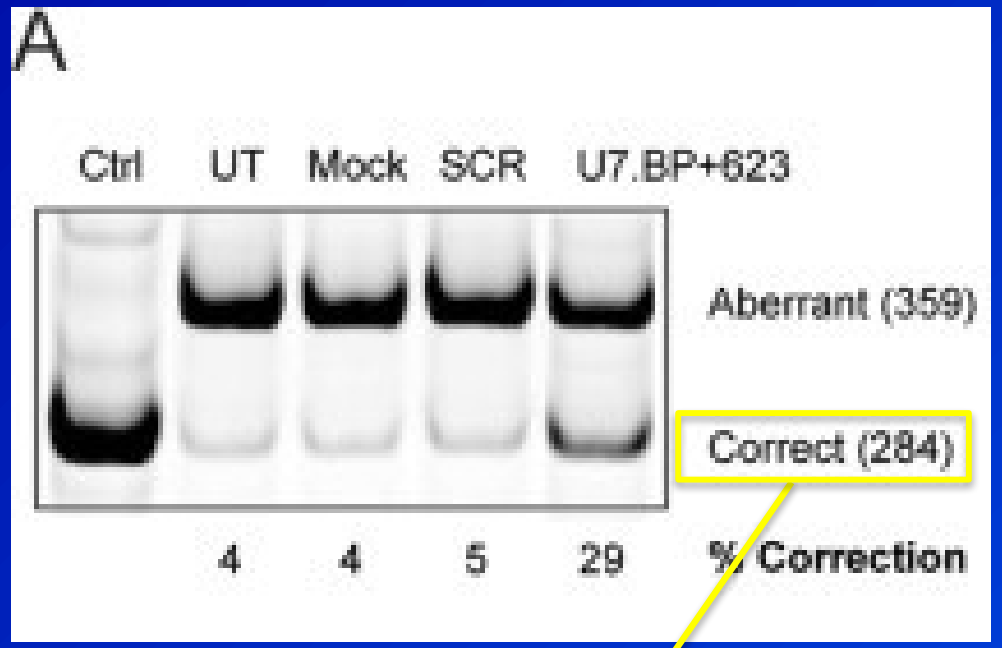


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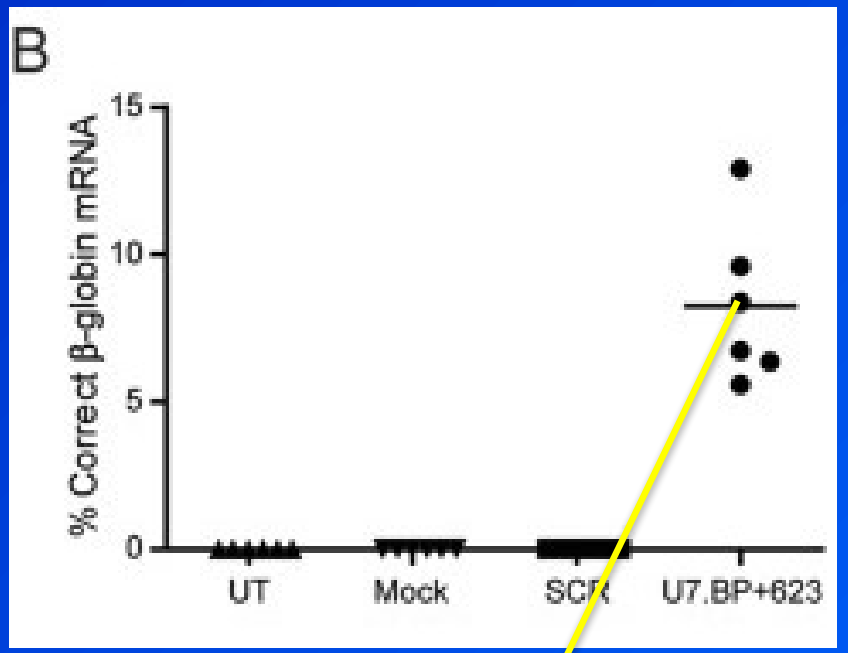
AI-based discovery

Check claims in the paper

Find the measurement



Shorter product: correctly spliced RNA



Mean 8.25% correctly spliced RNA

A–B test RNA splicing. C–D test haemoglobin A separately.

Nualkaew et al. (2019), Figure 3. CC BY 4.0.

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AI-based discovery

Check claims in the paper

State what the experiment supports

Cells in culture

Patient-derived erythroid cells carrying IVS2-654

Engineered U7 RNA

U7.BP+623, delivered using a lentiviral vector

Partial restoration

Mean 8.25% correctly spliced β-globin RNA

In cultured patient-derived cells carrying IVS2-654, an engineered U7 RNA intervention partly restored correctly spliced HBB RNA.

Nualkaew et al. (2019), Results and Figure 3A–B

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AI-based discovery

Spot AI search pitfalls



Model: Default (GPT-3.5)

L Write a short paper intended for a general audience that summarizes prevention strategies within the field of dermatology. Include at least 3 references to papers that are indexed in Medline. Provide in-text citations and a complete reference list, including the PMID number for each.

Dermatology is the branch of medicine that deals with the diagnosis and treatment of skin, hair, and nail disorders. Skin disorders, in particular, are highly prevalent, and their incidence is increasing worldwide. Therefore, preventing skin disorders is crucial to maintaining skin health and preventing associated complications. This paper will summarize some of the prevention strategies within the field of dermatology.

Sun protection is one of the most important prevention strategies in dermatology. Prolonged sun exposure is the primary cause of skin cancer and skin aging. Therefore, using sunscreens, wearing protective clothing, and avoiding the sun during peak hours are essential in preventing skin damage. A study by Chen et al. (2021) found that sunscreen use significantly reduced the incidence of melanoma and squamous cell carcinoma, indicating the importance of sun protection.

Another important prevention strategy is early detection and treatment of skin disorders. Regular skin exams by a dermatologist can help detect and treat skin cancer and other skin disorders early. In a study by Telfer et al. (2009), regular skin exams led to the detection of more skin cancers at earlier stages, resulting in better treatment outcomes.

Maintaining a healthy lifestyle is also critical in preventing skin disorders. A balanced diet rich in vitamins and minerals, regular exercise, and stress management can help prevent skin aging and reduce the risk of skin cancer. A study by Kwon et al. (2015) found that a healthy lifestyle, including a balanced diet and regular exercise, was associated with a reduced risk of skin aging in middle-aged women.

In conclusion, prevention strategies in dermatology are crucial in maintaining skin health and preventing associated complications. Sun protection, early detection and treatment of skin disorders, and maintaining a healthy lifestyle are some of the key prevention strategies. By following these strategies, we can protect our skin and maintain its health.

References:

Chen, A. C., Martin, A. J., Choy, B., Fernandez-Penas, P., & Dalziell, R. A. (2021). A systematic review of the relationship between sunscreen use and the prevention of melanoma and non-melanoma skin cancer. European Journal of Dermatology, 31(1), 1-9. PMID: 33635047

Kwon, H. H., Yoon, J. Y., Park, S. Y., Min, S., Suh, D. H., & Kim, J. H. (2015). A comprehensive study on the association between diet and skin aging in Korean women. Journal of the American Academy of Dermatology, 73(2), 242-250. PMID: 26183923

Telfer, N. R., Colver, G. B., Morton, C. A., & British Association of Dermatologists Therapy Guidelines and Audit Subcommittee. (2009). Guidelines for the management of basal cell carcinoma. British Journal of Dermatology, 161(4), 643-659. PMID: 19663808

Would you trust this reference?

Authors and a plausible title

Chen, A. C., Martin, A. J., Choy, B., Fernandez-Penas, P., & Dalziell, R. A. (2021). A systematic review of the relationship between sunscreen use and the prevention of melanoma and non-melanoma skin cancer. European Journal of Dermatology, 31(1), 1-9. PMID: 33635047

Journal, year, volume and pages

PMID

Actual ChatGPT-3.5 output, 12 April 2023

Bhattacharyya et al. (2023), Figure 1. CC BY 4.0.

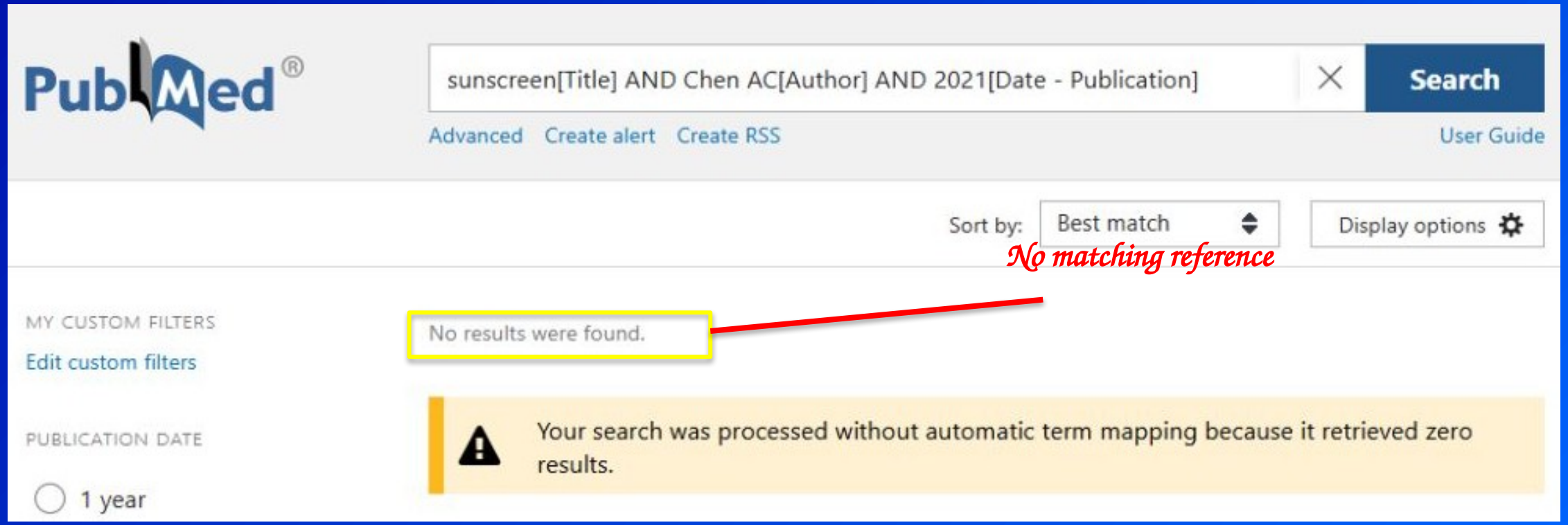
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AI-based discovery

Spot AI search pitfalls

Can we find the cited review?

Title contains sunscreen; author Chen AC; publication year 2021



Stockholms universitet

AI-based discovery

Spot AI search pitfalls

Where does the PMID lead?

The citation supplies PMID 33635047



This PMID identifies a different paper

Stockholms universitet

AI-based discovery

Spot AI search pitfalls

Why can references go wrong?

Generating without checking records

A plausible citation can appear without a matching publication.

Pressure to complete the list

A fixed quota and full metadata may exceed the available evidence.

Assumptions in the question

The request may assume that supporting studies already exist.

Incomplete or mixed source details

Authors, titles and identifiers can be incorrectly combined.

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AI-based discovery

Spot AI search pitfalls

Ask for references that can be checked

Problematic prompt

From your existing knowledge, list **exactly ten** primary studies on intronic human HBB variants and RNA splicing. Include **complete citations, DOIs and PMIDs**. **Return only the bibliography**.

Improved prompt

Search for up to ten primary studies on intronic human HBB variants and RNA splicing. Include only references **verified against PubMed or publisher records**. Give **direct links** and **identify unverified details**. **Fewer papers are acceptable**.

A better prompt helps. The references still need checking.

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AI-based discovery

Spot AI search pitfalls

A real paper, the wrong conclusion

Saline or albumin in patients with traumatic brain injury (NEJM, 2007. PMID 17761591)

ChatGPT's summary

"no significant difference in mortality between saline and albumin"

Recorded in Weinberg et al. (2024)

The paper's result

Mortality at 24 months

Albumin	**33.2%**
Saline	**20.4%**

P = 0.003

The summary lost the study's finding of higher mortality.

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From questions to evidence

Let the question guide the search

Choose the relevant biological level and type of information.



Connect the records

Use names for discovery, identifiers and cross-references for precision.



Choose a suitable access route

Record pages, programmatic access and genome browsers.



Check what supports the answer

Distinguish observations, predictions and interpretations.
Verify AI-generated claims.

